Table of Contents

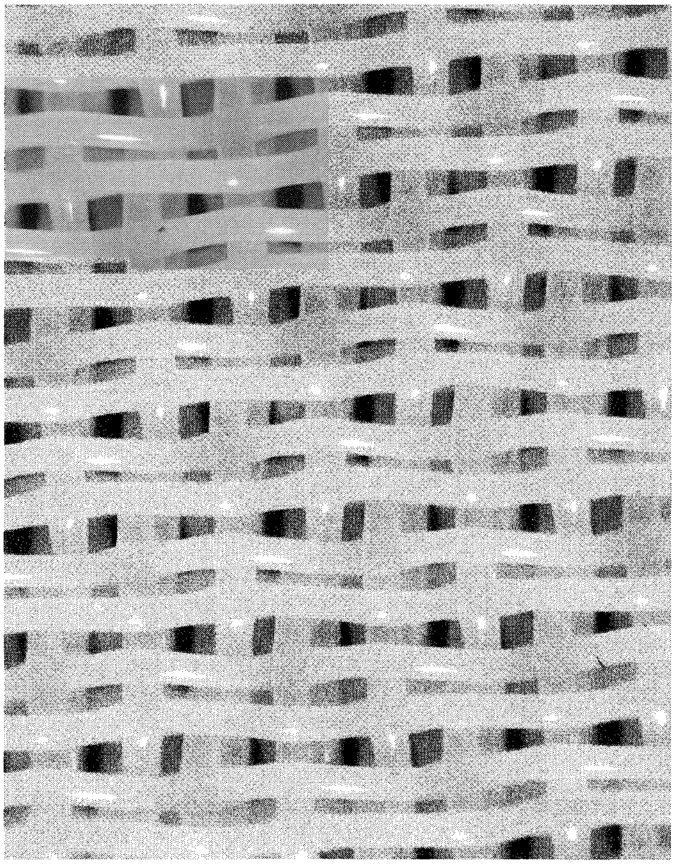


FIGURE 2.57. Twinning of top filling yarns in a two layer fabric.

2.2.4 *Properties and Testing of Forming Fabrics*

Many different factors are taken into account and are essentially built into the fabric when it is designed in order to satisfy the end use requirements. The following design and manufacturing parameters are calculated and recorded:

- material type
- mesh (warp and filling count)
- warp and filling sizes and types
- weave pattern
- plane difference
- percent projected open area
- air permeability
- void volume
- fiber support index
- drainage index
- number of holes per unit area
- hole size (length × width) and diagonal
- fabric thickness
- tensile strength and modulus
- yield as woven
- percent stretch in finishing
- maximum temperature and tension during heatset
- seam tensile strength

Mesh (Warp and Filling Count)

Warp count is the number of machine direction yarns per unit width of fabric and filling count is the number of cross machine direction yarns per unit length of the fabric. Generally, the finer the mesh, the finer the paper grade and the paper quality. Fineness has no direct correlation to fabric drainage. The absolute drainage of a fabric is related to the fabric's "openness" which not only is dependent on the weave pattern and mesh but also on the size (diameter) of the yarns used to weave the fabric. To maintain a level of "openness" for drainage, as fabrics are made finer and finer, the yarn diameters are reduced. Smaller diameter yarns result in reductions of wear resistance and stability.

Lunometers and counting glasses are used to count the number of warp and filling yarns in fabrics. As an approximate rule, a variation of ±1 in warp count and a variation of ±2 in filling count per inch are generally acceptable.

Plane Difference

In many instances, machine direction and cross machine direction monofilament yarns are not in the same plane on either side of the forming fabric. This topography is referred to as plane difference (Figure 2.58). In general, when the MD yarn is above the CD yarn, it is referred to as "warp runner" and when the CD yarn is higher than the MD yarn, it is referred to as "shute (filling) runner." When both are in the same plane, the fabric surface is "monoplane."

To obtain good fabric life, it is desirable to have the wear yarn over the other yarn to take the anticipated abrasion. For good fiber support on the sheet side, in many cases, it is desirable to be as close to monoplane as possible, thus giving support from both the CD and MD yarns to provide low wire marking. In tissue, plane difference is good for bulk generation.

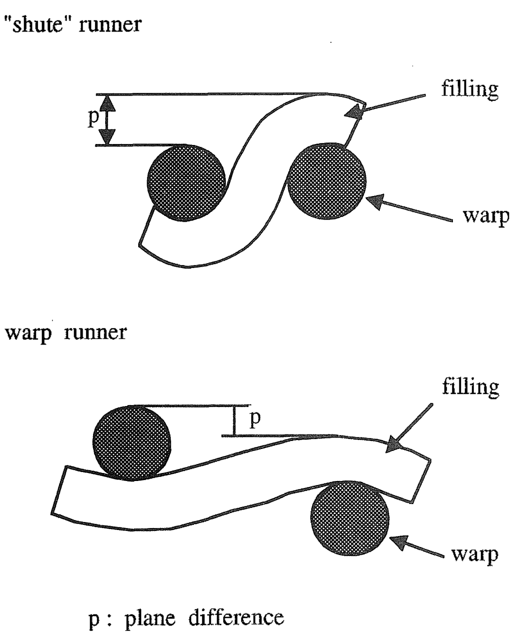


FIGURE 2.58. Plane difference.

With multilayer structures, some of which are inherently stable, material stiffness and heatsetting techniques are used to optimize plane difference levels. Optimization is possible for both the sheet and wear side for a particular application. Sometimes, forming fabrics that will be used for finer paper grades are polished with an abrasive roll. Polishing (sanding) removes the acute knuckle points on the fabric and provides a smooth, monoplane surface as explained earlier.

Percent Projected Open Area

It is a calculated number that describes, in a plane view, the percent projected open area. When comparing fabrics made with the same weave pattern and materials, it is a fair indicator of how each fabric will handle water. Open area is not a good indicator of drainage capability if changes are made to weave patterns, material or mesh. Open area does not take into account weave pattern, caliper, plane difference, material, level of crimp and wetted surface area. Open area is normally calculated based on mesh and yarn diameters.

$$\begin{aligned}\%\ \text{Open area} &= [1 - (\text{warp count})(\text{warp size})] \\ &\times [1 - (\text{filling count})(\text{filling size})] \times 100 \end{aligned} \tag{2.5}$$

For single layer fabrics, the percent open area is always larger than zero. For two layer fabrics, it ranges from 0 to a certain value. Figure 2.59 shows different levels of open area for different two layer forming fabrics. A 0% open area does not mean that there is no opening for water drainage. Figure 2.60 shows the top view and 30° angle view of a two layer fabric. Although the projected open area is zero, there is plenty of open area when seen from a 30° angle.

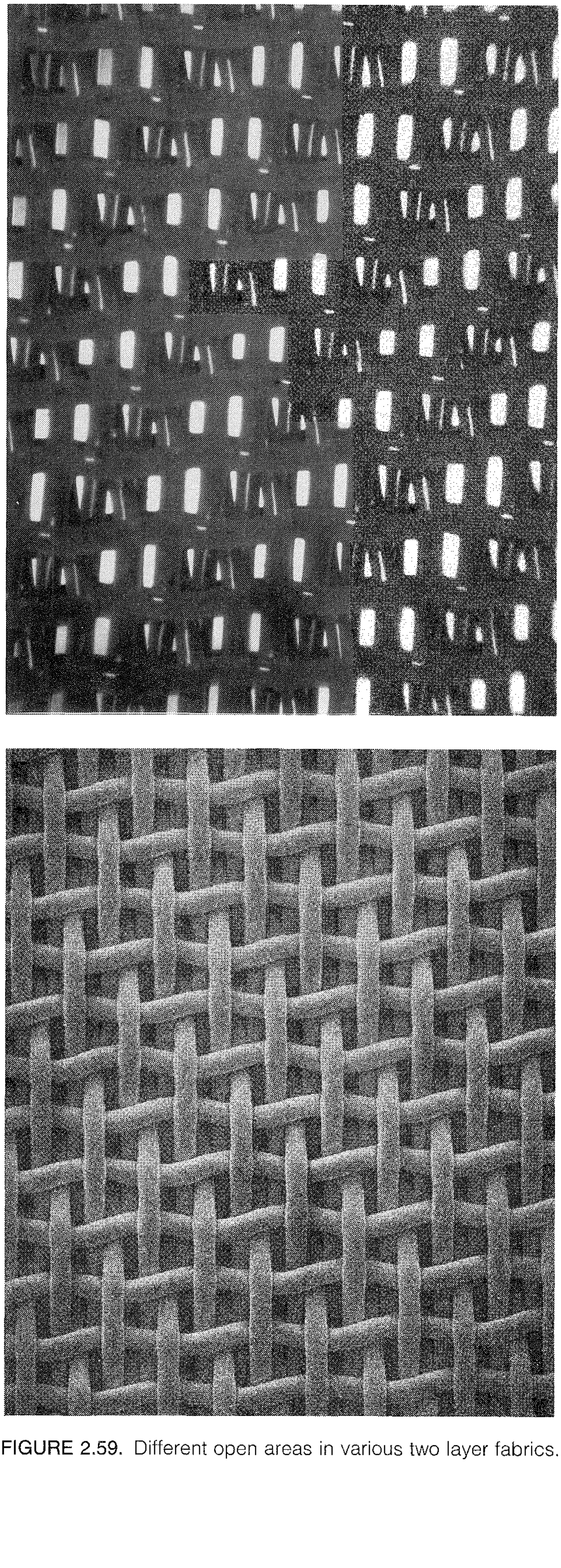


FIGURE 2.59. Different open areas in various two layer fabrics.

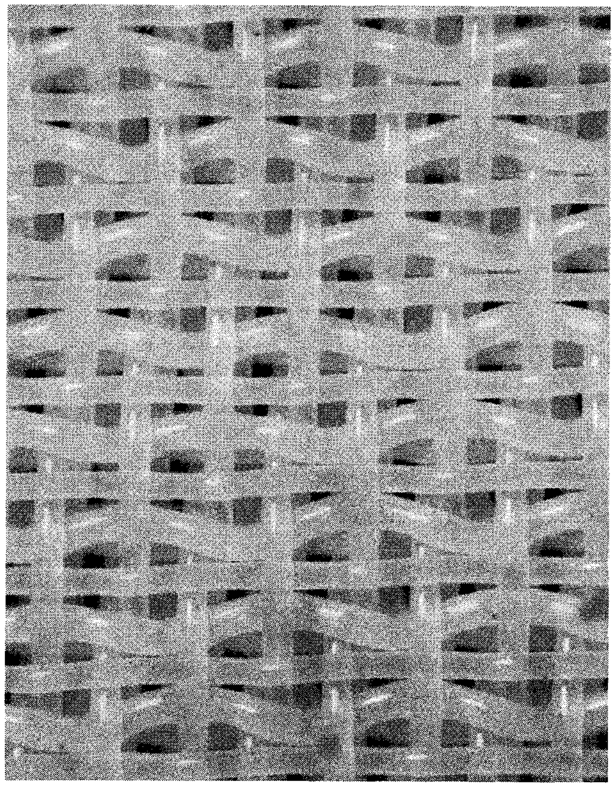
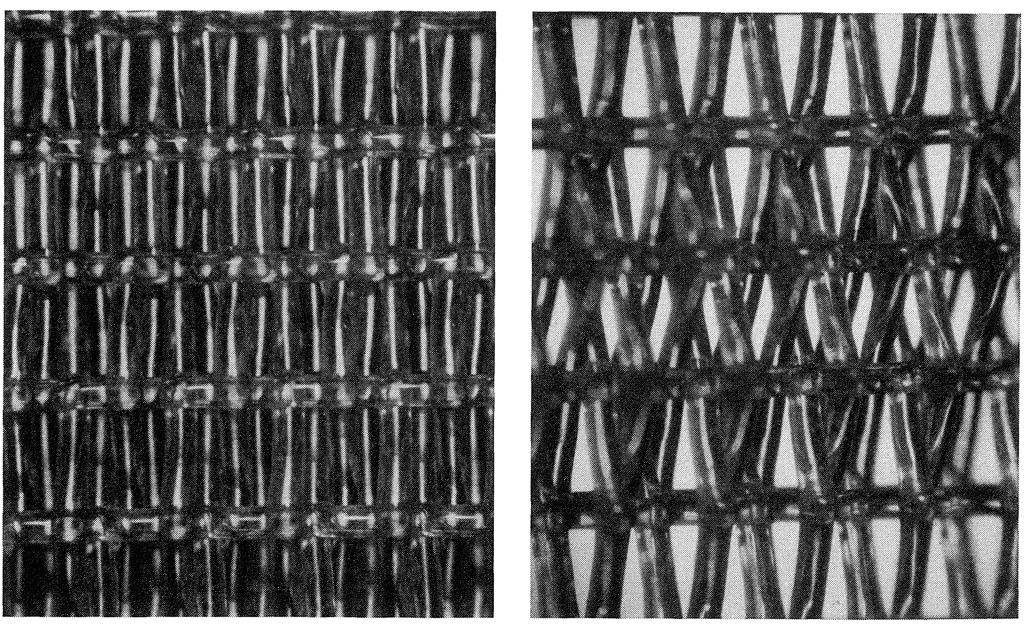


FIGURE 2.59 (continued). Different open areas in various two layer fabrics.



top

30° angle

FIGURE 2.60. Top view (left) and 30° angle view of a two layer fabric.

Air Permeability

It is a measure of air flow through a forming fabric at the standard pressure drop. The standard conditions are cfm/sq ft at 0.5 inch water column pressure.

Air permeability is of considerable value in predicting drainage characteristics of a fabric. It is, however, an air flow measurement, whereas the papermaker is interested in water flow. Within a given weave pattern and mesh range, air permeability, as a drainage indicator, may not correlate with actual machine data as sheet support characteristics also influence water handling capability. Nevertheless, air permeability is one of the most important properties of forming fabrics.

Air permeability is also a good indicator of fabric to fabric uniformity as long as weave and yarn diameters are constant.

Void Volume

Void volume, by definition, is the amount of space in a volume of fabric that is not occupied by solid material (Figure 2.61). Most single-layer and two layer fabrics have void volumes ranging from 60–70%. Both fabric thickness and internal structure affect absolute void volume.

Some studies have shown that having a high level of interconnected internal void volume could enhance dewatering over vacuum augmented boxes. On the other hand, a fabric with too high a void volume can lead to a wetter than normal sheet off the couch, due to the fabric having a high volume of air carried in the fabric.

Within a forming fabric, the void volume is not evenly distributed; some areas are more closed and it is the most closed area that best describes the water handling capabilities of a particular fabric. This area, called the "Minimum Average Hole Area (MAHA)" or "bottleneck," can be found by physically sectioning the fabric and measuring percent open area at the different planes through it.

The passage of water through a fabric is limited by the least permeable plane of a bottleneck within the fabric structure, but not by the projected open area. This bottleneck changes in shape and location when the material used to make the fabric is changed. It can also be altered by using different weave patterns. Void volume can be calculated as follows:

$$\begin{aligned}&\text{Total volume} = \text{length} \times \text{width} \times \text{caliper}\\&\text{Calculated void} = \text{total volume}\\&\quad - (\text{weight of sample/density of material})\\&\%\ \text{void volume}\\&\quad = (\text{calculated void/total volume}) \times 100\end{aligned} \tag{2.6}$$

Fiber Support Index (FSI)

The concept of fiber support index was developed by Beran [13]. His work, based on a statistical model, predicts the fiber support characteristics of forming fabrics based on

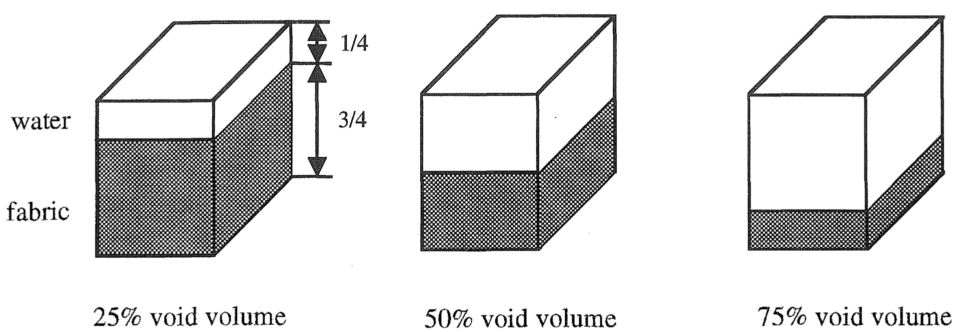


FIGURE 2.61. Void volume.

mesh, weave type and run configuration, i.e., long warp knuckle up or down (Section 2.3.2).

Beran made four major assumptions:

- fabrics woven were planar grids with no thickness
- support from yarns is based on a line with no width
- yarns are straight and parallel
- yarns are uniformly spaced

Having made these assumptions, Beran then devised a rather complicated mathematical formula for his model. This was, however, simplified to his basic formula:

$$\text{FSI} = \frac{2}{3}(a \times N_m + 2 \times b \times N_c) \quad (2.7)$$

where

N_m = number of MD yarns per inch
N_c = number of CD yarns per inch
a = support factor for warp yarns
b = support factor for filling yarns on sheet side

Support factors a and b are calculated as follows:

$$a = \frac{\text{number of filling yarns under warp knuckle} + 1}{\text{number of harnesses}}$$

$$b = \frac{\text{number of warp yarns under filling knuckle} + 1}{\text{number of harnesses}}$$

Table 2.6 lists the single layer FSI factors, a and b. In general, the warp factor, which contributes to the fiber support number, decreases with the increase in number of harnesses.

FSI calculation examples for single layers:

- 80 × 75 mesh, 1/4 weave with long warp knuckle down

$$\text{FSI} = \frac{2}{3}[0.4(80) + 2(1)(75)] = 121.3$$

- 80 × 75 mesh, 1/4 weave with long warp knuckle up

$$\text{FSI} = \frac{2}{3}[(80) + 2(0.4)(75)] = 93.3$$

- 80 × 75 mesh, 2/3 weave with long warp knuckle down

$$\text{FSI} = \frac{2}{3}[0.6(80) + 2(0.8)(75)] = 112$$

FSI factors for two layer and two layer extra designs are determined by viewing fabric samples and establishing warp and filling profiles. For two layer designs, half of the total filling count is used since filling yarns are stacked. For two layer extra, 2/3 of the total filling count is used. For three layer factors, the factors for top layer (sheet side) only are used.

Examples of FSI calculations for multilayer fabrics:

- two layer (warp count = 162, seven harness, number of filling yarns under warp knuckle = 2, total filling count = 120, number of warp yarns under filling knuckle = 5)

$$\text{FSI} = \frac{2}{3}[162(3/7) + 120(6/7)] = 114.9$$

- two layer extra (warp count = 150, seven harness, total filling count = 135)

$$\text{FSI} = \frac{2}{3}[150(2/7) + 180(6.5/7)] = 140.0$$

- three layer (top layer: plain weave, top warp count = 80, top filling count = 140)

$$\text{FSI} = \frac{2}{3}[(1)80 + (1)140] = 146.7$$

For fiber support index, a difference of between 5 to 10 is considered to be significant. There are several interpetations of Beran's work. In comparing index values, it is important that the same formulas are used. It should also be noted that Beran's work was done with single layer fabrics and the results are extrapolated to multilayer fabrics. There are

TABLE 2.6. Single Layer FSI Factors.

	LWU		LWD	
Weave	a	b	a	b
1,1	1	1	1	1
1,2	1	0.67	0.67	1
1,3	1	0.50	0.50	1
1,4	1	0.40	0.40	1
2,2	0.75	0.75	0.75	0.75
2,3	0.80	0.60	0.60	0.80
1,6	1	0.29	0.29	1
2,5	0.86	0.43	0.43	0.86
1,7	1	0.25	0.25	1
2,6	0.89	0.38	0.38	0.89

LWU: Long warp up.
LWD: Long warp down.

no known studies extending the theory to multilayers.

Wear Prediction

Several methods of wear prediction have been given in the literature. A general formula that has been used is:

$$\text{Wear Prediction Number } (W) = 0.5\, N^{0.5} L^{0.5} D^2 \qquad (2.8)$$

where

N = number of knuckles/square inch
L = knuckle length
D = cross direction (CD) yarn diameter

The number of machine direction yarns divided by harness number gives the number of knuckles along the CD yarn per one inch. When this number is multiplied by the number of CD yarns/inch, the result is knuckles/square inch. The reciprocal of the machine direction yarns multiplied by the float length in the CD yarn results in the knuckle length.

Example:

For a four harness (1/3), 86 × 60 design with an 8 mil diameter CD yarn

$$N = (86 \div 4) \times 60 = 1290 \text{ knuckles/in}^2$$

$$L = (1 \div 86) \times 3 = 0.035 \text{ inch}$$

$$W = 0.5\sqrt{1290}\sqrt{0.035} \times 8^2 = 215$$

The Wear Prediction Number, W, is a relative number and only has merit in comparison with other fabric design wear numbers. Higher W indicates better wear resistance. For example, a design with $W = 215$ should show better wear characteristics than a design where the wear number is 200. The wear prediction number should be used as a "tool" in the overall evaluation of fabric designs for machine applications. It should be noted that material type is a key factor for wear resistance.

Drainage Index (DI)

The Drainage Index was developed by Johnson as a tool for rating fabrics based on relative drainage rates [14]. Drainage Index takes into account air permeability in cubic feet per minute, filling count, and filling factor as follows:

$$\text{DI} = b \times P \times N_c \times 10^{-3} \qquad (2.9)$$

where

b = support factor for filling yarns on sheet side
P = air permeability
N_c = filling count

For two layer fabrics, only half of the total filling count is used for calculation since only half of the filling yarns support the sheet. For double layer extra fabrics, 2/3 of the total filling count is used in the calculation. For three layer fabrics, only the top layer filling count is used.

DI calculation examples:

- two layer (air permeability = 320 cfm, total filling count = 120, seven harness)

$$DI = (6/7)320 \times 60 \times 0.001 = 16.5$$

- two layer extra (air permeability = 400 cfm, total filling count = 135, seven harness)

$$DI = (6.5/7)400 \times 90 \times 0.001 = 33.4$$

- three layer (air permeability = 400 cfm, top filling count = 70, plain weave)

$$DI = (1)400 \times 70 \times 0.001 = 28.0$$

In general, a difference of 1 or greater in drainage index is considered to be significant.

Hole Size (Length and Width) and Diagonal

Hole dimensions are significant for predicting the fiber bridging ability of the fabric.

$$\text{MD dimension of the hole} = [1 - (\text{filling count})(\text{filling diameter})] \div \text{filling count} \tag{2.10}$$

$$\text{CD dimension of the hole} = [1 - (\text{warp count})(\text{warp diameter})] \div \text{warp count} \tag{2.11}$$

$$\text{Hole diagonal} = \sqrt{(\text{MD dimension})^2 + (\text{CD dimension})^2} \tag{2.12}$$

$$\text{Holes per unit area} = \text{warp count} \times \text{beat count} \tag{2.13}$$

Fabric Thickness

Fabric thickness is important since it affects the drainage characteristics. Too thick fabrics may cause water carryback. The standard test method for measuring thickness of textile materials is given in ASTM D 1777 [9].

Tensile Strength and Modulus

Fabric modulus is measured using ASTM Test Method D 885 [9]. Tensile properties of fabrics are measured using ASTM test method D 5034. Specifications of textile machines for tensile testing are described in ASTM D 76. The terminology of tensile properties of textiles are given in ASTM D 4848.

The tension measurement along the fabric width is done during manufacturing (on the heatset table) to determine if there is any length difference in the fabric along the machine direction.

Seam Tensile Strength

Seam strength is extremely important for proper functioning of fabric under high tension on the paper machine. Ideally, seam strength should be as good as fabric strength. However, seam strength is generally less than that of the fabric body. Seam efficiency is defined as

$$\text{Seam Efficiency (\%)} = (\text{Seam tensile strength} \div \text{Fabric tensile strength}) \times 100 \tag{2.14}$$

2.3 Applications of Forming Fabrics

The design of the forming fabric has an effect on sheet smoothness, retention characteristics and sheet integrity, which in turn affect the optical and physical properties of the sheet.

The forming fabric is just one parameter in the sheet forming process. The stock furnish, consistency, freeness, pH and fiber distribution are some of the other factors that will affect the final sheet properties. The retention of the fines and fillers must be optimized during formation. The sheet formation must be uniform. Fiber alignment in various directions (including the Z-direction) must be carefully controlled to ensure that there are not significant differences. The Z-direction changes will cause curl which will put constraints on other parts of the paper machine.

Sheet formation is affected by the bridging of the open areas in the forming fabric by the long fibers that are placed onto it. This bridging will subsequently affect the retention of the fines and fillers that are inherent in the sheet. Fabric hole size and shape affect the sheet characteristics. By changing the weave pattern, mesh and beat count, different hole sizes and shapes can be obtained as shown in Figure 2.62. In general, optimum bridging takes place over a square open area. This allows the fibers to interact and form a lattice that will in turn affect the fines and fillers retention.

In a fabric structure, the point of initial fiber bridging in the fabric is the point where a plug of fiber-filler-fines begins to form. As the fabric structure is changed by altering yarn diameters, the depth in the fabric of initial fiber bridging is changed. Using finer yarn diameters and higher mesh counts while maintaining or increasing the percentage of open area decreases the depth of the probable point of bridging. This in turn results in a plug of less total mass because it is not formed as deeply in the fabric. This shallower plug mass comes closer to matching the average mass of the sheet. In other words, the height of the hills on the fabric side of the sheet becomes less. A lower hill means more uniform calendering and a smoother sheet.

In Figure 2.63, $d_1 > d_2$. The fiber length a is the same in both cases which is equal to the distance between two adjacent yarns. As it can be seen from the figure, the depth of the fiber bridging is less in the case of small diameter yarns because $b > c$.

Wire mark is the result of optical and physical characteristics of the sheet caused by point to point density differences and surface differences caused by fabric topography. By minimizing the plug density effect and the height of the knuckle above the plane of the fabric, the wire mark will be minimized. Surfacing of fabrics to obtain monoplanarity is done to reduce knuckle height.

Differences in tension across the fabric can cause hills and valleys to appear parallel to the machine direction in the forming area of the fourdrinier table. Since the stock will tend to run off the hills and into the valleys, streaks of varying basis weight or mass will result.

Forming fabrics have an average running life of three to six months on the paper machine. However, it should be noted that the life of a fabric depends on many things including the installation. Even a small damage to the fabric during installation may make the fabric useless, reducing the life to zero days. On the upper extreme, some forming fabrics, especially coarse fabrics, can run more than a year on the new, modern machines. A vast majority of fabrics are removed due to damage; very few are actually worn out.

2.3.1 *Fabric Design Considerations*

There are two major sets of considerations for forming fabric applications: paper type and paper machine design as shown in Table 2.7. Each of these factors in itself is extremely important. A survey of the machine is required prior to the manufacture of the first fabric. The survey should be updated periodically (Chapter 5).

Paper Considerations

The grade of paper is the first consideration. Table 2.8 lists recommended typical designs for common paper grades. Some machines run several different grades of paper and, generally, one design is required to run them all. It is then important to know what the predominant grade is as well as the most costly grade. The fabric may have to be designed to favor the highest quality grade, yet can only be designed to run the remaining grades adequately.

The type of furnish is important in determining how large a hole size can be tolerated. The makeup of hardwood, softwood, recycled and rag, as well as the combinations made from these greatly affect the decision

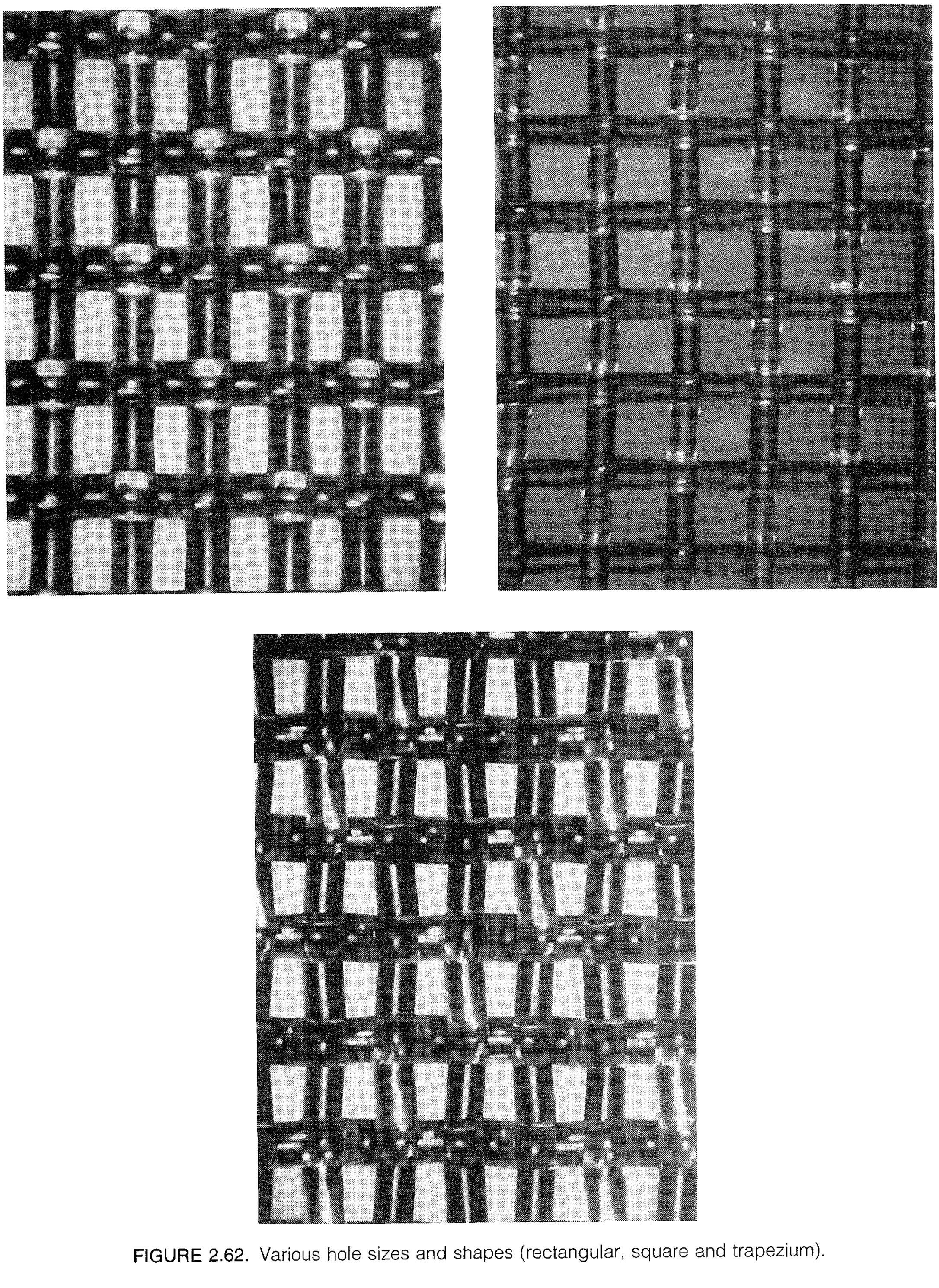


FIGURE 2.62. Various hole sizes and shapes (rectangular, square and trapezium).

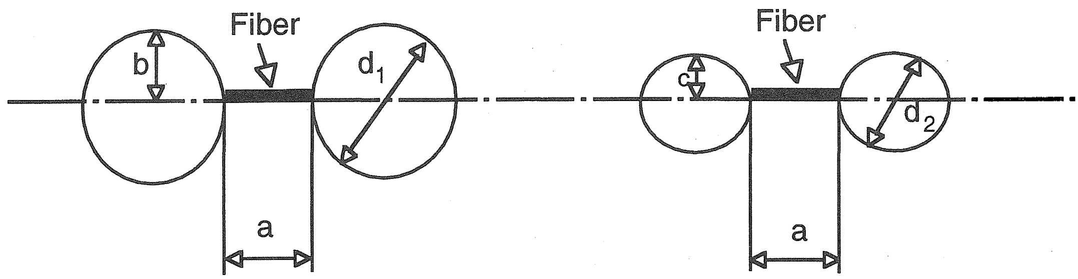


FIGURE 2.63. Fiber bridging.

on the number of cross machine yarns in the design as well as the fineness of the design. Due to the rising cost of pulp and the trend to make things recyclable, more and more recycled furnish is being used. This presents a finer fiber to the furnish and is more apt to bleed. Finer mesh designs are required for a furnish containing recycled material. The weight range as well as the predominant weight is important and requires the same considerations as the grade of paper to be run.

Paper Machine Considerations

Type and manufacturer of paper machine indicates what basic machine is being dealt with and along with width and length, indicate if a design has to be coarse or fine. Knock down type machines generally have more problems with fabric sizing than do cantilever machines. Twin-wire machines have two design considerations (Section 2.5.2). The speed range as well as the predominant speed is critical. Most generally the fabric will be designed to run best at the predominant speed and run adequately at the various speed changes.

In fourdrinier machines, the table configuration indicates how drainage takes place on the table as well as the sheet formation. A table consisting of all rolls could require a more open fabric design than a combination of rolls and foils or all foils (Section 2.5.1).

The operating tension and the available take-up are extremely critical. Long machines with short take-up, operating at high tension, will experience difficulty in fabric installation. This becomes a more serious problem if a fine mesh fabric design is required to run the product. The solution would be to install adequate take-up, but generally a design tradeoff for a higher modulus coarser mesh fabric is made to compensate for the lack of takeup, resulting in quality somewhat lower than what would be expected from a finer fabric.

2.3.2 *Design Selection*

Proper choice of fabric design can provide benefits in minor variations and proper manufacture of the chosen fabric will relieve more serious problems. Fabric history is a major consideration in eliminating trial and error in fabric design. It establishes a good sense for

TABLE 2.7. Information Needed to Select the Proper Fabric Design for a Given Application.

Paper	Paper Machine
• Grade of paper to be run	• Type and manufacturer
• Type of furnish (recycled stock content)	• Size (width and length)
• Weight range of product line	• Speed range
• Wire mark considerations	• Type of pickup (open draw, suction or shoe)
• Type of forming (pressure or velocity)	• Table configuration, rolls versus foils, etc.
• Fabric history	• Operating tension and available take-up
• Similarity to other successful applications	

TABLE 2.8. Typical Recommended Designs for Common Paper Grades.

	Mesh (warp count × shute count); Air Permeability (cfm)			
	Single Layer Four Harness	Single Layer Five Harness	Double Layer	Three Layer
Corrugated Med.		46 × 37; 600	126 × 90; 550	78 × 72; 500
Linerboard				
lightweight		45 × 45; 580	126 × 96; 500	78 × 70; 400
medium/heavyweight		46 × 37; 605	126 × 90; 550	78 × 72; 550
Bleachboard		62 × 46; 650	144 × 135; 470	78 × 70; 400
Bag	58 × 48; 600	64 × 40; 550	144 × 135; 470	78 × 70; 400
Bond		92 × 70; 560	144 × 135; 400	72 × 80; 450
Duplicating		92 × 70; 560	144 × 135; 470	72 × 80; 450
Writing		92 × 70; 560	180 × 126; 450	72 × 90; 400
Directory		92 × 70; 560	144 × 135; 470	72 × 80; 450
Newsprint		92 × 70; 560	180 × 126; 450	72 × 70; 500
Rotogravure		92 × 70; 560	180 × 140; 400	72 × 80; 450
Coated book		92 × 70; 560	144 × 135; 470	72 × 80; 450
Publication		92 × 70; 560	144 × 135; 470	72 × 80; 450
Glassine		92 × 70; 560	144 × 135; 470	72 × 80; 450
Wrapping		92 × 75; 515	144 × 135; 400	72 × 90; 400
Kraft		84 × 75; 675	144 × 120; 525	78 × 70; 450
Fine		92 × 70; 560	144 × 135; 400	78 × 75; 400
Tissue		86 × 82; 680	200 × 165; 500	80 × 120; 550
Towel		84 × 75; 675	144 × 120; 600	80 × 120; 500
Napkin		84 × 75; 675	144 × 120; 600	80 × 120; 500
Wadding		84 × 75; 675	144 × 120; 600	80 × 110; 550
Carbonizing		110 × 90; 500	180 × 140; 400	78 × 75; 300
Cigarette		110 × 90; 400	180 × 165; 325	72 × 90; 400
Nonwovens		36 × 32; 700	84 × 66; 600	78 × 75; 400

fine tuning designs. The similarity to other successful applications is important. A successful design on a specific machine is a plus factor to be utilized with all of the previously mentioned points.

The design of the fabric for a paper machine must take into account grade, furnish, machine layout and operation. The furnish and grade influence the selection of design; machine layout and operation govern the actual drainage characteristics required within the design parameters of mesh and yarn diameters.

The design selection process encompasses two major concerns: the design of the fabric itself and the run attitude.

In a fabric design, there are some independent variables that control various dependent variables as shown in Table 2.9. The weave type specifies the number of harnesses (plain, four harness, five harness, etc.), warp and filling profiles (number of cross-over points and knuckle length) as well as the number of layers in the fabric (single layer, two layer, two layer extra, three layer). Any change in one of the independent variables will affect the controlled parameters. Each is weighed for its effect on the design.

The second part of the selection process

TABLE 2.9. Independent and Dependent Fabric Design Parameters.

Independent Parameters	Dependent Parameters
Weave	Open area
Mesh	Air permeability
Strand diameter	Hole size
Strand material	Hole shape
	Modulus
	Caliper
	Drainage index
	Fiber support index

deals with selection of the run attitude. Run attitude describes whether the fabric is run with the cross machine knuckles toward the sheet or toward the machine elements as shown in Figure 2.64. Wire mark considerations are important in determining the run attitude of the fabric.

There are advantages and disadvantages to running in a specific attitude as shown in Table 2.10. The advantages for a specific attitude are generally disadvantages for the opposite attitude. The type of formation is also important in determining the run attitude of the fabric. Pressure formation (Section 2.5.1) can lead to release problems. It also affects the cleanliness of the fabric as well as the drainage ability of a fabric over a long period of time.

The effect of running attitude on fiber support (fiber bridging) and drainage is shown in Figure 2.65. Long CD knuckle to the sheet side gives shorter distances for fibers to bridge resulting in higher fiber support. In the case of the long CD knuckle down, the protruding ("proud") knuckle is in machine direction, which is the direction of water flow. Therefore, the proud knuckles do not disturb the water flow as much. When the long CD knuckle is up, the proud yarns are perpendicular to the water flow direction. Water hits these yarns and loses its velocity and drains faster.

The fiber supporting characteristics of the fabric are primarily dependent on the following factors:

- number of support points
- size of support points
- distribution of support points
- orientation of support points
- surface topography
- hole size and shape

For grades that utilize intense dewatering forces (linerboard) and/or light in weight (tissue), fabrics are generally run long MD knuckle down. This allows better drainage and sheet release. The main disadvantage of forming on the long cross-direction knuckle is that the life potential is not as great as it would be if these CD yarns were the wear members. In fine paper fabrics, the hole is generally longer in MD direction.

Figure 2.66 shows a two layer fabric along with its top and bottom surfaces. The knuckle impressions of both surfaces are also shown which indicate the surface topography for initial fiber bridging.

In the event that a fabric is supplied in one run attitude and it becomes desirable to turn the fabric inside out, it is possible to do so before installation on the machine. Figure 2.67 shows a typical procedure for turning fabrics inside out with the least amount of damage.

In general, there is little choice of run attitude with multilayer fabrics. Single layer fabrics may make paper on either side.

Effect of Harness Numbers on Two Layer Fabric Properties

Assuming the same warp and filling counts and the same size yarns and that fabrics are woven in approximately the same weave styles, the harness number has several effects on fabric properties and performance on the paper machine.

Since there are fewer number of crimps in the warp of higher harness number design, it should have a higher modulus and, therefore, is less susceptible to stretching on the machine.

There is less interlacing of warp and filling in higher harness number design, which results in a higher internal void volume. The higher void volume causes higher air permeability. Consequently, as compared to a lower harness number design, the higher harness number design can have a higher filling count for better sheet support or it can have larger filling yarns for better wear for the same air permeability. However, larger filling yarns will make the fabric thicker in caliper, which might make it more difficult to remove the water.

Since there are fewer number of crimps and less interlacing of warp and filling in higher harness number design, it is not as rigid or has as much lateral stability as a lower harness number design. Therefore, it is more susceptible to skewing if the machine is misaligned. Fabric rigidity enters into guiding and fabrics

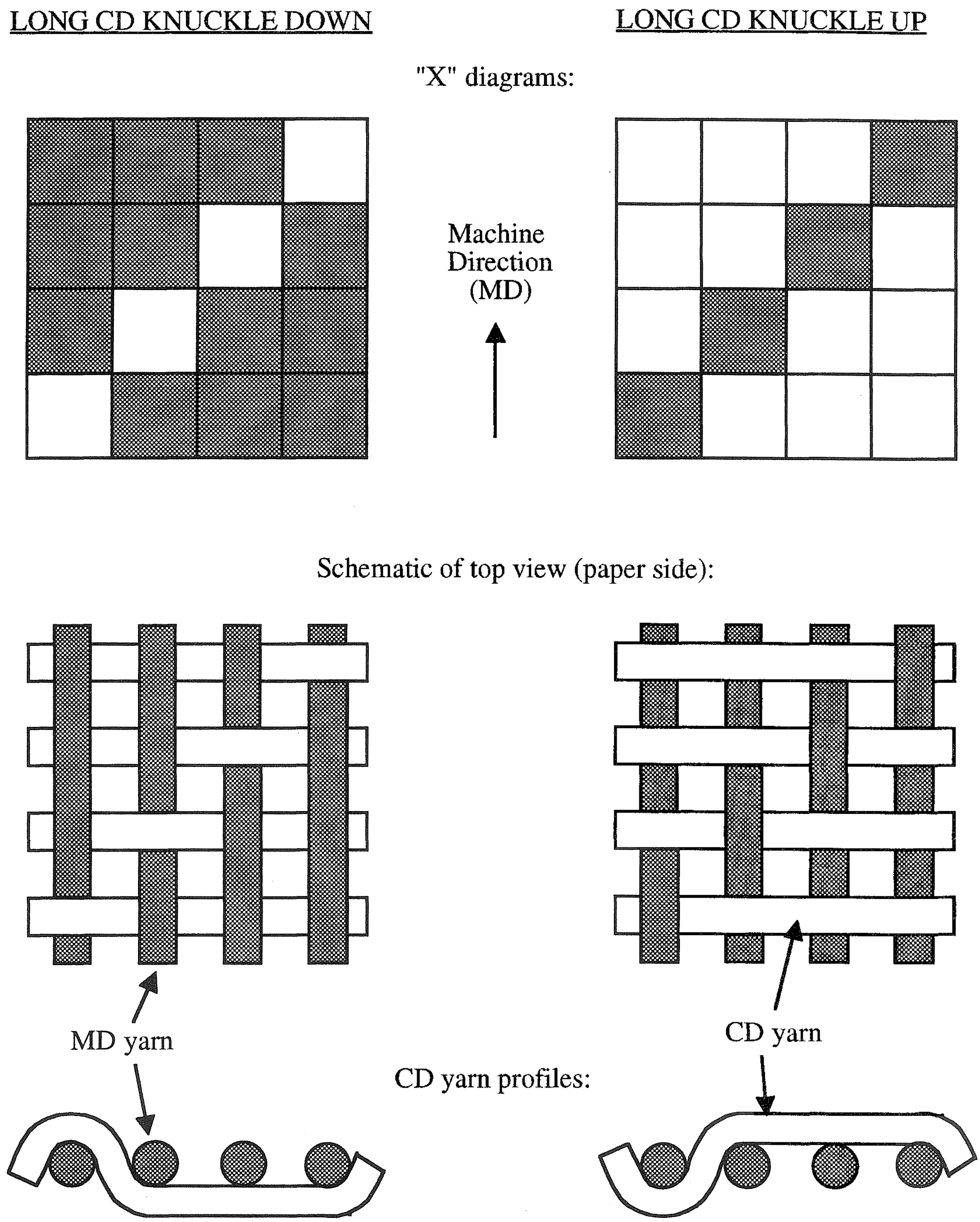


FIGURE 2.64. Fabric running attitudes for 1/3 single layer fabric.

TABLE 2.10. Effects of Fabric Running Attitude for Single Layer Fabrics.

Advantages	Disadvantages
1. Long Cross Machine Knuckle to the Sheet	
1. Earlier formation • good release • easier drainage • lower load 2. Sheet rides higher • less bleed tendency • easier cleaning 3. Better fiber bridging • better sheet formation due to good fiber bridging	1. Shorter life—less wear knuckles 2. Less stability—due to early wear on warp strands 3. More pronounced wire mark 4. More edge curl potential
2. Long Cross Machine Knuckle to the Machine Elements	
1. Better wear resistance, longer life • more strand surface against machine elements • not wearing the warp strands 2. Better stability • less edge curl potential • smoother surface 3. Less wire mark	1. Greater loads—more wear surface 2. Poorer sheet formation due to less fiber bridging 3. Poorer release—due to sheet following fabric 4. More bleed potential—fibers passing through elongated holes 5. More difficult to clean

which are too firm have more of a tendency for guiding problems. In general, the angle of the twill line is not affected by the harness number and, from this aspect, the guiding would be the same.

2.4 Service for Forming Fabrics

In today's industry, major paper machine clothing manufacturers generally provide service to paper companies. Services are designed to assist the papermaker in obtaining a maximum level of efficiency from the paper machine at the lowest cost per ton (see also Chapter 5, Paper Machine Auditing).

2.4.1 *Maintenance of Forming Fabrics*

Optimizing forming fabric life, thereby reducing machine clothing costs, can be achieved by preventive maintenance, good housekeeping and a good understanding of the care and handling of fabrics.

The proper maintenance of forming fabrics and of the machine as it relates to fabrics, can increase fabric life and productivity. Periodic checks of the machine and proper care and handling can maximize a fabric's useful life and machine performance.

It is necessary to prepare the machine area prior to fabric installation. The following steps or precautions should be taken:

- clean area for fabric unrolling
- remove all obstacles
- wash machine prior to installation
- pull fabric evenly during installation (prior to rotation)

Paper Machine Checkpoints

Inspecting the wet-end of the paper machine between forming fabric changes can drastically help to reduce damage during installation and improve fabric life. The following checkpoints have been compiled as a result of the most common points which are overlooked and are the major contributor to early

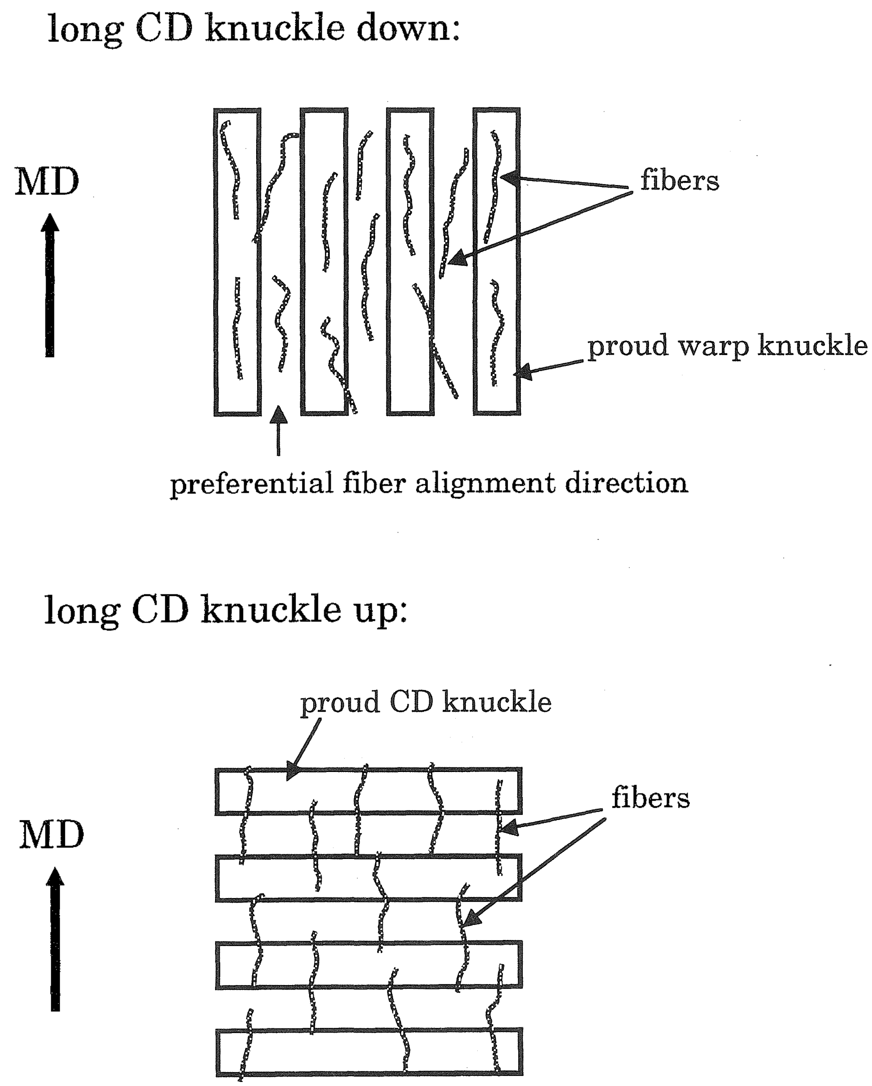
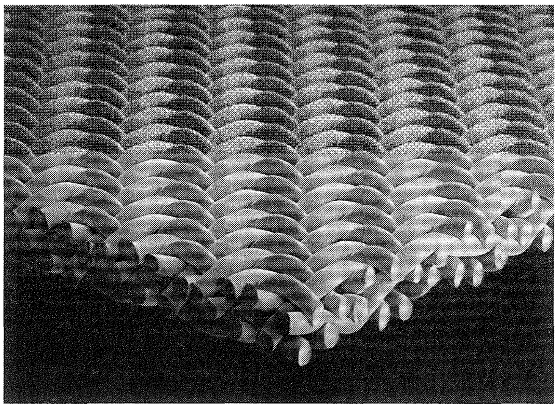


FIGURE 2.65. Effect of fabric running attitude on fiber support.

forming fabric removal due to damage or wear. It should be noted that some of these suggestions are machine specific and therefore should be followed if applicable.

(1) Check foils and forming board edges for rough, thin or chipped areas and for proper alignment. These can cause fibrillation to the underside of the fabrics. The end result is a marked sheet or severe damage to the fabric causing its premature removal. Poor alignment will cause the fabric to guide improperly.

(2) Check table rolls and return rolls for tight bearings, burrs and proper alignment. Tight bearings and burrs will cause considerable damage to the underside of fabrics as well as increase the energy required to drive the fabric. Tight bearings will cause the fabric to slip over the roll surface and fibrillate, closing the holes and restricting the drainage. Burrs on the roll surface will pick at the underside of



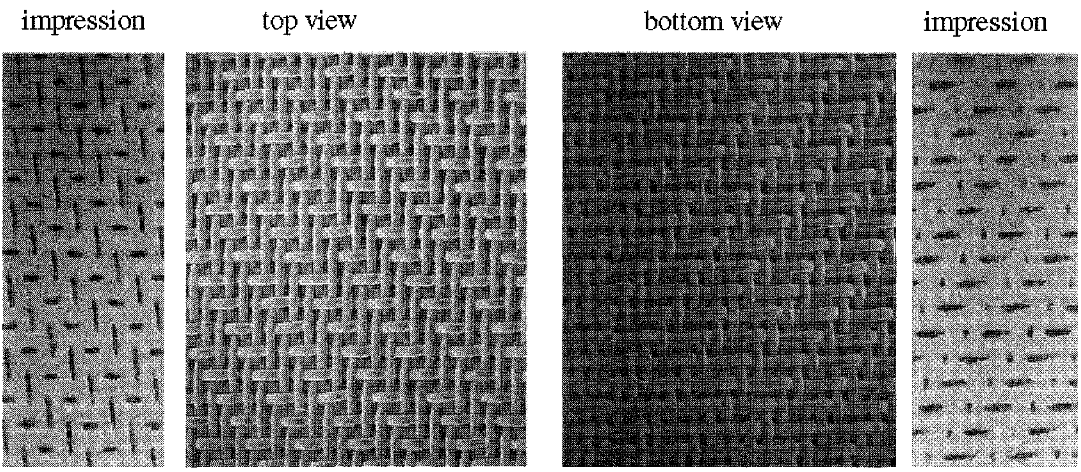


FIGURE 2.66. Two layer fabric knuckle impressions.

the fabric causing a fibrillation streak. This could result in sheet mark and possibly fabric ridging.

(3) Check shower nozzles for proper operation. All showers must be in proper operating condition with no plugged nozzles (Section 2.4.6). A plugged nozzle will cause a dirty streak in the fabric. The dirty streak will restrict the drainage and show up as poor sheet formation. All high pressure needle showers must oscillate when in operation and must only be operated when the fabric is moving. Fibrillation streaks will occur when high pressure showers fail to oscillate during the fabric run and will cut a hole in the fabric if left on when the fabric is stopped. Shower pumps should automatically shut down when the fabric is stopped.

(4) Check return roll doctor blades for grooves and nonuniform pressure. Tight doctor blades will slow a roll down, causing fibrillation to the underside of the fabric. Grooves in doctor blades will allow stock buildup, causing streaks in the fabric and possibly in the sheet. This condition can also cause a fabric to ridge.

(5) Check suction box covers for evenness and correct cut. Suction box covers should be maintained with a regular

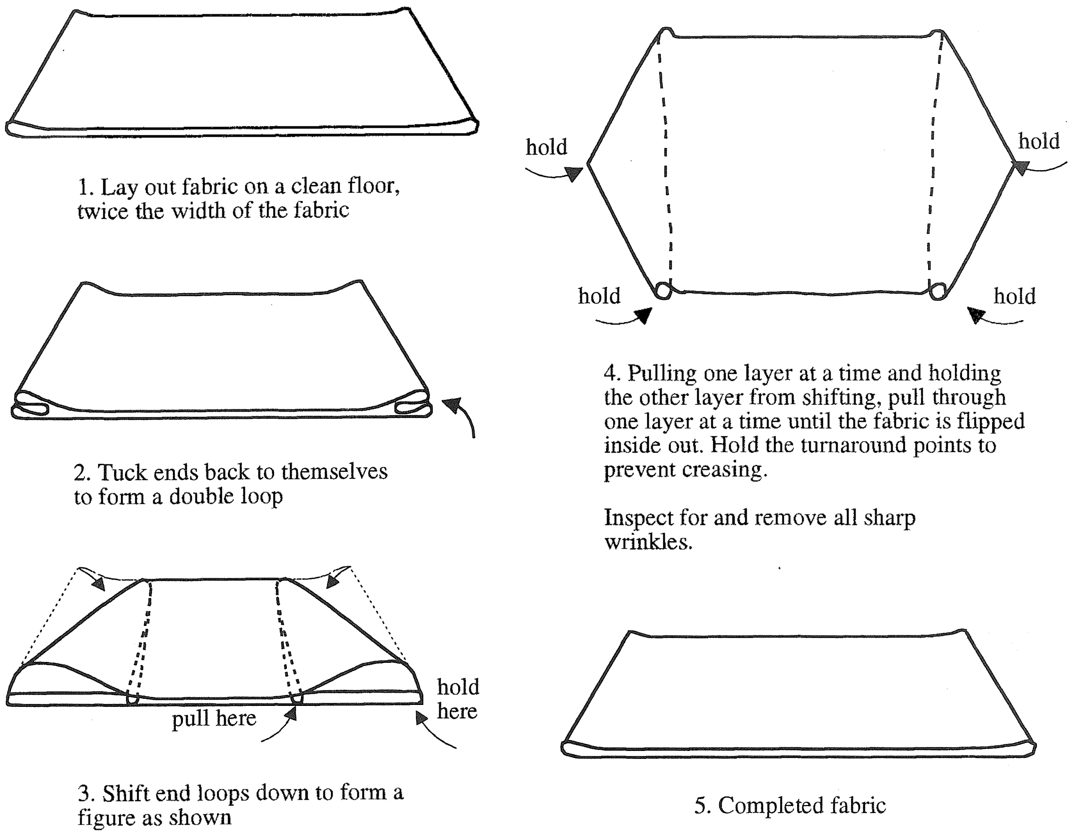


FIGURE 2.67. Turning fabrics inside out.

grind surface. If the surface is too smooth, the fabric will tend to seal off, causing higher loads, bleed and premature wear. If the chamferred edge is worn off, it could possibly cause fibrillation to the underside of the fabric. This condition becomes more serious as the fabric wears in its later stages, whereby the underside of the fabric is monoplane.

(6) Check couch roll, breast roll and wire turning roll for corrosion and pitting. This condition can cause fibrillation on the underside of the fabric leading to poor drainage, premature removal and poor sheet quality.

(7) Check fabric alignment before applying tension and rotation. An improperly aligned fabric can be ridged when tension is applied and the machine rotated. It is also important that the machine alignment front to back is plus or minus, a maximum of 1/4″ (0.63 cm) differential.

(8) Maintain optimum slice clearance. Be sure slice is set properly and also clean. It is common for stock to build up on the underside of the slice. This can be knocked loose during cleanup when a fabric is being installed and can cause serious damage if it becomes lodged between the fabric and the slice during a startup. It is better to try to get minimum clearance and insure good cleanliness under the slice with good showering. Delivery will be better with clearance 0.100″–0.125″ for fine paper up to 0.200″ for brown papers.

(9) Smoking around fabrics during installation or shutdowns should be avoided. Extreme care should be taken when welding near a fabric. A hot spark from a pipe, cigar or cigarette can burn a hole

in a fabric faster than it can be reached to put out. Hot sparks from a weld unit can also burn holes in the fabric as well as embedded particles in the fabric which will cause holes.

(10) Cover all sharp edges during installation of fabrics. Be sure to cover all sharp protruding edges to prevent damage to fabrics during installation. If the fabric is to be socked on, it is recommended that an old fabric be used to cover the floor as well as to cover as much of the front edges of the machine as possible. Minor damage such as wrinkling can be removed by gently folding a wrinkle back in the opposite direction. Cross machine wrinkles will generally disappear when the tension is applied to the fabric. Some damage can require a refinishing of the fabric.

(11) Check guide palm for free movement. Make sure air is turned on. Be sure the guiding surface is not gouged or cut by the fabric edge.

Fabric Tension

Maintaining the proper fabric tension is important in optimizing fabric life and web quality. If a fabric is too loose, premature wear can result from slippage at the drive roll. Poor release is often seen, since the web tends to ride deeper into the mesh, on a slack fabric. In addition, sheet picking may occur if the seam ends are projecting above the plane of the fabric surface. Excessively high tension can cause stretching in the machine direction and narrowing in the cross machine direction. If the tension is high enough, the seam ends can be pulled from their locked position at the seam joint.

It is recommended that a tensiometer, an instrument for measuring tension in forming fabrics, be used to maintain proper tension. It is important that the fabric be run at the tension it was designed for and manufactured at.

The following method is recommended for taking fabric tension reading.

(1) Position
- with tensiometer on inside of fabric no closer than one foot to an outside roll
- no closer than two feet to an ingoing nip
- minimum of 15″ in from the fabric edge

(2) Direction
- Always align long direction of tensiometer parallel with fabric run.

(3) Tension
- Align tensiometer correctly with fabric.
- Gently press tensiometer down until the pointer stops.
- Read indicator. Using the conversion chart convert to PLI (pounds per lineal inch).

Rolls and Table Elements

Fabric life can be significantly reduced by worn or defective rolls and table elements. Thus, a preventive maintenance program, which involves periodic checks of all surfaces which come in contact with the fabric, is highly recommended. Periodic checks will identify potential problems in the early stages, which will not only help to eliminate these problems, but will increase productivity and fabric life.

It is imperative that accurate records be kept of periodic checks, in order to document defects and excessive wear. A defect documentation chart, with diagrams of the machine components, is very helpful to track data as they are collected. Table 2.11 shows the inspection frequency of various parameters.

TABLE 2.11. Recommended Inspection Frequency of Fabric and Machine Parameters.

Checkpoint	Frequency
Nick and burrs	Every fabric change
Bearings	Once per month
Wear	Once every two months
Alignment	Once every two months
Roll covers	Once every two months

2.4.2 *Potential Operational Problems of Forming Fabrics*

There are several areas of potential problems with fabrics on paper machines.

Abrasion

- definition: striations on a wear surface. Usually a narrow groove, channel, fine streak or line associated with a number of other parallel lines. Lines run parallel to machine direction.
- classification: dependent upon physical appearance of striations on wear surface. Abrasion can be classified as light, moderate and heavy.
  —light abrasion: wear surface nearly smooth. It would have striated marks that would be barely visible or shallow in appearance.
  —moderate: wear surface has deeper, more pronounced striations. Tendency of wear surface to appear more wavy and nonsmooth.
  —heavy: wear surface characterized by deep striations. Displays nonuniform, very jagged appearance.
- cause: rough stationary machine components such as suction box covers, fiberglass roll covers, sand, corrosion of breast and couch rolls.

Bow and Skew (Figure 2.68)

The condition in which the CD yarns lie in the fabric in the shape of an arc is called fabric bow. Bow is defined as the greatest distance, measured parallel to the selvages, between a CD yarn and a straight line drawn between the points at which this yarn meets the two selvages. It is expressed as a direct measurement of AB in Figure 2.68, or as a percentage of the fabric width:

$$\% \text{ Bow} = AB/CD \qquad (2.15)$$

Fabric bow can be symmetrical or non-symmetrical. The causes of bow are fabric profile, stock buildup, improperly installed bowed roll and roll deflection. There are several possible actions to correct fabric bow:

- Install a bowed roll.
- Clean stock buildup.
- Reduce fabric tension.
- Install larger diameter rolls.

The condition in which the CD yarns in a fabric do not lie perpendicular to the MD is called skew. Skew is defined as the distance, measured parallel to and along the edge, between the point at which a CD yarn meets one edge, and perpendicular from the point at which the same CD yarn meets the other edge. It is expressed as a direct distance (EF in the figure) or as a percentage of the fabric width:

$$\% \text{ Skew} = (EF/EG) \times 100 \qquad (2.16)$$

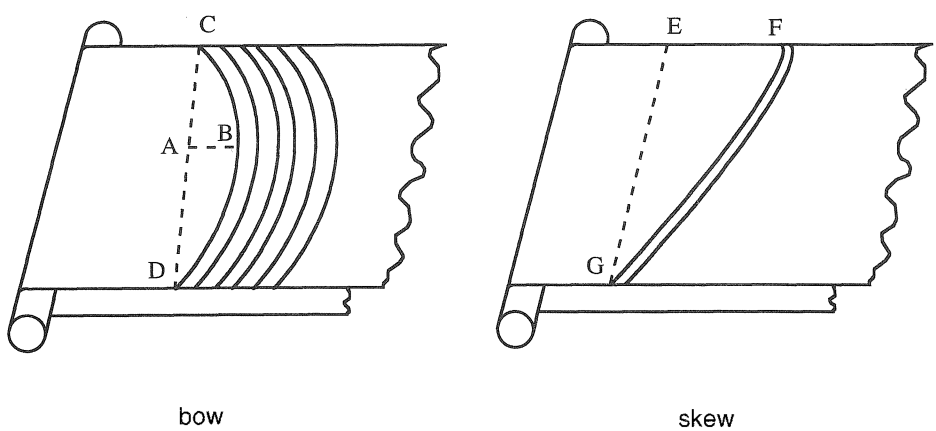


FIGURE 2.68. Fabric bow and skew.

Causes for skew:

- roll misalignment
- stock buildup on roll
- nonuniform roll diameter
- suction box or foil misalignment
- fabric slackened on one edge

Burring

- definition: burring occurs in plastics as a slight extension or flattening of the wear surface edge occurring opposite the fabric run direction.
- classification: dependent upon physical appearance of edge on wear surface, burring can be classified as light, moderate and heavy.
  —light: very little extension of the edge. Edge is nearly in line with edge of wear surface.
  —moderate: more pronounced extension of wear surface. Tendency of extension to curl slightly.
  —heavy: severe extension of wear surface. Extension displays moderate to heavy curl.
- causes: roll slippage, off-speed table rolls, high load operating conditions, high speed applications.

Stretching

Stretching is the extension of the fabric on the paper machine. Low fabric modulus and/or excessive run tension are the cause of fabric stretching. There are two sets of variables which affect the stretching of fabrics on a paper machine: manufacturing variables and paper machine variables.

Manufacturing Variables

(1) Nature of materials: Most forming fabrics are made from polyester. Polyester fabrics will stretch a given amount based on the load applied. Using higher modulus yarn materials improves the situation.

(2) Type of weave: The type of weave has a great deal to do with how much modulus can be incorporated into a fabric. A five harness generally will have a better modulus than a four harness and a four harness better than a three harness. As yarn crimp in the MD is increased, the modulus decreases. Modulus is a measurement of the fabric elongation at a given load (Section 2.2.4).

(3) Yarn diameter: Larger yarns have a greater resistance to stretching than smaller yarns.

(4) Temperature and time at temperature: The temperature at which a fabric is manufactured as well as how long it is at that temperature has an effect on how much modulus is incorporated into the fabric. Proper heatsetting increases fabric modulus drastically. Using higher stretch tension during heatset improves the situation. This is because the crimp in the MD yarn will be decreased. The straighter the MD yarn, the higher the fabric modulus. Yarn has a higher modulus than the fabric. Crimp in the yarn causes modulus to drop under load.

(5) Yield: Yield is the final length of a machine direction yarn in the fabric, compared to its length before starting manufacture of the fabric. Generally a fabric having higher yield has less tendency to stretch.

Paper Machine Variables

(1) Operating tension: Fabrics are sized to be at a given length for a given tension. The fabric length will follow the tension if it is increased or decreased.

(2) Drainage components: The drainage components on the table affect the tension to which a fabric is subjected.

Fabrics are processed to modify the amount of stretch they will yield for a given load. Figure 2.69 shows typical stretch amount a fabric will yield when subjected to a specific load. Referring to the figure, to find the fabric stretch, select a desired load, trace to modulus

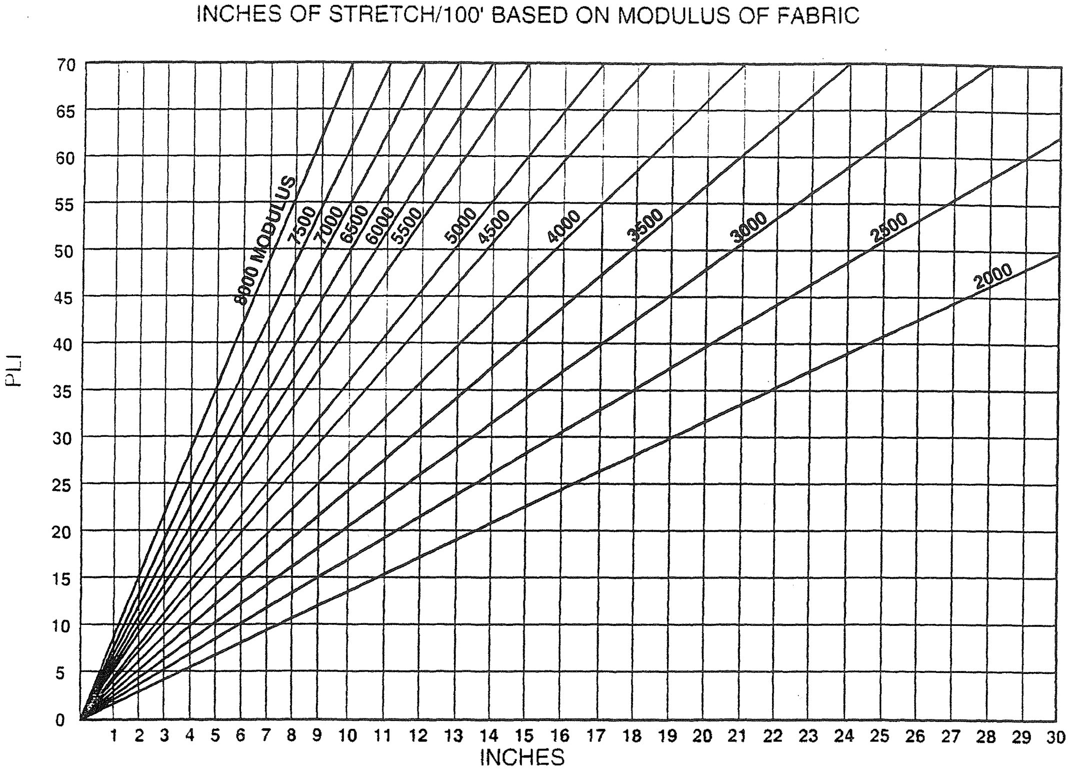


FIGURE 2.69. Fourdrinier fabric stretch chart.

and drop perpendicular to find fabric stretch in inches. For example, at a running load (tension) of 25 pli, a fabric with an 8000 pli modulus will stretch approximately 3.5″ per 100′ whereas a fabric with a 2500 pli modulus will stretch 13.5″ per 100′.

Then the total fabric stretch (TFS) is calculated by:

$$\text{TFS} = (\text{machine length in feet}/100') \times (\text{inches}) \tag{2.17}$$

Ridging

Ridging is caused by five major conditions.

(1) The flatter a fabric is, the less susceptible it will be to ridging.
(2) Damage: A fabric can be ridged by severe damage in the machine direction as well as through stock buildup on rolls.
(3) Loading: Loading is extremely critical. If the power ratio between the couch and wire turning roll is incorrect, a fabric will tend to ridge.
(4) Machine misalignment: Machine misalignment is a prime suspect when a fabric creases at an angle as opposed to a straight machine direction line.
(5) Table shake: Excessive table shake can cause a fabric to ridge.

Using a crayon tracing technique, it was shown that ridges in forming fabrics could be the cause of ridges in finished rolls of paper. If a ridge develops in a fabric, it can usually be removed, or lessened, if done properly. When attempting to remove a ridge, the fabric should be held under tension.

Fabrics are generally flattest at their heatset tension. If the fabric is being run at higher tension, ridging can be reduced by reducing tension. If run tension is less than heatset tension, flatness will be improved by increasing tension.

One method to remove a ridge is with the use of a household iron with variable heat settings. Using a thermometer on the flat surface of the iron, set the temperature to 350°F. If the fabric is wet, blow it dry with an air hose before attempting to remove the ridge. While rotating the fabric, move the iron in a back and forth motion over the ridge. To prevent puckers, it is important to keep the iron moving. It is important to use caution in the seam area, to prevent the seam ends from backing out. Lower the setting of the iron to 250°F when removing the ridge in the seam area.

An alternate method of removing a ridge is with the use of a hot air gun, or steam hose. However, extreme caution must be used so that the temperature does not exceed 350°F. As with the iron, keep the heat gun, or steam hose, moving at all times.

Bleed-Through

Bleed-through is a condition where fines or fillers pass through a fabric and collect on machine elements inside the fabric loop. Associated with bleed-through are the conditions of stapling, where fines become embedded in the fabric mesh and cannot be knocked off by showering; and carryback, where the fines collect on the return rolls and doctor blades.

Causes of bleeding, stapling and carryback:

(1) Fabric too open
- A fabric in which the hole size is too large will allow fibers to filter through and be lost upstream. While manifestations may not be readily evident in the sheet produced, the condition loads up the system with fines which, in many cases, eventually becomes a problem. This is especially true with furnishes consisting of a high percentage of hardwood or recycled fiber.
- A fabric having excessive drainage will allow the sheet to set too rapidly or seal. If this occurs, residual water cannot be effectively removed at the suction boxes and fines will be pulled through and be deposited on the covers. Some of the fines may be trapped in the mesh after the sheet releases and these may then cause stapling and/or carryback.

(2) Fabric too closed: A fabric having insufficient drainage does not allow enough water to filter through before passing over the suction boxes. Removal of the excessive water at the boxes without a proper mat being formed allows fines to be sucked through and collected on the boxes. Here again, some of the fines may be trapped in the mesh after the sheet releases and these may cause stapling and/or carryback.

(3) Fabric attitude with the long warp knuckle either on the sheet side or machine side may influence whether or not carryback occurs, especially on open draw machines. Fabrics run with the long warp knuckle on the machine side allow the fiber to ride higher on the fabric and the sheet releases more readily. As a consequence, carryback occurs less frequently with fabrics run in this attitude.

Remedies to reduce bleeding, stapling and carryback:

- Drop some of the flat boxes and redistribute the vacuum.
- Cut down the vacuum on the boxes to move the wet line ahead.
- Use fresh water in the knockoff system. It is very possible that the bleed problem is not bleed, but contaminated shower water. This is easy to tell by the amount of buildup on a fabric when the fabric is stopped and the low pressure showers remain in operation.
- Lubricate the leading edge of the flat boxes through a low pressure shower.

Guiding

If not properly guided, fabric may move off the machine during operation. Figure 2.70 shows the schematic of guiding. When the

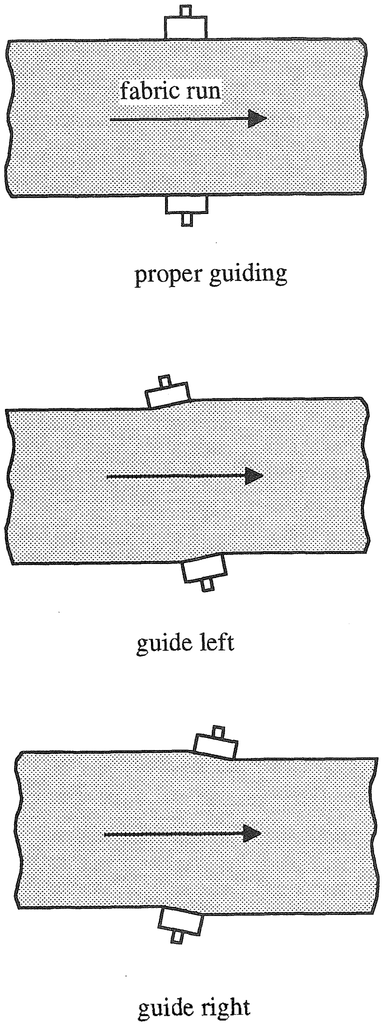


FIGURE 2.70. Schematic of guiding.

guide roll pivots to the left, the fabric will be shifted to the left. When the guide roll pivots to the right, the fabric will be shifted to the right. Guiding problems are generally caused by:

(1) Low tension: Guide palm does not pick up movement.
(2) Insufficient wrap on the guide roll: A minimum of 25° is recommended as sufficient wrap on the guide roll (Figure 2.71). Lead-in and lead-out distance are critical. Figure 2.72 shows the recommended lead-in and lead-out setup.
(3) Improper guide palm sensitivity
    - too sensitive: If the guide palm is too sensitive, it may pick up the normal oscillation of a fabric and transfer to the guide roll causing the fabric to shift unnecessarily.
    - Insufficient sensitivity: If the guide palm is too insensitive, it may not make the necessary corrections if the fabric walks on the machine.
(4) Nonuniform fabric tension. Nonuniform tension across the width may cause a fabric to tend to walk off the machine. Nonuniformity can be caused by a problem in the manufacture of the fabric, making one side longer than the other or a machine misalignment having the effect on the fabric as though one side was longer than the other.
(5) A strong twill pattern in the fabric may also cause guiding problems. The possible solutions to this are:
    - Break the twill pattern and even out the protruding knuckles.
    - Reverse the twill pattern to be opposite to the hole pattern on rolls.
(6) Tension difference in MD direction along the fabric width. Fabric moves toward the shorter side. Nothing can be done on the fabric to solve this problem. The fabric movement needs to be aligned using guide rolls.

Fibrillation

Fibrillation is a shredding of the monofilament which can occur uniformly across the entire width of the fabric or nonuniformly in streaks (Figure 2.73).

Uniform Fibrillation

Uniform fibrillation can be caused by:

- extremely high or low tension: High tension will tend to rub the underside of the fabric excessively against the machine elements. Low tension will

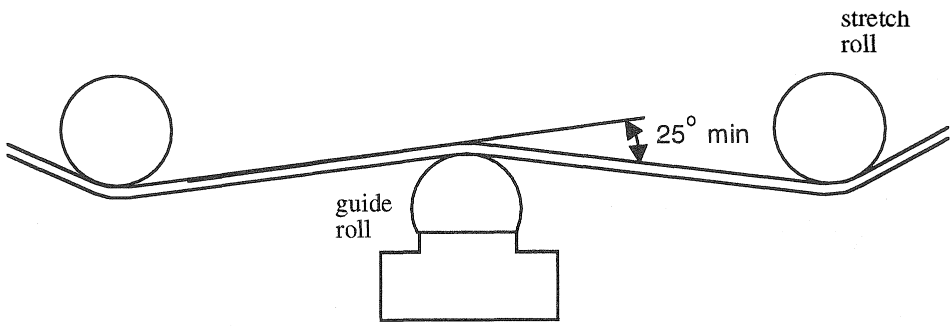


FIGURE 2.71. Guide roll angle of wrap.

allow the fabric to slip causing the turning rolls to fibrillate the underside of the fabric.

- insufficient wrap on the return rolls: Insufficient wrap will allow the fabric to slip across the rolls.
- defective table roll bearings: This creates heavy drag loads.
- high vacuums at the flat boxes: High vacuum will cause high drag loads.
- incompatible flat box material: Incompatible material will wear the fabric prematurely.

Nonuniform Fibrillation

Nonuniform fibrillation can be caused by:

- metal roll corrosion or pitting
- burrs on suction boxes, foils or forming boards
- chipped or porous areas in hard box covers
- hard box cover segments out of alignment
- foreign matter caught in slice: This problem is common in startups. Dry stock will break loose during a cleanup and get caught between the fabric and the slice, causing damage to the fabric.

Holes

Holes are very common cause for fabric removal. Causes of holes include:

- foreign matter from the machine that has built up due to long intervals between fabric changes
- hot sparks from cutting and welding being done near a machine
- hot ashes from cigars, cigarettes and pipes
- tools and implements dropping from pockets of personnel working nearby
- rust from inside the fabric run

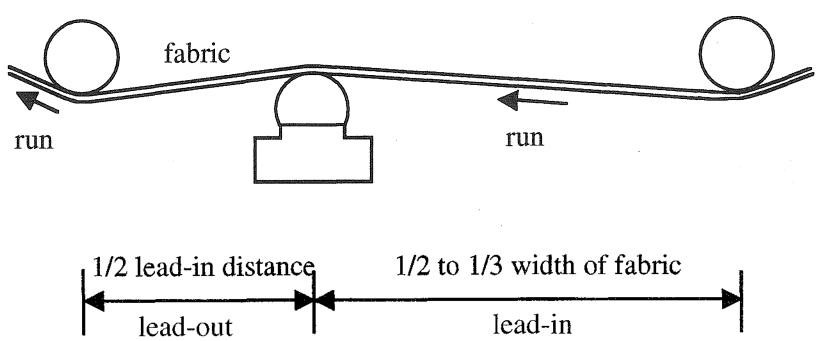


FIGURE 2.72. Fabric lead-in and lead-out ratio.

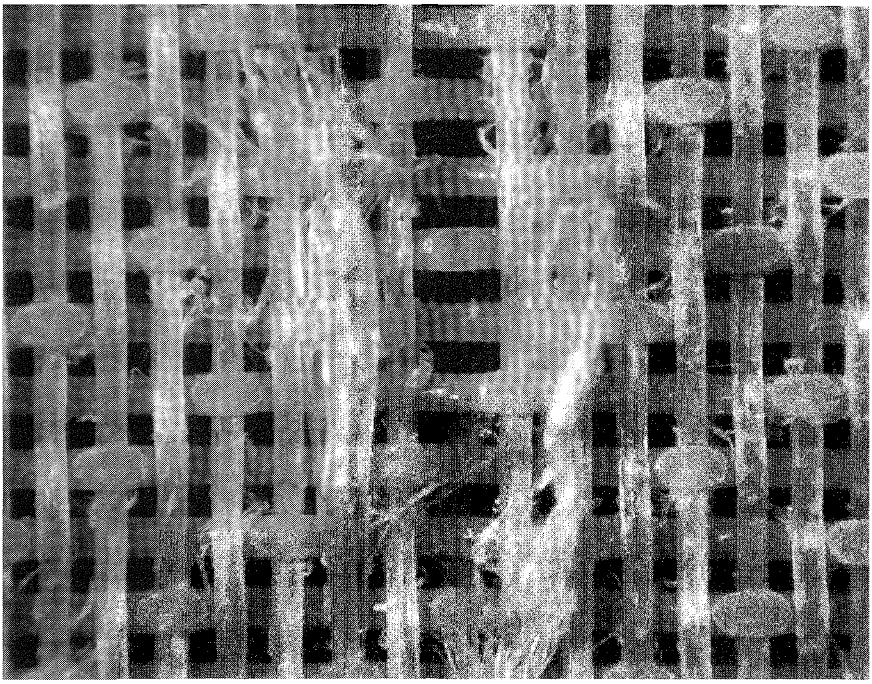


FIGURE 2.73. MD yarn fibrillation.

- foreign material knocked into fabric run by wash-up hoses (broken concrete from the floor, rust from the frames)

Wear

Forming fabrics may be worn due to friction between the fabric and various machine elements. Figure 2.74 shows levels of a two layer extra fabric wear. Improper graduation of flatbox vacuums is a large contributor to wear. The wear prediction of new fabrics is given in Section 2.2.4.

Determination of Single Layer Fabric Wear

As a guide in determining the amount of wear that has taken place in a single layer fabric, the following calculation may be used (Figure 2.75).

Percent Worn:

$$W = \frac{C - C_1}{0.58d} \times 100 \qquad (2.18)$$

Worn out:

$$C_2 = C - 0.58d \qquad (2.19)$$

where

W = percent worn
C = original caliper of the fabric
C_1 = caliper of the worn fabric
C_2 = caliper when approximately worn out
d = diameter of the wear yarn given by the fabric manufacturer.

The wear rate of a fabric decreases considerably as it continues to run, especially when the maximum point of potential wear life is being approached. Even at this maximum point, the fabric may retain sufficient strength to sustain the normal tensions encountered in operation. Therefore, in the final analysis, other factors should be considered in determining whether the fabric should be removed because of wear. The major effects of fabric wear are:

- a dramatic change in drainage that usually causes excessive bleed-through

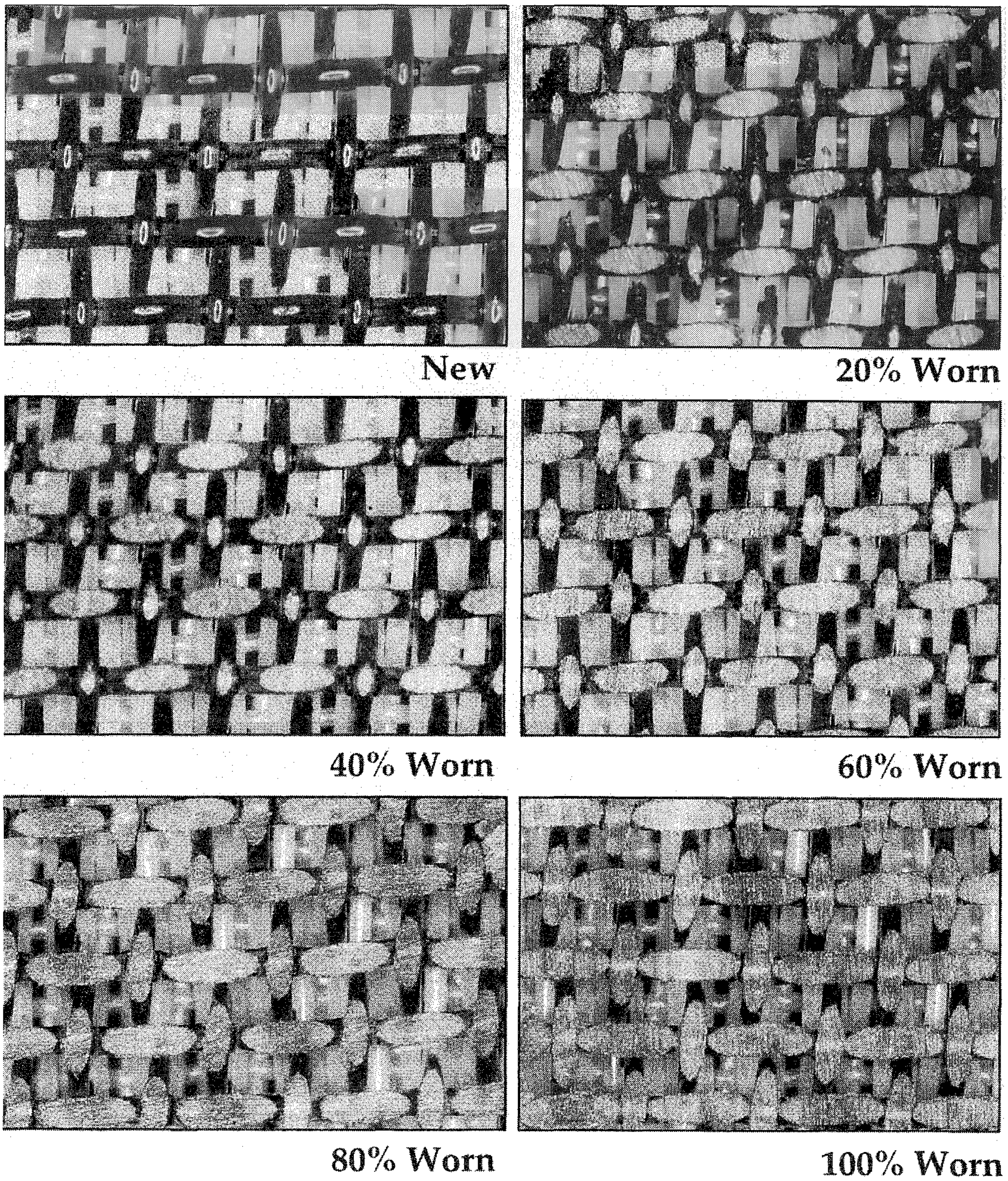


FIGURE 2.74. Various degrees of a two layer extra fabric wear.

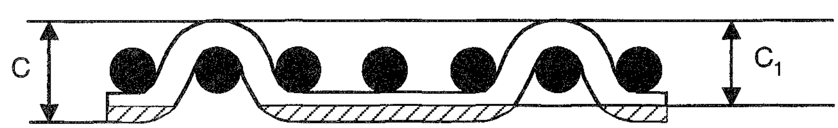


FIGURE 2.75. Calculation of single layer fabric wear.

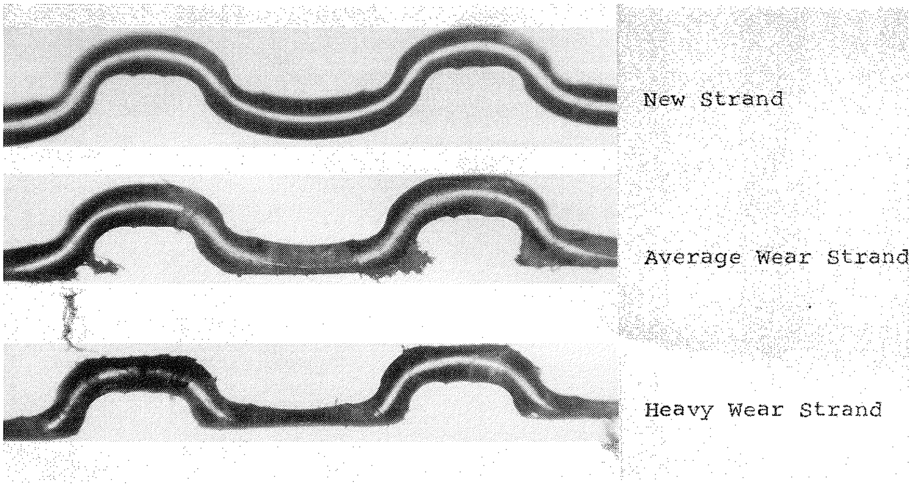


FIGURE 2.76. Comparison of a new yarn with worn yarns.

- development of pin holes in the fabric, which reflects in the quality of the sheet
- fabric stretching

Figure 2.76 shows a comparison of a new yarn to an average wear yarn and a yarn taken from a heavily worn edge area. The fibrillation and degree of top wear can also be seen.

Seam Picking

Occasionally, a seam end may be pulled from its locked position in the seam joint, and project above the plane of the fabric surface as shown in Figure 2.77. This may cause picking of the fibers in the web, which may result in pinholes, or poor web release from the fabric. This problem can generally be eliminated by gently sanding the seam ends until they are flush with the fabric surface. Starting with a 320 grit abrasive paper, sand in a machine direction, back and forth manner, to eliminate the formation of burrs on the fabric knuckles. Finish with 400 grit abrasive paper, in the same fashion. A simple pick test can be performed by rubbing over the seam ends with a cotton ball, or a handful of synthetic fiber, such as polyester or rayon. The seam ends should not snag any of the fibers when rubbed across them.

Edge Sealing

If the fabric edges begin to fray, they can be resealed while the fabric is on the machine. The machine, however, should not be running when sealing edges. A standard soldering gun with a flat tip can be used. For best results, the tip should be bent into a slight V. This will help guide the gun when sealing the fabric edge. Trim off any rough edges with a scissors prior to sealing. When sealing the edges, caution should be used not to melt into the fabric. Begin by sealing in a fast motion along the edge. If the edge does not seal, slow the motion down slightly. After the edge is heat sealed, an adhesive can be brushed on.

Sheet Release

Monoplane fabrics with long MD yarn up may have sheet release problems. Sheet does not release itself to be transferred to the press section (Section 2.7). The solutions are to use long top CD yarn or not to have monoplane surface.

2.4.4 *Fabric Cleaning*

Forming fabrics may be cleaned with most solvents and cleaning agents used in the paper industry. Inasmuch as these cleaners and their

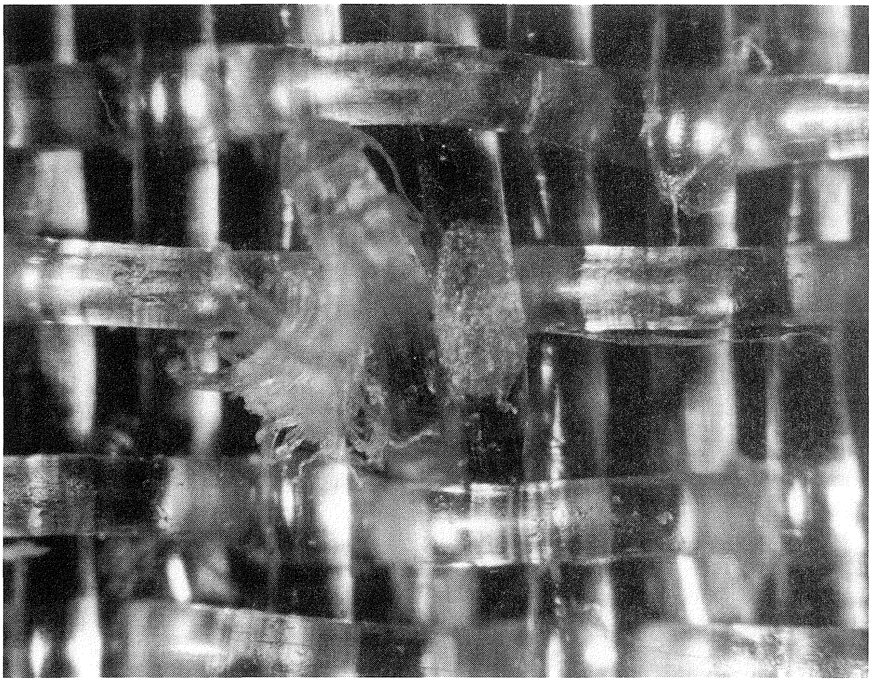


FIGURE 2.77. Seam picking.

usage varies from mill to mill, it is not possible to specify standard cleaning methods. The following points should be considered when cleaning forming fabrics:

- Concentrated acid, concentrated alkali, phenolic compounds, and strong bleaching solutions should be avoided.
- Due to their toxicity and flammability, caution should be exercised when handling organic solvents.
- A metal wire brush or stiff bristle brush should not be used for scrubbing. Also, the solvent should not affect the bristles in the brush used.
- High pressure showers to rinse off cleaning solutions should be used only if the fabric is rotated.
- High pressure steam is not recommended. When used, the hose nozzle should not come in contact with the fabric and should be moved constantly.

The question of chemical resistance of forming fabrics is of high importance for many applications. As it cannot be answered generally, evaluations are necessary on an individual basis.

The chemical resistance decreases with higher temperatures. Besides this, the concentration of chemicals is important in many cases. Often there is in the end use only a short term influence of a certain product or temperature on the monofilament. This can be of high importance in estimating the lifetime. Last but not least, the mechanical influence also has to be considered. Projected open area in a fabric makes it easier to clean.

2.4.5 *Fabric Contamination and Contaminant Resistance*

Contaminants such as pitch and tar stick to forming fabrics causing contamination. When recycled paper is used, the situation gets worse due to latex. The polyester yarn is susceptible to contaminant buildup due to the chemical groupings in the molecules. Contaminant buildup fills the fabric causing drainage and release problems. One way to prevent contamination is to treat the surface of yarns or fabrics with anticontaminant finishes or coatings which reduce the fabric's attraction for contaminants. Coatings can be used on either the individual yarns or the finished fabric. Coating can be applied during manufacturing of the fabric or on the paper machine. Coating also may "weld" the MD and CD yarns at crossovers preventing the mechanical attachment of the contaminant to the fabric. It is claimed that coated fabrics increase production due to speed increases, fewer sheet breaks and less downtime due to changing or cleaning filled fabric. It is also reported that coating increases fabric life and improves fabric stability and stiffness. However, as the coating wears out or washes off due to high pressure showers, the performance characteristics of fabrics may change over time.

Figure 2.78 shows wetting angle comparison of two different monofilament yarns. The standard filling yarn has a contact angle of approximately 25°. The contaminant resistant yarn shows a contact angle of 47°.

Mechanisms of Contamination

(1) Chemical bonding between yarn and contaminant. To prevent bonding between the yarns and contaminants, yarns with contaminant resistant properties are used. Contaminant resistance in a yarn can be achieved in three ways:
- using contaminant resistant (CR) additives in the yarn polymer: This method is the most efficient because the contaminant resistance property is inherent in the yarn.
- coating the yarn with a contaminant resistant solution: The drawback of this system is that the coating may be lost partially or totally during operation of the fabric.
- combination of the two methods

(2) Mechanical locking of contaminants at yarn crossovers: Fabric coating is used as a remedy to this kind of contamination. However, fabric coating requires capital investment and environmental issues may also be a concern.

Recycled stock is the major source of contamination. In general, recycled fibers are shorter than virgin fibers. Due to short fibers in recycled stock, higher mesh fabrics are used to increase fiber support index and improve retention. The fabric should provide a surface that is easy to bridge with short recycled fiber.

regular yarn
contact angle = 25°

contaminant resistant yarn
contact angle = 47°

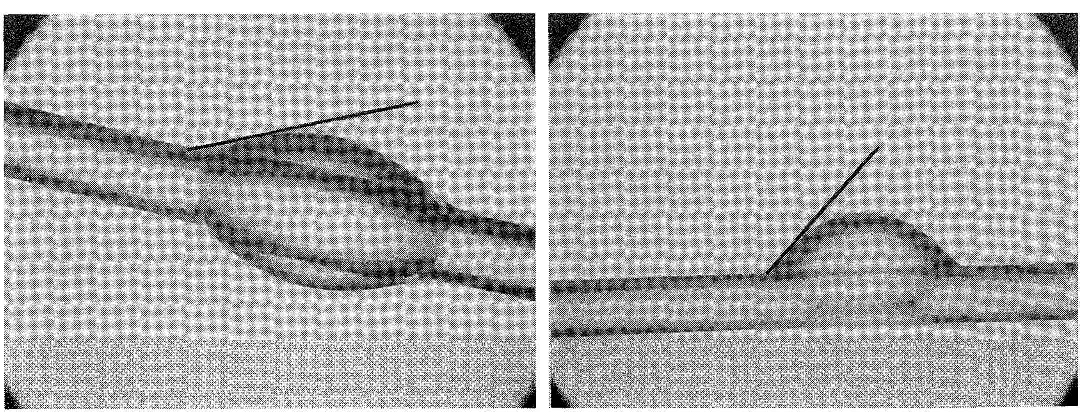


FIGURE 2.78. Wetting angle comparison of regular and contaminant resistant materials.

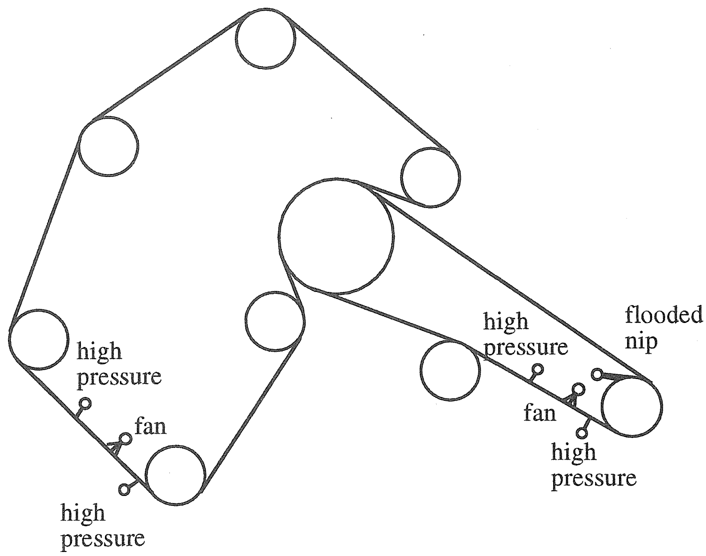


FIGURE 2.79. Shower positions on twin-wire formers.

Fabrics used in recycled furnish should allow easy cleaning with existing shower systems. Low caliper in general allows ease of cleaning.

Fabric showering is an effective method to prevent contamination buildup on the fabric during operation. High pressure showers are used for this purpose. Due to the high pressures involved, fabric stability and seam strengths may become an issue in fabric design. With more open area (higher permeability) and straight through drainage, the showerability of fabrics is improved.

2.4.6 *Fabric Showering*

Properly operating cleaning showers are necessary to remove fibers and contaminants from the fabric during operation to optimize fabric life. Showers are used on wet-laid machines for fabric cleaning, sheet knockoff, trimming and lubrication. In any showering system, it is critical to maintain uniform coverage and open flowing nozzles. Another consideration in shower design should be that nozzles cannot fall into the fabric run, creating a major fabric/roll accident.

The major types of showers are high pressure needle and fan showers, flooded nip showers and knockoff showers [15]. High pressure fan showers are generally used in tissue manufacturing. Figure 2.79 shows the typical positions of the showers on a twin-wire former. Table 2.12 summarizes the forming section showers.

High Pressure Needle Showers

High pressure showers are the most effective device for contaminant removal on the wet-end of the machine. The high energy jet can cause fabric vibrations, resulting in premature failure if not located close to a support roll. The shower should be located as close to the outgoing nip of a support roll as possible. High pressure needle showers are needed to penetrate the voids in the fabric to optimize cleaning.

There are different opinions about shower distance from the fabric surface. One should consider the fabric structure and production needs before making this decision. Published data indicate that very close showering distances can clean contaminated fabrics well, minimizing damage. As the shower is moved away from the fabric, it begins to entrain air which results in a nonuniform pounding of the surface. While this can be utilized to remove contaminants, it should be reserved for inside

TABLE 2.12. Forming Section Showers.

Application	Nozzle Type/Size	Spacing	Distance	Pressure	Gallons/ Inch
Flooded nip (machine side)	40° Fan stationary	3″	8″–16″	100–150 psi	RVV × 1.1
Inside high pressure (machine side)	0.040″ Needle oscillating	3″	6″–8″	350–500 psi	.18–.23
Outside high pressure (sheet side)	0.040″ Needle	3″	2″–4″	200–500 psi	.18–.23
Fiber/sheet knockoff (machine side)	40° Fan stationary	3″	4″–7″	200–400 psi	1.0–2.0
Suction roll	0.040″ Needle oscillating	3″	4″	350–500 psi	.21–.25
Wire return roll	40° Fan stationary	6″	8″	40 psi	.07–.09

single layer showering or multilayer structures.

Oscillating showers are typically 3 inch spaced needle orifices with a 6 inch stroke. Oscillating speed must be set to machine speed to insure total fabric high pressure cleaning. The speed of shower is calculated as follows:

$$OS = \frac{MS}{FL} \times NS \qquad (2.20)$$

where

OS = oscillating speed (m/min)
MS = machine speed (m/min)
FL = fabric loop length (m)
NS = nozzle size (m)

The high pressure showers should have an automatic kick-off system to shut off when the machine shuts down. There should also be a relay that will not allow the high pressure showers to be started until the machine is put into the run mode.

Pressures to be used in the high pressure showers are dependent upon the amount of cleaning and the speed at which the machine is running. At faster speeds, higher pressures are needed to penetrate the voids. Conversely, operating at high pressures and low speed can and will fibrillate the fabric. Optimizing pressures need to be determined.

Inside High Pressure Shower

Inside high pressure showers are generally used on forming fabrics utilizing a blend of secondary fiber and virgin fiber or 100% virgin fiber. This shower is very effective in cleaning contaminants from the void volume of the fabric. It is critical to have an oscillating needle jet-type shower with a recommended operating pressure of 350 to 500 psi. The nozzle size is generally 0.040 inch installed at a distance of 6 inches to 8 inches from the fabric. When very fine single layer fabrics are required, inside showering is recommended.

Sheet Side High Pressure Shower

This shower is more effective in removing fiber and other contaminants from the surface of the fabric. For two layer fabric design applications with no projected open area, it will be necessary to have the ability to adjust the showers to a 30° angle (CD) with respect to the fabric (Figure 2.80). The 30° nozzle allows penetration of the high pressure showers through the angular voids, which is a characteristic of two layer design fabrics (Figure 2.60). It should be noted however that this technology is relatively old.

Since seam ends on single layer fabrics are generally terminated to the sheet side, it is recommended that operating pressure be kept

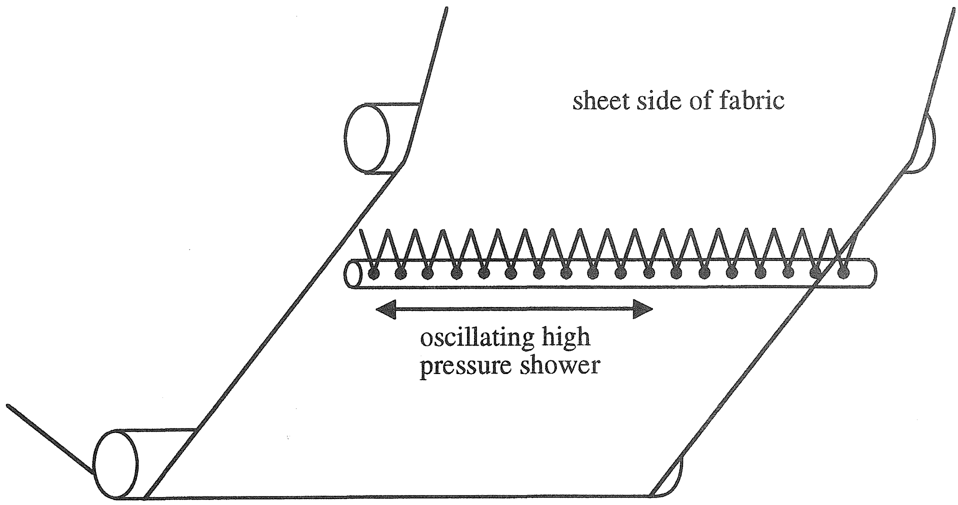


FIGURE 2.80. Two layer showering.

at 250 psi or lower. Yarns smaller than 0.005 inch can break at high pressure showers above 250 psi. When running a multilayer fabric, because of design structure difference and depending on yarn sizes, pressure should be set from 200 to 500 psi and run perpendicular to the fabric surface. Oscillating needle shower utilizes nozzle sizes of 0.040 inch and is generally set at a distance of 4 inches or less to the fabric.

To effectively prevent the fabric from plugging, most cleaning applications require both inside and outside high pressure oscillating showers.

Suction Roll Showers

On suction roll machines, a high pressure needle shower is located perpendicular to the roll surface. The nozzles should be spaced at 3 inch centers, 4 inches from roll surface, at a pressure of 350–500 psi.

Flooded Nip Showers

On a standard fourdrinier with wire turning roll, the flooding nip shower is placed on the inside of the fabric run at the ingoing nip of the wire turning roll (Figure 2.81). They should be of a fan type and provide complete coverage into the nip. The nozzle distance, from the nip, should be 3–4″ away. The jet from the fan nozzle should hit the roll slightly before the nip. The angle of the fan shower jet should be approximately 30° with respect to the fabric. It is commonly recommended that this shower nozzle be of a self-cleaning type. The pressure and flow rate are largely dependent on the void volume within the two-layer constructed fabric. When expelled through the opposite surface, the water will affect the separation of the fabric and the sheet. The main criteria used to determine rates are machine speed, number of nozzles, forming fabric caliper and void volume.

In high speed twin-wire formers, flooded nip showers have dual purposes of internal cleaning and sheet knockoff. Sheet knockoff can occur running a gallons per minute rate 75% of the fabric's total running void volume at speeds above 3000 fpm. This result is created by the centrifugal force as the water is thrown through the fabric due to the wrap on the roll. Flooded nip cleaning flow is 10% more than the calculated running void volume. The roll chosen for this shower should have a minimum wrap of 30%. A stationary 40 to 45° fan nozzle shower should be installed to the roll/fabric nip and is much more effective in a pressure range of 100 to 150 psi. The flooded nip shower should be located 12 to 16 inches from the roll/fabric nip to allow

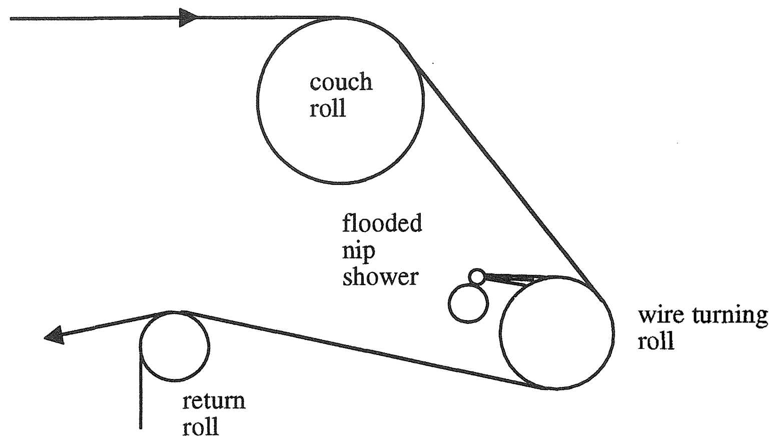


FIGURE 2.81. Flooded nip shower on a standard fourdrinier with wire turning roll.

water to directly enter the nip. In several instances, knockoff showers have been eliminated by application of the flooded nip with good results. Many shower manufacturers reduce water volume as speed is increased above 5000 feet/min.

Minimum flooded nip shower requirement to fill fabric void:

$$Q = 0.055 \times C \times W \times MS \times V \quad (2.21)$$

where

Q = water requirement to fill fabric void (GPM)
C = caliper of fabric (inch)
W = fabric width (inch)
MS = maximum machine speed (fpm)
V = void volume factor (usually 0.6)

Normally, Q is multiplied by 1.1 to obtain water requirement for cleaning.

Fiber/Sheet Knockoff (Fan) Showers

The sheet knockoff shower or fan shower is used to supply fairly large quantities of water at a relatively low pressure across the full width of the fabric. The application of higher pressure significantly improves the efficiency of this type of shower. The effectiveness of lower pressure showers is questionable when moving to multilayer structures. The bottom surface of multilayer shears the fan jet, creating machine direction flow before water passes through the plane of the fabric.

This shower is generally a stationary 40 to 45° fan on 3 inch nozzle spacing at a pressure of 200 to 400 psi. Since distribution of water by a fan nozzle is uneven, a possible second pipe with offset nozzles is recommended. The maximum impact force is obtained by placing the nozzle 4 inches to 7 inches from the surface of the fabric.

The actual impact of the shower depends on the distribution of water across the face of the fabric. This is controlled by the fan angle which is chosen. Table 2.13 shows the effect of changing fan angle and fabric distance on coverage.

Shower Maintenance

Periodic inspection of the shower system is recommended at every fabric change. Excessively high pressure can fibrillate the individual yarns in the fabric, leading to a shortened fabric life. Low pressure can prevent optimum cleaning, possibly decreasing its drainage capacity. On abrasive fiber grades, such as glass, a decrease in fabric life can result from fibers embedded into the mesh. Table 2.14 lists the recommended inspection frequency of showers.

It is critical to inspect shower nozzles visually on each machine down. Plugged nozzles

TABLE 2.13. Effect of Changing Fan Angle and Fabric Distance on Coverage.

Fan Angle	Distance to Fabric (inch) 4	5	6	7	Impact Factor
30°	2.0	2.5	3.0	3.8	.16
40°	3.2	4.2	5.0	6.0	.12
50°	4.0	5.2	6.5	7.4	.10
60°	4.8	6.2	7.5	8.8	.08
80°	6.8	8.8	10.4	12.0	0.5
	Surface coverage (inch)				

can cause MD bands of varying tension inducing drainage problems and premature loss of the fabric. Shower nozzles should routinely be replaced on a 3–12 month cycle because tips erode over time, changing the water distribution pattern. All shower pipes should be set up to allow flushing on the run. Self-cleaning nozzles are recommended where possible.

Forming Section Doctors

Doctors are an integral part of cleaning and conditioning of forming fabrics. Contaminants naturally transfer to the smoother roll surface from the rougher forming fabric surface and must be doctored away from the roll or they will build up and cause operational problems. Roll doctors are generally made out of poly or fiberglass. Generally, doctors do not oscillate in wet-end installations, but is an option provided by some vendors. It is recommended that all rolls have a wire return roll shower to aid in doctoring efficiency and provides lubrication between the roll and doctor.

Brush Cleaning

Brush cleaning has been tried on several machines and is utilized to a limited extent as a final cleaning device. Fiber must be flushed from the surface of the fabric before contacting the brush to avoid brush plugging problems. A properly operating brush will allow stickies to be removed from the machine white water system.

TABLE 2.14. Recommended Inspection Frequency of Showers.

Checkpoint	Frequency
Plugged nozzles	Every fabric change/on run
Pressure check	On run
Purge shower header	Every fabric change

2.4.7 Returned Fabric Analysis

Returned fabric analysis, which is extensively done in the industry, gives valuable information about the performance of fabrics. This information can pinpoint any potential problem areas on the machine. It is also useful to improve fabric design and structure.

A typical returned fabric analysis includes the following:

- wear profile and degree of wear
- air permeability profile
- abrasion
- burring
- fibrillation
- contaminants
- edge condition
- seam condition

Air permeability test done on the returned fabric is to determine how fabric wear has affected drainage quality. To do this, it is necessary to remove as much contamination in the returned fabric as possible. Some contaminants such as dried binders, wax or tar-like specks commonly found in secondary furnish cannot be removed without destroying the polymer yarns. However, many contaminants found in returned fabrics such as pulp, calcium carbonate, paper dyes, ink or grease can be easily removed. For these contaminants, a mild alkaline liquid (containing potassium hydroxide, water, tetra-sodium pyrophosphate, sodium heptagluconate, tetra-potassium pyrophosphate) solution can be used with a water content of 90%.

First, the air permeability of the test sample is measured as it is received. The test sample is then completely submerged in the solution. The time the fabric is submerged depends on the amount of contamination. The average time can range anywhere from 20 minutes for lightly contaminated test samples to 2 hours for heavier contaminated fabrics.

After the test sample has been soaked, it is rinsed with water and scrubbed with a soft bristle hand brush. The test sample is then blown dry with a compressed air nozzle. The test sample is then retested for air permeability. At this point, any loss in the cleaned test sample can be attributed to:

- burring, which causes the fabric mesh to close up resulting in less air flow
- the development of a wear pad, which increases the area in which the air must travel around also reducing drainage quality

The air permeability measurements, taken before and after cleaning, are compared to the new value and reported in the form of percentage retained.

2.4.8 *Wet-End Surveys*

It is important to measure the performance of paper machine clothing on paper machines. Some paper machine clothing manufacturers provide extensive technical support that may include:

- showering surveys
- crew training
- startup assistance
- lab reports
- chemical analysis
- roll abrasion studies
- drainage studies

Machine surveys are commonly done in the paper industry to assess the performance and productivity of paper machines and papermaking process (Chapter 5). The collected data are analyzed and used to make recommendations to the papermaker to improve his process. Appendix B gives a guide for forming section troubleshooting.

2.5 Forming Section Configurations on Paper Machines

The most important part of choosing a paper machine is probably selecting the forming section. This is because sheet forming mechanism has a significant effect on sheet properties. Since the invention of the first paper machine in the 1790s, there have been major improvements and inventions in forming sections of the paper machines. As a result there are virtually endless varieties of paper machines in existence today. A comprehensive coverage of the paper machines is out of the scope of this book. A brief introduction to each major group will suffice for the convenience of the reader.

There are five main types of formers in industry today: fourdriniers, twin-wires, two-wire formers, multiwire formers and Crescent formers. Although multicylinder machines are not manufactured anymore, many are still in use.

2.5.1 *Fourdriniers*

Fourdrinier machines are considered the conventional paper machines. However, their use is declining. Figure 1.10 shows the schematic of a typical fourdrinier machine.

There are many elements on a fourdrinier machine to help with the formation and drainage. These elements include forming board, foils, table rolls and suction boxes. These elements are located under the portion of the fabric where sheet is formed and transported. There are other machine elements that are used to control the fabric such as stretch rolls, guide rolls and fabric turning rolls.

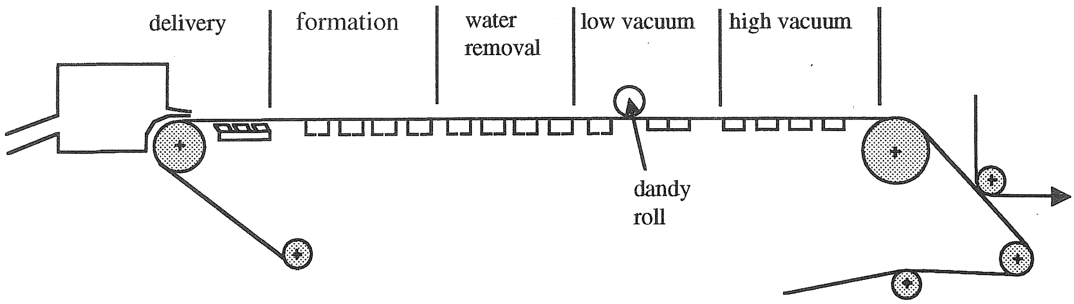


FIGURE 2.82. Sheet forming process on a fourdrinier machine.

Operation of the Fourdrinier

Fourdrinier machines are very useful to explain the formation of the sheet. A rather detailed explanation of the operation of a typical fourdrinier paper machine is given below.

The operation of a fourdrinier wet-end is very complex. There are many interrelated components and actions, each of which is important to good operation and quality paper. Attention to the details of the whole operation is essential to having a well run machine.

In modern fourdriniers, the sheet forming process on the machine can be divided into several regions as shown in Figure 2.82: stock delivery, sheet formation, water removal, low vacuum and high vacuum. To get the best possible sheet of paper to the couch, several different conditions are established on the wet-end of the machine to form the sheet and to remove water. Each of these conditions has an impact on the efficient operation of the machine and the quality of the sheet being made [1].

Stock Delivery onto the Fabric

The delivery part of the process is made up of the headbox and slice, the breast roll and the forming board. All are related to the performance of the forming fabric.

Headbox

The headbox is one of the key parts of the machine as it determines the uniformity and stability of the basis weight of the sheet being made. It also has a major impact on the formation and strength properties of the sheet. It affects the way the forming fabric performs with regard to sheet physical properties and sheet finish and it can help even out problems with uniformity of the sheet which may originate in the stock preparation area, the approach flow system, and the headbox manifold.

Headbox consistency is one of the primary concerns for good formation. The fibers must be dispersed as uniformly as possible so as to minimize floc formation in the headbox.

Slice

The slice controls the angle of impact of the jet onto the fabric. The angle is governed by the relation of the top lip to the bottom lip or apron (Figure 2.83). If the top lip is moved downstream of the apron, then the jet is said to have a tendency toward pressure forming. If the top lip is moved upstream into the headbox from the apron, then the jet is said to have a tendency toward velocity forming.

Within practical limits, sheet formation is always improved as headbox consistency is lowered and the slice operated at the largest opening that the table, fabric and furnish are capable of draining.

The ratio L/b describes the relative location of the top lip with respect to the edge of the apron or bottom lip. It relates directly to the angle of impact of the jet. While it is an important number to be aware of, it is more important to observe the actual landing spot and

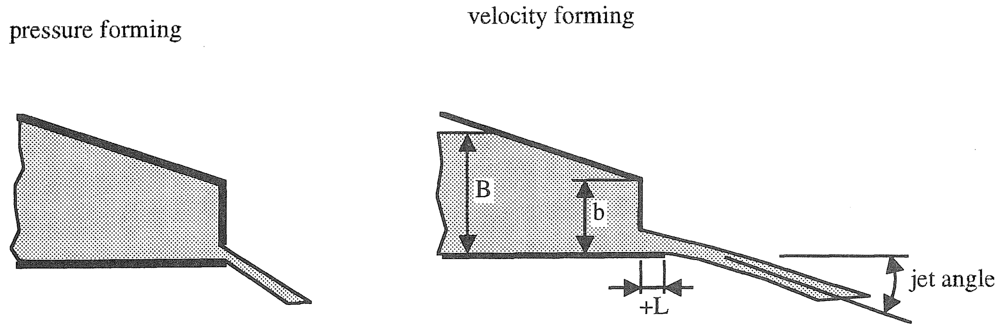


FIGURE 2.83. Control of jet angle with slice opening.

attempt to have it in the right position for the best formation and drainage. L/b has a minor effect on fiber orientation except as it relates to excessive turbulence and stock jump that may occur with too much pressure forming. In practice, pressure and velocity forming are defined based on the L/b ratio as follows:

$$\frac{L}{b} \cong 0.5 \text{ pressure forming} \qquad (2.26)$$

$$\frac{L}{b} \geq 1 \text{ velocity forming} \qquad (2.27)$$

In pressure forming, the jet angle to the fabric is steeper. In velocity forming the jet angle is low and the jet travels relatively parallel to the fabric a longer distance. In modern machines, velocity forming is preferred. The jet should land on the fabric just before the leading edge of the forming board.

Pressure Forming

Pressure forming tends to move the impact point upstream toward the headbox, breast roll and slice. If the contact is too close to the breast roll, the nip vacuum may act upon the initial fiber mat. The drainage force can pull fiber, filler, fines and water through the forming fabric. As speeds get higher, the drainage forces get greater, deflecting the fabric downward and creating severe flow disturbances on the table. Fibers can become tangled in the mesh of the fabric, creating high draws at the couch. The higher the early drainage forces, the worse the wire mark.

Pressure forming is very sensitive to any defects or irregularities in the slice opening or headbox flow patterns. These defects become set into the fiber mat because of the high drainage rates and are virtually impossible to remove or correct on the table. Pressure forming also tends to seal the sheet on the fabric, making efficient drainage very difficult to achieve.

Velocity Forming

Velocity forming is more desirable because there is more latitude to impact the jet correctly onto the fabric and there is a greater possibility of evening out flow irregularities from the slice or headbox if they exist.

Velocity forming, at the extreme of landing the jet completely on the forming board, will eliminate any possibility of drainage until the slurry is over an open area of the board where gravity alone will provide drainage force. Delivery may be smooth but the large amounts of water being carried down the table may give skating and other formation defects.

Ideal delivery lies somewhere between the two extremes of pressure forming and velocity forming. If the jet is directed so that it lands 1/4″ to 1/2″ before the forming board and the balance lands on the board thus draining 10–20% of the flow between the slice and the trailing edge of the forming board, some of the initial drainage will penetrate the forming medium, filling the interstices or holes of the fabric. Water will replace the residual air in the fabric, giving a uniform layer of water in the fabric rather than an unknown mixture of air, water and fiber.

The slice delivery is important in making a success of a fabric. With pressure forming, the fibers will be driven toward the bottom of the holes, making drainage difficult and creating draw problems at the couch. Using velocity forming, the fibers will not penetrate the fabric as easily at contact and will have a chance to become partially cross machine direction oriented over the forming board. As they turn, the fibers have a better chance of bridging across adjacent MD yarns, staying much further up toward the fabric surface and thus keeping the drainage paths more open.

The jet speed to wire speed (i.e., forming fabric speed) ratio is very important. This ratio can be changed to obtain a rush or drag effect during formation. Rush/drag changes are made to develop grade qualities. For example, if MD tensile is of top importance then the sheet may be dragged to a great extent to align the fibers in the machine direction. Other properties such as burst may need more rush to get some cross-direction orientation. When formation is of great importance, it is best to drag the sheet slightly to prevent the rolling of fiber which sometimes comes from a rush. The rolling fibers will make fiber clumps which result in poor formation. Rush/drag has essentially no effect on the angle of impact.

Breast Roll

In setting up the forming table, the center line of the breast roll is generally used as the point of reference from which the table layout is made. The roll must be very accurately positioned horizontally and vertically so that the slice lip and all other table elements are precisely aligned to one another. The geometry of the slice, breast roll and forming board arrangement determines the area where the jet will contact the fabric.

The breast roll is directly beneath the headbox. Since this is a roll with a high amount of wrap, it must be large enough and strong enough to prevent deflection from the tension forces generated by the fabric. It should have a properly adjusted doctor.

Any roll on the paper machine is an example of a very powerful drainage force. Foil blades are a second example of the same force.

A vacuum is generated in any nip where a roll surface separates from the fabric. This vacuum induces drainage through the fabric. In general, the amount of vacuum induced in the nip is dependent on the machine speed. The higher the machine speed, the higher the vacuum. The drainage rate, Q, is calculated as follows:

$$Q = \frac{DV^m}{R^2} \tag{2.24}$$

where

D = diameter of the roll
R = drainage resistance factor of the furnish
V = speed of the machine
m = coefficient related to the furnish being used

The nip where the fabric leaves the breast roll is the first area on the table where drainage of the headbox slurry can occur. This drainage should be avoided because of its effect on the sheet.

Forming Board

As machines became faster and wider, the diameter of the breast roll became larger. The span of unsupported fabric from the center line of the breast roll to the next supporting roll became longer and the forming medium sag or deflection became greater under the force of the jet and the lack of support. The forming board reduced the sag by enabling the support point to be moved in closer to the roll, and improved the landing zone conditions for the jet by allowing some control of the drainage of the sheet (Figure 2.84).

The board is used to provide a well-defined landing place for the jet. It also controls the drainage at the entry zone of the table. Drainage at the forming board is largely controlled by gravity and is dependent on the open area of the board. The greater the open area, the greater the drainage. Doctoring by the blade nose also contributes to dewatering.

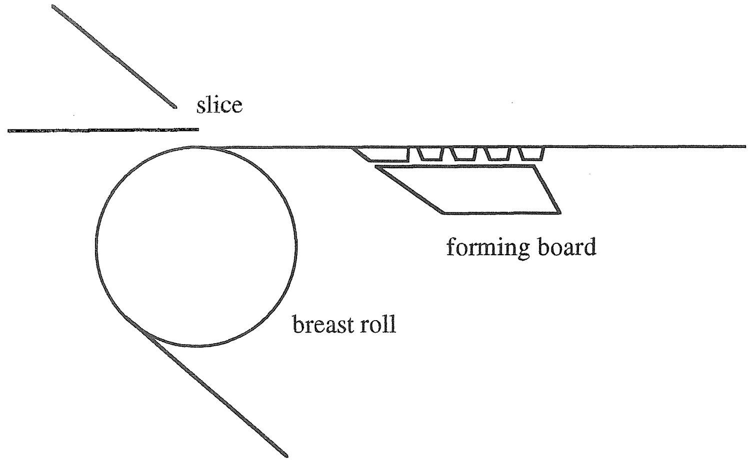


FIGURE 2.84. Forming board.

The forming board body must be extremely rigid. The water forces that result from the jet impact are very high. If the body is not rigid enough, there will be vibrations which will affect the sheet quality. The body also supports the blades and keeps them correctly aligned for uniform drainage. Usually there are three, five or seven blades on a forming board. Ceramic blades have replaced early forming boards made of polyethylene. Most forming board blades have no built-in drainage angle (they are flat). Drainage over the board comes from the velocity of the jet as it impacts the fabric before and over the early part of the lead blade. No drainage can take place over the blade itself and gravity provides the balance of the drainage. Since elapsed time over the open area is short and the height of the water column is not high, little drainage can occur. What does occur is related to the ability of the blade nose to doctor water from the underside of the fabric. There are some signs that the more blades (and noses) that there are, the more dewatering will occur.

The forming board nose can be located such that it is easy to adjust the top slice lip to get good jet impact. The forming board also may need to be located at the point on the table where the desired jet angle will put the jet in the right position with respect to the forming board.

Sheet Formation Region

In this region, the sheet is formed with the help of turbulence and shear.

Turbulence

In the stock approach system and the headbox, the consistency of the furnish is run at as low a level as possible to disperse the fibers as completely as possible. Turbulence generation in these systems ensures dispersion. On the forming table, dispersion must be continued and turbulence is the primary tool for maintaining dispersion. Shear is the primary means of providing turbulence for dispersing the fibers. Shear is simply causing controlled flow disturbances which disperse fiber flocs or bundles.

Shear

As the jet exits the headbox, shear may continue for a very short time due to surface friction between slurry and the slice surfaces. In addition, the way the jet impacts the fabric may be very turbulent depending on the entry zone setup. At speeds up to 2000 feet/min, dispersion may be accomplished by using the shake. On some machines, the shake may affect only the breast roll and on other machines, part of the table may also be shaken. The shake creates a cross machine direction shear in the fiber flow. As speeds increase from the very low to the 2000 fpm range, the time between the jet landing on the fabric and the sheet set point being reached becomes very short, decreasing the effectiveness of the shake.

As the fabric travels down the table, the foils create shear by deflecting the fabric downward. The change of direction of the fabric causes the water to attempt to travel vertically upwards from the fabric. This movement creates shear. If the angle is high and the consistency low, droplets may actually separate from the surface and travel through the air.

Collapse of ridges also provides a powerful source of shear and turbulence. A dirty slice or a damaged slice will exhibit streaks on the surface of the sheet at the defect. Slices can be purchased serrated which generate a CD series of streaks or ridges which collapse and reform. These streaks can be likened to valleys and peaks of hills. The hills collapse and are replaced by peaks. This cross-direction flow provides shear. At foil noses, peaks become valleys and valleys become peaks. This will continue down the table for some distance depending mainly on consistency and spacing. Formation showers have the same effect as serrated slices.

Formation Shower

These showers are used to help form the sheet properly on the forming fabric. They are particularly helpful on heavier weights and may be harmful on lightweight sheets if the pressure and volume are not controlled to prevent complete penetration of the sheet on the fabric. The showers are stationary and use low pressure water to create ridges in the stock furnish on the forming fabric.

Quite often, these showers are simply a drilled hole shower which will work quite well with white water if the burrs are removed inside the pipe and around the circumference of the holes. The shower is positioned 5 to 10 inches above the fabric over the forming board or first foil unit. The shower is angled downstream 10–15° from perpendicular (Figure 2.85). Suspending the shower from the machine room crane for trials will allow easy adjustment for determining the optimum operation. Permanent brackets can be designed to locate the shower at the best location.

Water Removal Region

In this section of the table, formation is essentially completed and the consolidation

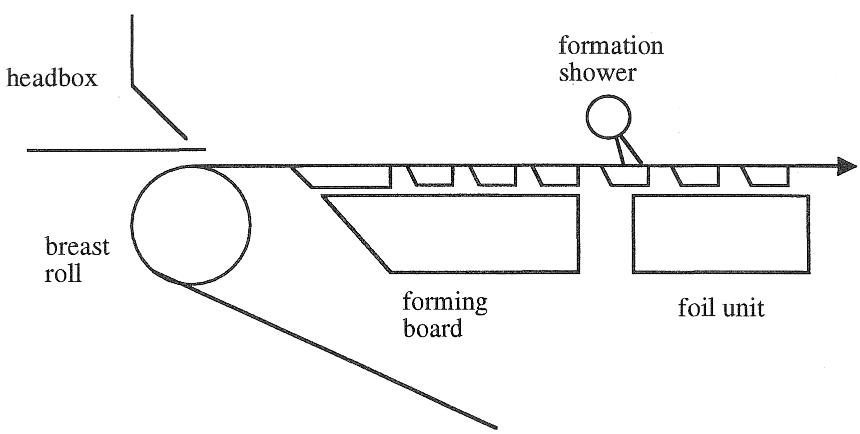


FIGURE 2.85. Location of formation shower.

of the sheet begins. The foils do the work of water removal and the downward flow of water causes the fibers to be brought into closer contact with each other.

As the consolidation of the sheet is taking place, it is important to diminish the turbulence level constantly. If the sheet is disrupted by high turbulence this far down the table, it will not reform into an acceptable sheet.

Foil Precautions

Plastic foils will wear under the action of the fabric passing over the surface. Since the amount of water removal is dependent on the amount of time that vacuum can act on the underside of the fabric, a worn foil will not be able to dewater a fabric as well as a new foil.

If the dewatering rate appears to be reduced, it is always a good idea to check the condition of the blades. Ceramic blades or partial ceramic blades will retain their shape for long periods of time. However, great care must be used when handling these blades as they will chip and break rather easily. Damage to fabrics may occur if blades are chipped.

If a pair of new foils are put onto a foil unit on each side of a worn unit, the worn foil will not contact the fabric, no nip will be formed and no doctoring or dewatering can take place. The same thing can happen if the total table is not accurately aligned. All foils should be at the plane of the fabric and level in all directions.

Low Vacuum Region

Dandy Rolls

Dandy rolls were originally used to impart special marks in the sheet for decorative and identifying reasons. Many dandies are used today to improve the formation of the sheet. This improvement comes from a shear action of the roll relative to the sheet and the fabric. The shear is intended to rearrange fiber in the sheet, breaking up large flocs and reducing their size. In order for this shear to be effective, it must take place when the sheet has enough water in it to permit the disruption of floc and the rearrangement of fiber. If the sheet is too dry, the dandy will have no effect.

Special flooded boxes are sometimes used under the dandy to help with rearranging the fiber to improve formation and watermark clarity if a watermark is used. In the absence of these boxes, some dewatering can take place at the dandy but this is not a major function of the dandy.

It is generally accepted that 3% consistency is the highest consistency where good dandy operation will take place. This is somewhat grade dependent. It is best to run consistency trials to determine the optimum effectiveness of the dandy. In most cases, the dandy surface speed will exceed the wire speed by up to 1.5%.

Spray from the outgoing nip of the dandy can cause water spot defects in the sheet if the drop hits the sheet after it is too dry. Sometimes an internal fog shower with a water based defoamer in the spray will reduce or eliminate the spray.

Vacuum

Foils are capable of drying the sheet out to about 1.5% to 3.0% consistency, depending on the grade of paper being produced. Vacuum assistance of the foil drainage is necessary for further drying.

In the forming section, the sheet is being compacted and consolidated to create smaller and smaller flow paths to promote efficient dewatering. After the foil section, the sheet is a completely saturated fiber mat. As long as there is free water in the fiber mat, vacuum is useful in removing that water.

To remove the saturating water, low level vacuums are very effective. After the water film or saturating water has been removed, then high vacuums will remove much of the rest of the water.

Low Vacuum Equipment

The typical low vacuum unit consists of a stainless steel box which can be fitted with foil blades. There are several different designs of foils which can be used on the box. Each supplier of boxes has his own theories on what works best. All styles of units have deckle pieces which completely close the top of the box with a vacuum seal.

A vacuum inlet is provided to permit the application of vacuum to the underside of the fabric in the spaces between the foils. In many cases, this inlet is independent of the water drain of the box although a few units exist where the water comes out the same opening through which the vacuum is applied. In all cases, the water drain must be set up so that water can drain freely from the box without air leaking back in.

For best operation, the vacuum source should be different than the flat box vacuum source. One of the best arrangements is to use a low vacuum level blower with the vacuum line being connected to the inlet of the blower.

Low Vacuum Operation

Low level vacuum boxes are effective at removing water which is saturating a fiber mat. The water acts as a seal, allowing the sheet to drain into the box, evacuating the free water from the fiber mat. The water flow not only dries the sheet but the flow velocity creates drag on the fibers that gradually compacts the mat into a denser sheet. The effectiveness of the low vacuum level is gone when air breaks through the sheet. This happens at the dry line.

Early on the table, the sheet is liquid enough that the deflection caused by the foil action and the applied vacuum provides additional turbulence for better formation. Late on the table, the sheet is no longer mobile enough to have any formation improvement. At this point, the effect is solely to remove water.

Low vacuums dry the sheet to 7 to 9% consistency, again depending on the grade. Vacuum augmented foils typically operate at 1 to 20 inches of water vacuum.

Low Vacuum Troubleshooting

A vacuum foil body must drain water freely to work the way it should. There should be sufficient clearance under the seal leg. As a rule of thumb, the surface area of the imaginary cylinder beneath the seal leg should be at least equal to the cross-sectional area of the seal leg.

The vacuum lines connected to the unit should be large enough in diameter and as short as possible to ensure the total effect of the vacuum source is carried to the unit. Small diameter piping will cause large pressure drops in the piping. There should also be no low spots in the piping to create a water pocket which will cause vacuum surges and uneven water removal in the machine direction.

A vacuum gauge or a manometer tube should be installed on each unit. The maximum effective vacuum that can be applied to the unit is equal to the distance from the water in the seal pit to the bottom of the vacuum inlet. If this distance is exceeded, the vacuum level in the unit will vary and cause uneven drying.

High Vacuum

The water remaining in the sheet after the dry line is the water remaining on the fiber surface, in the fiber crossovers, and the water in the fibers themselves. The efficient way to remove the first two is to use high velocity air to strip the water off the surfaces and out of the crossovers. The last must be removed in the presses and dryer sections. Flat boxes with various types of open area (slots or holes) and high vacuums generate the high air velocities necessary to accomplish the stripping of water.

A flat box must be rigid enough to withstand the vacuum source applied and support the cover to prevent cover collapse. The number of slots and the width of the slots depend on the grades being produced, speed of operation and sequence of boxes on the table.

With polyethylene covers, the plastic of the cover will wear away and somewhat protect the fabric. The cover wear will however be uneven and eventually uneven water removal takes place over the cover. Profile problems and drying streaks will result.

Ceramic covers have a lower coefficient of drag than polyethylene covers. A good cover will last indefinitely and cause no fabric wear. However, the edges of the holes and slots must be very smooth to prevent fabric wear. Nicks, chips or rough margins will eat fabrics up.

High Vacuum Operation

Flat boxes are the greatest source of wear to a forming fabric. Covers for flat boxes have either drilled holes or narrow slots over which the fabric travels. The high vacuums cause the yarns to be distorted and abraded on the hole edges or by abrasive particles trapped between the fabric and the cover.

Typical fabric wear patterns show high wear rates at the end of the slots on both the drive side and the aisle side of the machine. This wear area is generally the place where the fabric will wear and split in the machine direction. Several options such as lubrication showers, graduated vacuums and staggered slot ends will reduce the problem. It can hardly be eliminated.

Fabric wear can also be reduced by the use of wear resistant yarns containing nylon or other special materials, by applying special resin treatments to the fabric and also by applying fabric wear beads to the high wear area of the fabric outside the trim squirts.

High power loads on the drive motors on the wet-end can be caused by poor flat box operation. In any vacuum application under the forming fabric, higher dryness and lower loads will result from graduating both the low and high vacuums from low to high, reserving the highest vacuum for the last box. This method of operation will also help reduce fabric wear.

The flat boxes should dry the sheet within the range of 16 to 18% if they are operated efficiently. Flat boxes operate generally at 1 to 6 inches of mercury vacuum.

Steam boxes are used over flatboxes on some grades to heat the sheet water, lowering the water viscosity, making it easier to remove.

Couch

Some styles of fabrics will cause a loss in couch efficiency. Thick, open bottom fabrics will carry air or permit radial air leakage into the couch box, bypassing some of the high velocity air under rather than through the sheet. However, the most common cause of low drying efficiency is the couch itself, either from worn seal strips, poor internal lubrication, or a vacuum source not operating at design standards.

The couch is the final opportunity to dry the sheet to the maximum prior to transfer to the presses. On most machines, a 4 to 6% increase in consistency can be expected over the couch.

Couch rewet results from excess water at the couch, particularly the water from internal and external couch showers. It is helpful to install a flow meter on the internal shower and seal water lines. By reducing the flow rate until a slight drag increase is noted and then increasing slightly will ensure the minimum of excess water and reduce rewet possibilities. In addition, locating the internal suction box so that there is a very small airflow into the couch at the point of sheet release will remove the entrained water in the fabric, keeping that water from rewetting the sheet.

Lump breakers are used to compact the sheet over the couch. By compacting or densifying the sheet in this area, pore size is reduced which increases air flow velocity for more efficient dewatering.

2.5.2 *Twin-Wires*

Modern twin-wire machines were developed to overcome the following disadvantages of fourdrinier machines [16]:

- two sidedness of the sheet
- lack of fine scale formation
- nonuniform profiles
- length and speed limitations

Twin-wire forming is the type of forming in which sheet is formed between two rotating forming fabrics. Although the idea of forming paper between two screens dates back to 1875, the credit for the first modern twin-wire machine is given to Daniel Webster of Consolidated Paper who had a working model in 1953.

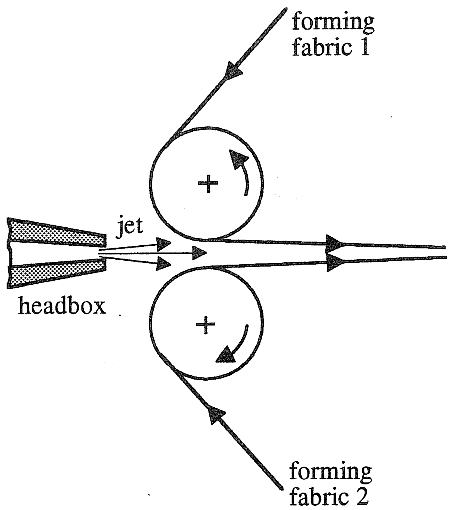


FIGURE 2.86. Schematic of gap former.

Since then, many variations of twin-wire formers have been developed. Delivery is through a nozzle which may have layering capabilities. A detailed discussion of these machine types is out of the scope of this book. Therefore, only a brief description of the machine types will be given below.

Twin-wire formers can be classified as gap formers and top wire formers.

Gap Formers

In a gap former, the slurry is injected from the headbox nozzle between two converging forming fabrics as shown in Figure 2.86.

Depending on the dewatering mechanism, there are three major types of gap formers:

1. Gap roll formers: roll wrap is the major dewatering mechanism (Figure 2.87). Since there are no stationary parts, fabric life is usually longer in these machines compared to fourdriniers. Other advantages of gap roll formers are low drive power, reliability, speed capability, reduced two sidedness, less linting and improved printability. The disadvantages are higher headbox consistencies, potential for pinholes, lower internal bond and grainy formation.

Neglecting the centrifugal and vacuum forces, the following simplified formula has been accepted to determine the water removal force in gap roll formers:

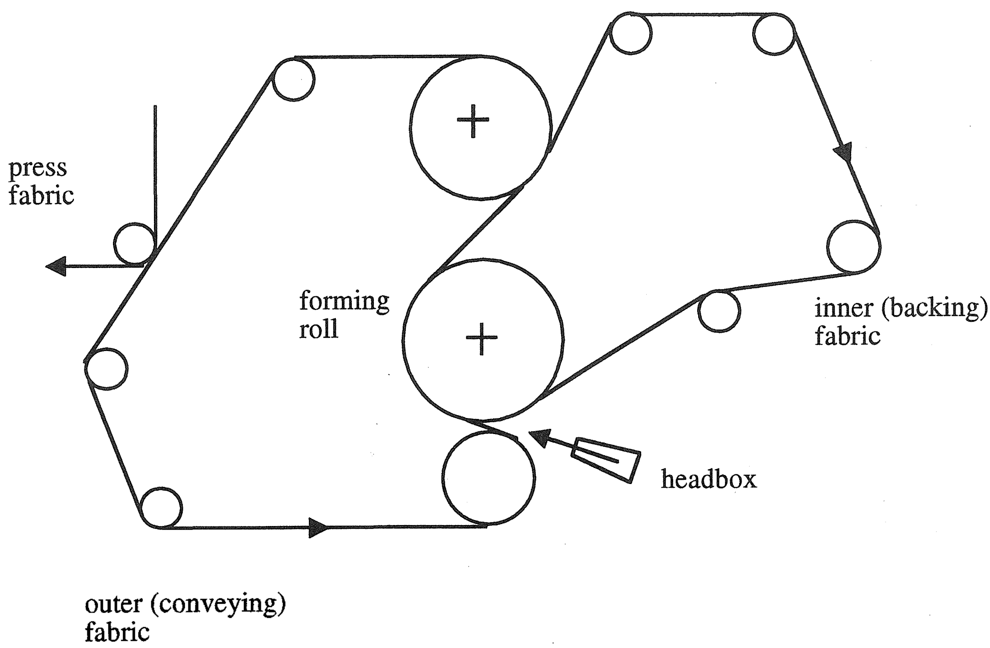


FIGURE 2.87. Schematic of a typical gap roll former ("C" wrap).

$$P = \frac{T}{R} \qquad (2.25)$$

P = dewatering pressure
T = outer wire tension
R = radius of curvature of the forming roll

Gap roll formers were initially used for newsprint grades. Today, their uses include brown paper, newsprint and high speed, lightweight sheet manufacturing such as tissue. The two common tissue formers are called "C formers" or "S formers" with respect to the shape of the drainage zone. The "C former" has a "C" shape drainage zone as shown in Figure 2.87. The forming roll is either solid for one-sided dewatering or perforated with suction for two-sided dewatering.

Figure 2.88 shows the schematic of an "S former" with an "S" shape drainage zone. Forming roll can be solid or with suction. In either case, two-sided dewatering takes place in this machine due to its configuration. The outer, conveying fabric also provides primary drainage.

2. Gap blade formers: blade action is the major dewatering mechanism (Figure 2.89). The drainage zone can be straight or curved. If the drainage zone is curved, then Equation (2.25) can be applied for curves that are approximately 500 cm in radius.

Gap blade formers have better formation, higher drive requirements and lower retention compared to gap roll formers [16].

3. Gap roll/blade formers: both wrap and blade actions are used for dewatering (Figure 2.90).

Top-Wire Formers

In top-wire formers, a forming section with an additional fabric is placed on top of the fourdrinier table, which carries the bottom forming fabric (or conveying wire). There are a variety of top-wire formers which are called retrofits, hybrids, preformers or on-top formers. The common characteristic of top-wire formers is that they all have a preforming zone of "open-wire" papermaking like a fourdrinier machine. Following open-wire preforming, dewatering can be done with roll, blade, roll/blade or adjustable blade. Figure 2.91 shows

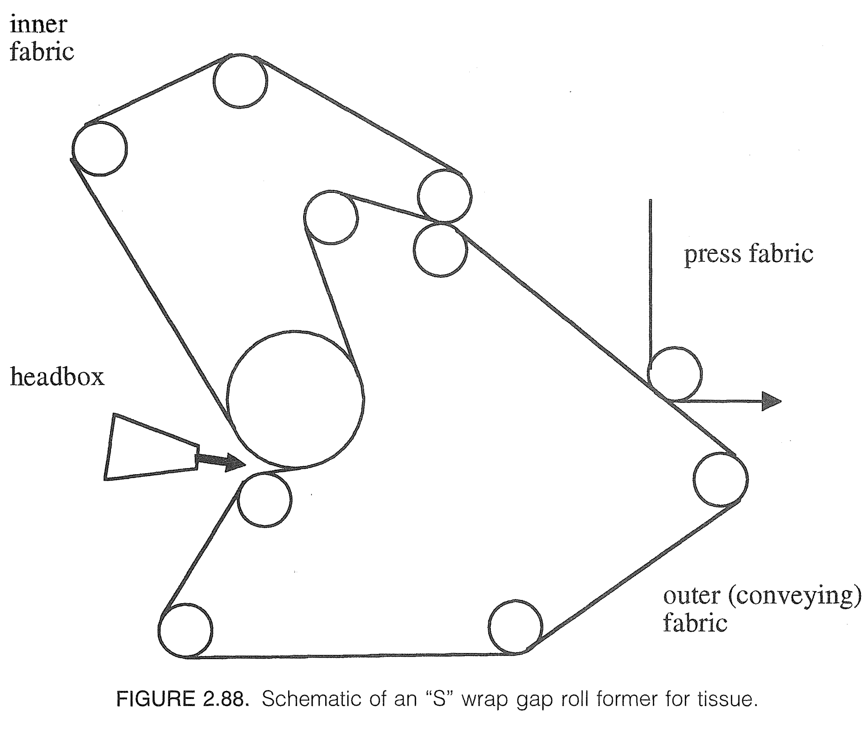


FIGURE 2.88. Schematic of an "S" wrap gap roll former for tissue.

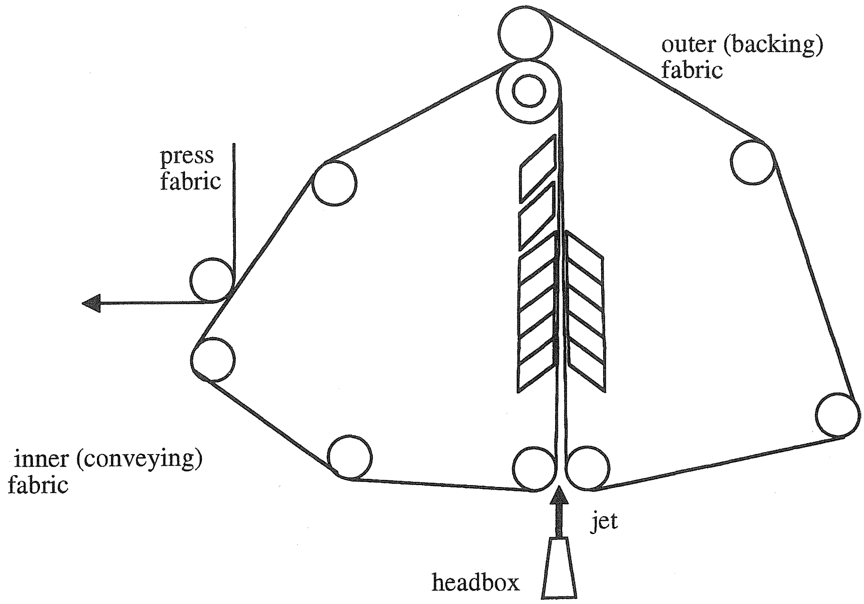


FIGURE 2.89. Schematic of a typical gap blade former.

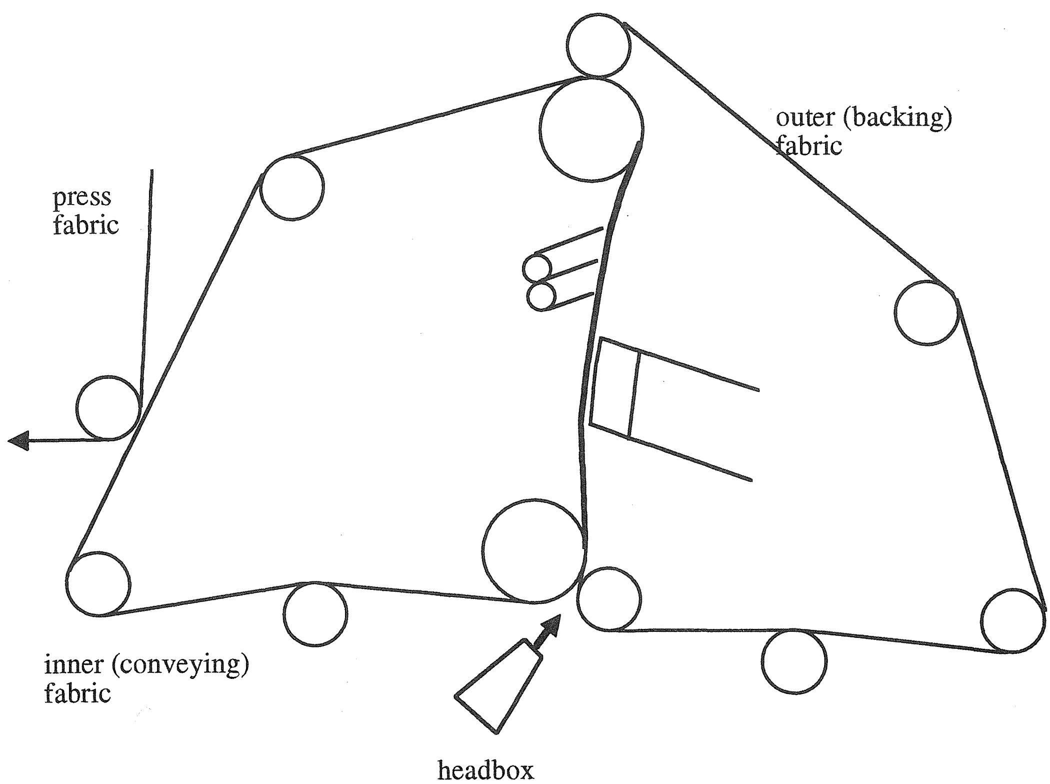


FIGURE 2.90. Schematic of a typical gap roll blade former.

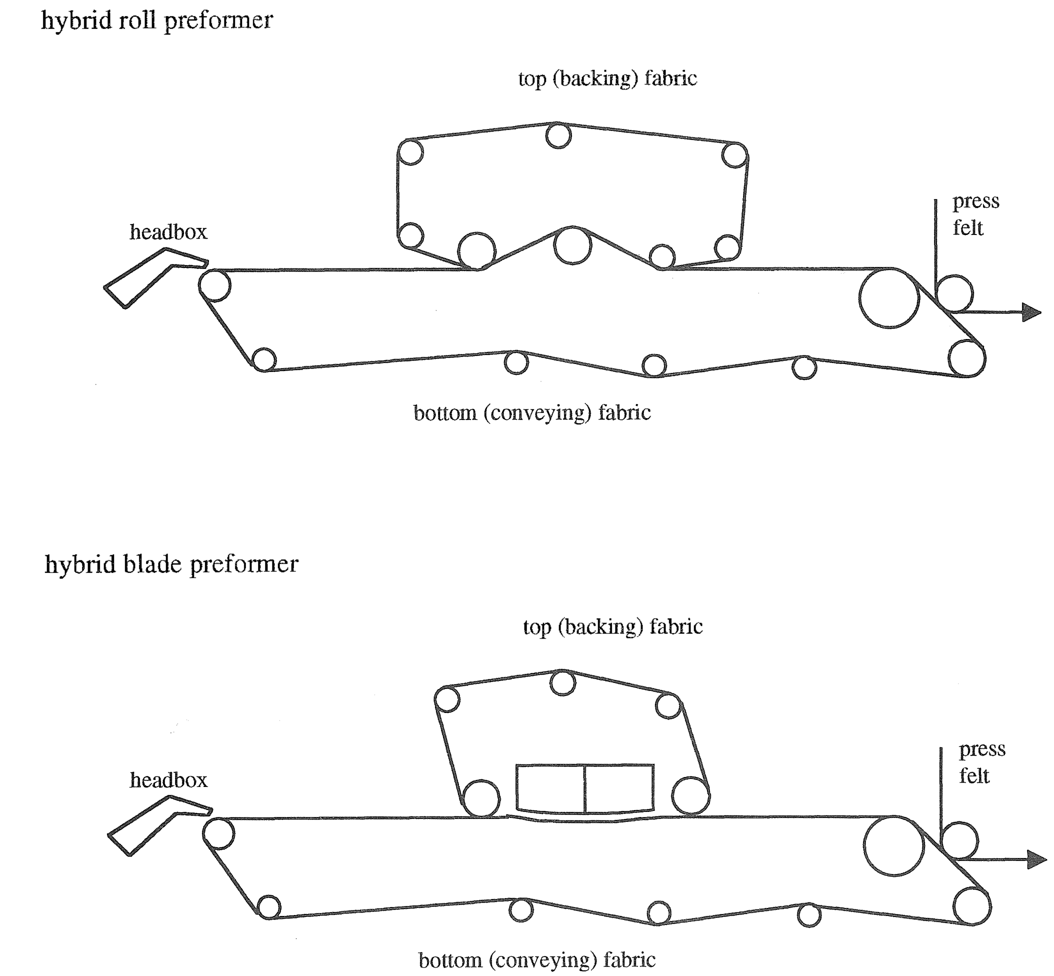


FIGURE 2.91. Examples of top-wire formers.

examples of top-wire formers. These formers are generally used for lighter weight sheets. It is reported that they improve two sidedness, reduce linting and improve formation compared to fourdrinier machines. Compared to gap formers, preformers have less streaking, good formation with improved first pass retention, better cleaning and drainage capability.

2.5.3 *Two-Wire Formers*

In two-wire formers, two sheets are formed separately and combined later on the forming section. Therefore, there are two headboxes on the paper machine as shown in Figure 2.92. These machines are usually used for heavier weight grades.

2.5.4 *Multiwire Formers*

In multiwire machines three or more fourdrinier tables are used to form the sheet. As a result, there are at least three headboxes and three forming zones as shown in Figure 2.93. Proper bonding between the sheet layers is critical. Bonding is controlled by the consistency of the various layers of fibers being combined.

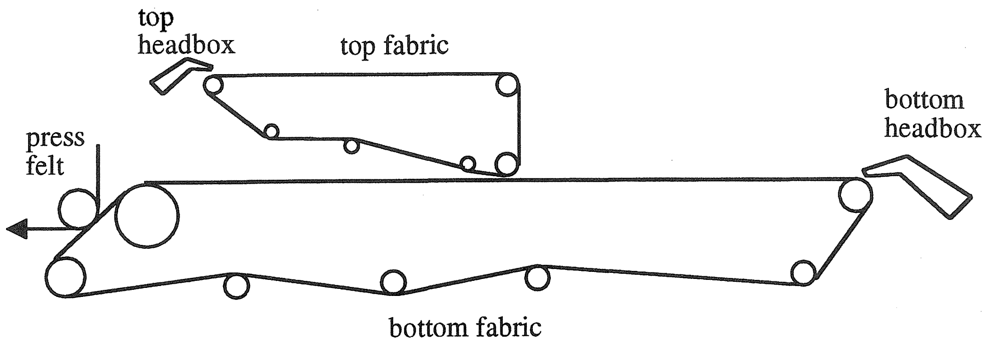


FIGURE 2.92. Schematic of a typical two-wire former.

2.5.5 *Crescent Formers*

A Crescent former is basically an inverted "C" gap roll former that is used for tissue manufacturing (Figure 2.94). Felt and forming fabric are used in the Crescent former which are unique to this machine.

2.5.6 *Cylinder Machines*

Cylinder machines were especially suitable for heavyweight multiple board grades. Although cylinder machines are still running in some applications such as tube stock, core stock, jigsaw puzzle and other grades that are too heavy to be practical for fourdrinier production, their design has become obsolete and therefore they are not manufactured anymore. Figure 2.95 shows the working principle of these machines. The sheet is formed on the wire mesh of a rotating cylinder mold, partially submerged in a vat of fiber stock. Aided by a couch roll, the fiber mat transfers from the cylinder to a making felt, which is a woven fabric combined with nonwoven batt. The flow of slurry into the vat can be in the same direction of or opposite to the rotation direction of the vat. Several cylinder vats are placed in a sequential manner which allows formation of a multilayer sheet structure. These machines were slow and had some formation problems such as uneven basis weight.

2.6 Manufacture of Specific Paper Grades

The sheet formation principles of different paper grades such as tissue, newsprint, fine paper, brown paper, etc., are basically the same. Nevertheless, there are some differences in manufacturing of some grades. This section explains the major manufacturing

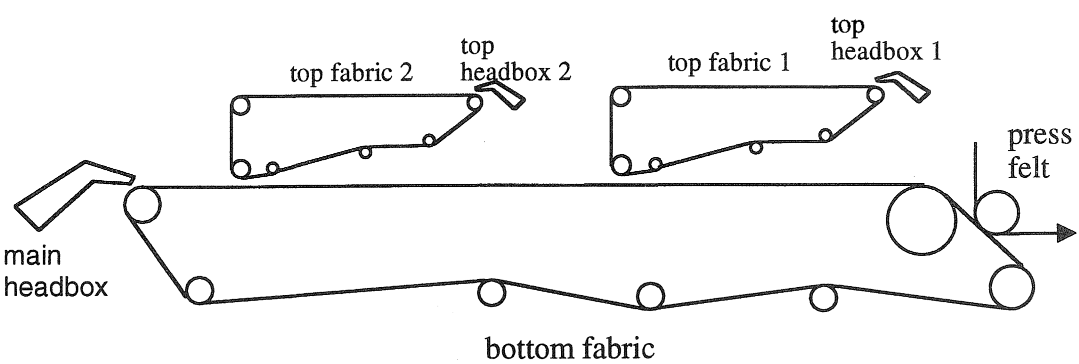


FIGURE 2.93. Schematic of a typical multiple forming machine with three forming fabrics.

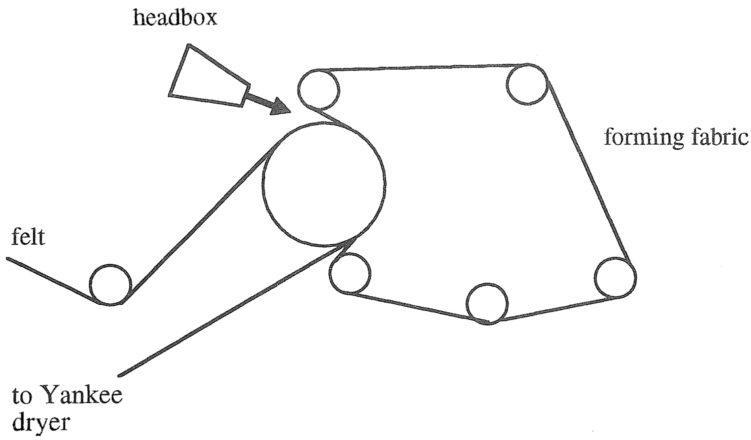


FIGURE 2.94. Schematic of a Crescent former.

characteristics that are specific to each paper grade. The manufacture of nonwovens is quite different than the other paper grades. Special methods and equipment have been developed to manufacture nonwovens. A more detailed description of these methods is given.

2.6.1 *Tissue Manufacturing*

The tissue market covers a variety of low-weight sheets used primarily in sanitary (facial and bathroom tissue, towel, napkin, etc.) applications. A much smaller segment includes specialized grades such as condenser, carbonizing and wrapping tissue [17]. Most sanitary tissue fabrics run CD knuckle proud to orient the fibers in the CD direction. The hole is longer in the CD direction.

Due to its low basis weight (as low as 5 g/m^2) or loose structure, special machines have been developed to manufacture tissue. One machine that has been used consists of a fourdrinier forming section and Yankee dryer. The Yankee dryer is a large steam cylinder with a polished surface to dry the tissue. Tissue is pressed against the Yankee surface and transferred to it. Through-air dryers are used before or after Yankee dryer to preserve bulk and increase drying efficiency as shown in Figure 2.96 [17–19]. Modern tissue formers include "C" wrap, "S" wrap and Crescent former (Figure 2.97). The wet tissue sheet must be supported in forming, pressing and drying sections of the paper machine.

In many cases, creping is done to increase the softness of tissue paper. As a result of

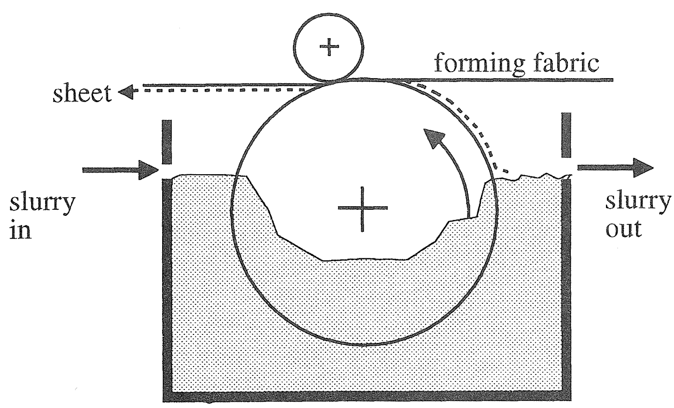


FIGURE 2.95. Schematic of a cylinder vat.

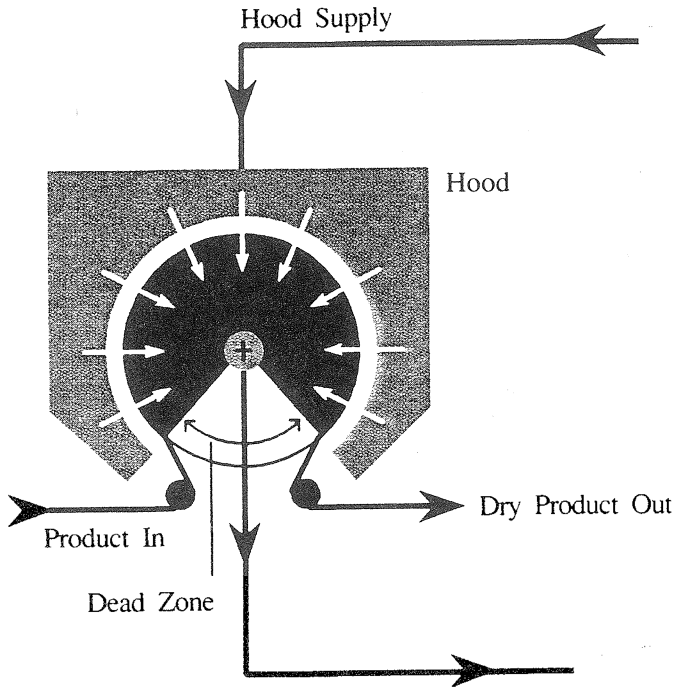


FIGURE 2.96. Through air dryer (courtesy of Valmet Paper Machinery).

creping, the MD and CD strengths of the paper are reduced. Extensibility is increased which increases the rupture resistance of the sheet. Creping also increases the bulk of paper.

Creping is done on the Yankee dryer cylinder using a doctor blade (Figure 2.98). The web follows the dryer surface to hit the doctor blade. The degree of crepe has an effect on the bulk. The degree of crepe is defined as the difference in peripheral speed between the dryer and the pope divided by the speed of the dryer. Around 50% crepe, the bulk reaches a maximum. Crepe formation is affected by doctor blade geometry, adhesion between the paper and dryer surface and mechanical properties of the paper [20].

2.6.2 *Newsprint and Fine Paper Manufacturing*

During formation of newsprint and fine paper, it is preferred that fabric speed is higher than the jet speed.

During printing, newspapers ran in the direction of reading which is the machine direction on the paper machine. There is considerable amount of tension on the sheet

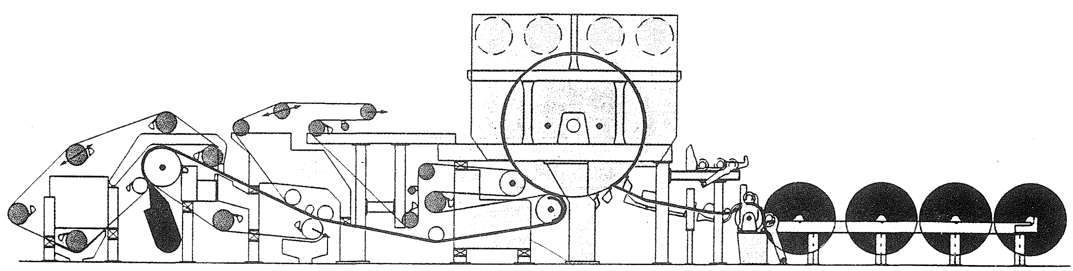


FIGURE 2.97. Schematic of a tissue machine with Yankee cylinder (courtesy of Voith Sulzer).

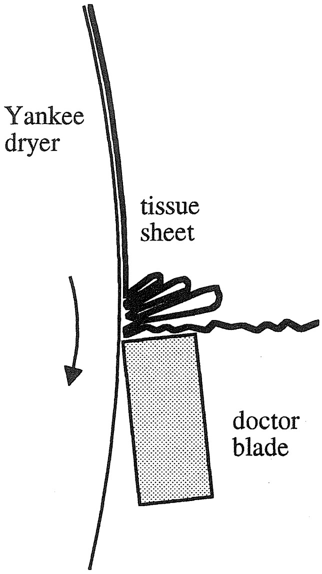


FIGURE 2.98. Schematic of creping process.

during printing; therefore, the paper should be strong in this direction. This is achieved by orienting the fibers in the machine direction. In fine paper fabrics, the hole is longer in MD direction.

The major additives in the fine paper grades are fillers which give improved opacity and other optical properties to the sheet. The two major types of fillers are clays and carbonates, both of which are very small particles compared to fibers. To be effective, they must be retained in the sheet.

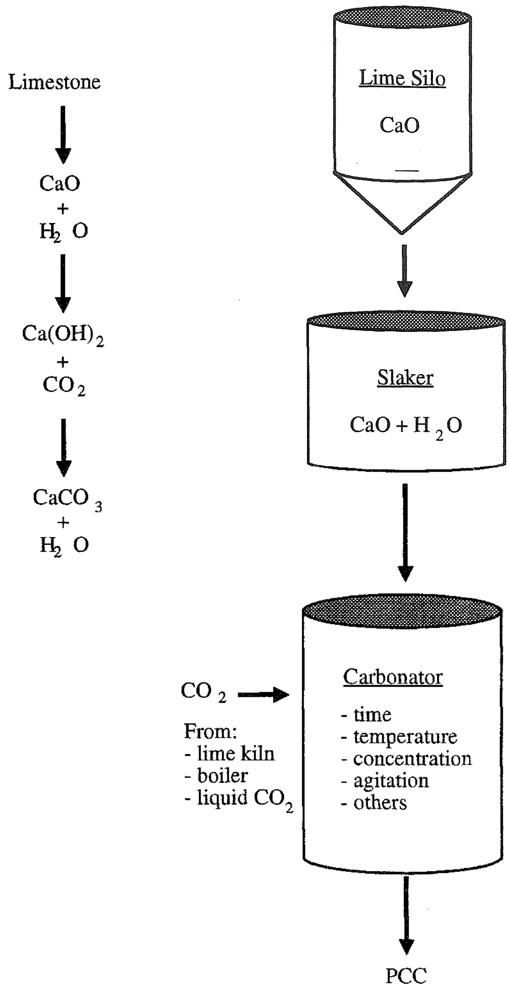


FIGURE 2.99. Schematic of PCC production.

Acid versus Alkaline Papermaking

Fine paper manufacturing can be done either in acid or alkaline environment. Alkaline papermaking results in longer paper life. Therefore, the use of alkaline papermaking is increasing. Ground Limestone (GL) is used in alkaline papermaking.

Contents of Precipitated Calcium Carbonate (PCC):

(1) calcite
- rhombohedron (barrel shaped)
- scalenohedron (rosette clusters)

(2) aragonite
- needle shaped (3:1 aspect ratio)

Figure 2.99 shows the schematic of PCC production. The acid and alkaline papermaking systems are compared in Table 2.15. Table 2.16 lists the effects of alkaline papermaking. For better abrasion resistance, nylon yarns are used along with polyester in forming fabrics that are used in alkaline conditions. In general, PCC produces less fabric wear than GCC (ground calcium carbonate).

2.6.3 *Nonwoven Manufacturing*

A nonwoven product is made directly from a web of fiber. Although paper also fits this

TABLE 2.15. Acid and Alkaline Papermaking Systems.

	Acid	Alkaline
pH	4.5–5	7.5–8
pH control	Alum	Calcium carbonate Soda ash
Fillers	TiO_2 Clay Extenders	Calcium carbonate TiO_2 Clay, extenders
Size	Rosin	Alkyl ketene dimer (AKD) Alkenyl succinic anhydrides (ASA)
Cationic starch	Tertiary amine	Quaternary amine
Points of addition		
Fillers	Machine chest/blend chest	Machine chest/blend chest
Size	Machine chest	Stuff box
Cationic starch	Machine chest/blend chest	Machine chest/blend chest

definition, paper is not considered to be a nonwoven product [21]. Due to manufacturing and product similarities between nonwovens and paper, a nonwoven product is usually considered as an intermediate form between paper and conventional textile fabrics.

Comparison of major textile fabric and paper properties are shown in Table 2.17. Nonwoven fabric properties generally fall in between those of paper and conventional textiles. Therefore, nonwovens frequently compete with textiles or paper in already established markets. Table 2.18 compares the production speeds of various processes.

Paper is a relatively low priced commodity compared to nonwovens which may be a motivational factor for paper companies to move toward nonwovens.

Five major types of nonwoven processes that use forming fabrics are wet laying, air laying, spunbonding, meltblowing and spunlacing. All of these processes are very

TABLE 2.16. Effects of Alkaline Papermaking.

Terms	Effects
Deposits	Generally less than with acid conditions, some press problems with size pick out in press section.
Drainage	Since $CaCO_3$ does not contain hold water, drainage is more rapid under alkaline.
Draws	May require slight changes.
Drying	Alkaline sheets dry easier than acid sheets due to less water bonding.
Formation	Table set up is critical. Refining is usually less and furnish has more long fiber.
Pressing	Equal to acid conditions.
Retention	Extremely important for filler and sizing efficiency. Early retention very important. Retention aids a must.
Runnability	Providing good wet-end stability, runnability is equal to or better than acid. Deposits and holes are most common problem.
Sheet dryness at couch	Usually better under alkaline conditions. $CaCO_3$ and lower fiber mass drain easier.
Table set up	Generally set up to carry water down the table. Grading of flatboxes is critical. Forming zone also critical.
Fabric wear	Proper fabric design and table setup minimizes wire wear. Equal to acid configurations under proper setup.

TABLE 2.17. General Characteristics of Paper and Textiles [21].

Paper	Textiles
Less expensive	More expensive
High production rate	Low production rate
Short fiber	Long fiber or filament
Simple structure	Complex structure
Smooth surface	Textured surface
Dense	Bulky
Inextensible	Some extensibility
Able to hold sharp folds	Noncreasing
Stiff, little drape	Flexible, drapes relatively easier
Relatively nonporous	Porous
Low wet strength	Wet strength is essential
Low tear strength	High tear strength

TABLE 2.18. Productivity of Various Manufacturing Methods [21].

Process	Linear Speed (m/hour)
Weaving	5–40
Knitting	100
Mechanical web formation	3,000
Spunbonding	4,500
Papermaking	50,000

different from one another and may require different designs of forming fabrics.

Wet Laying

In the wet-laying process, a mixture of fibers and water are laid down on the moving forming fabric (Figure 2.100). The water drains through the fabric, leaving a web or sheet of entangled fibers. The fibers can be wood, cotton, glass, polyester, rayon or any mixture of these. The function of the fabric in this process is to provide a carrier that is adequate to support the fibers, keeping them from penetrating into and through the mesh. Good web release and exceptional wear resistance are required in the fabric. An example of a wet-laid product is fiberglass mat used in the manufacture of roofing shingles which is shown in Figure 2.101 along with the forming fabric used to make it. This mat is impregnated with asphalt and provides the strength in the shingle. Table 2.19 lists the requirements of forming fabrics for wet-laying processes. Adhesives are required to replace the hydrogen bonding formed in papermaking.

Air-Laying

In the air laying process, wood pulp is ground up, mixed in an air stream and fed onto the moving forming fabric (Figure 2.102). A variety of fiber types can be used, but wood pulp is very common. An example of an air-laid product is a wipe which is characterized by being very soft, bulky and absorbent.

Forming fabrics used in air-laying must have good fiber support for retention of fibers and relatively high air permeability to ensure fiber hold-down. Due to the very high static buildup in this process, antistatic cross-direction yarns are used.

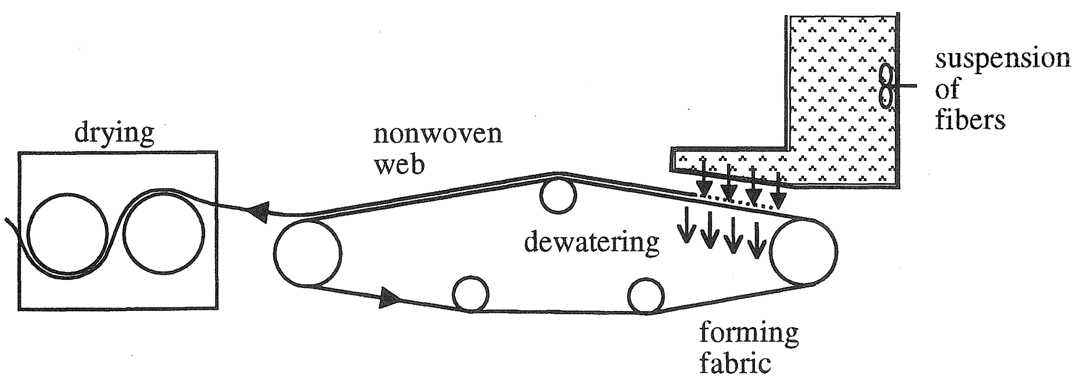


FIGURE 2.100. Schematic of wet-laying process.

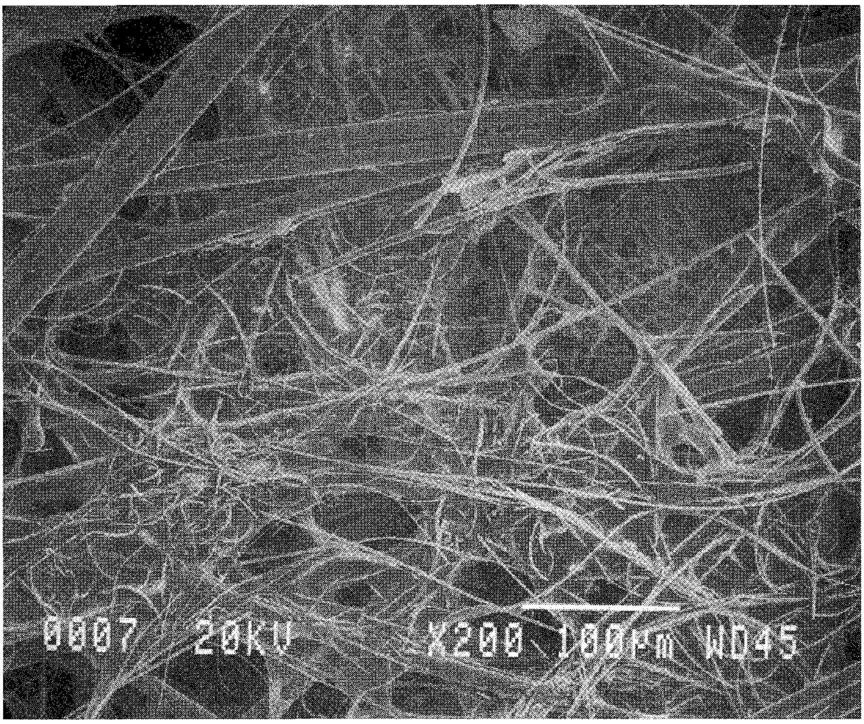


FIGURE 2.101. Structure of wet-laid fiberglass mat for roofing.

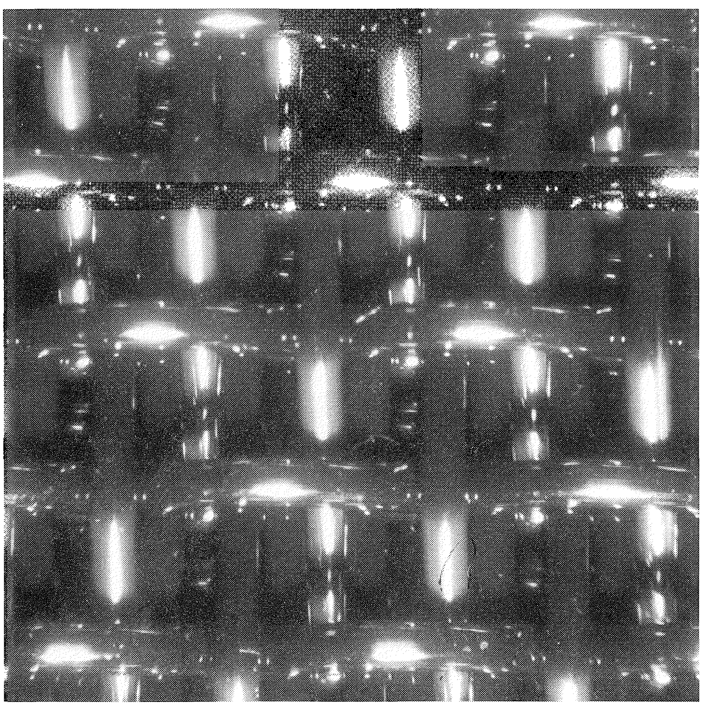


FIGURE 2.101 (continued). Forming fabric used to make wet-laid fiberglass mat for roofing.

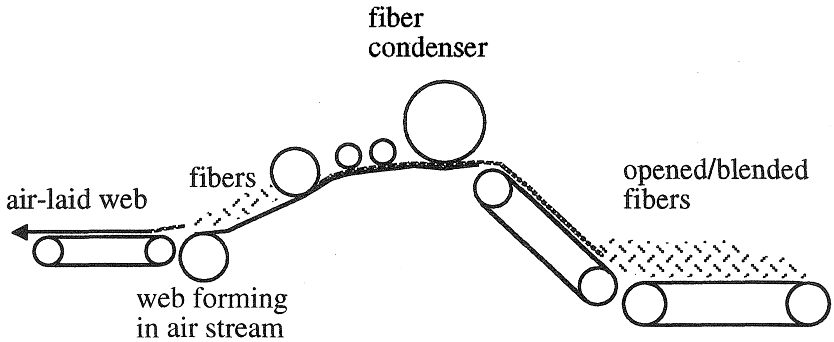


FIGURE 2.102. Schematic of air-laying process.

Forming fabric characteristics for the air-laying process:

- good fiber support
- fiber retention
- high air permeability
- medium mesh range
- two layer designs
- antistatic properties

Spunbonding

In the spunbonding process, pellets of polymer are melted down and forced through a spinnerette, a device with tiny holes like a shower nozzle (Figure 2.103). Continuous filaments are extruded through the spinnerette, are blown about and spread on the moving forming fabric.

The forming fabric used in spunbonding must have good fiber support characteristics to stop fiber penetration, provide good web release and not mark the sheet in any way. Figure 2.104 shows a polypropylene spunbonded medical product along with the forming fabric used to make it.

TABLE 2.19. Forming Fabric Requirements for Wet-Laying Process.

Binder Position	Forming Position
Stability	Stability
Durability	Durability
Good drainage	Good Drainage
Fiber retention	Fiber retention
Single layer designs	No wire mark
Good seam joints	Good seam joints
	Two layer designs
	Medium-fine mesh fabrics

Forming fabric characteristics for spunbonding:

- stability
- durability

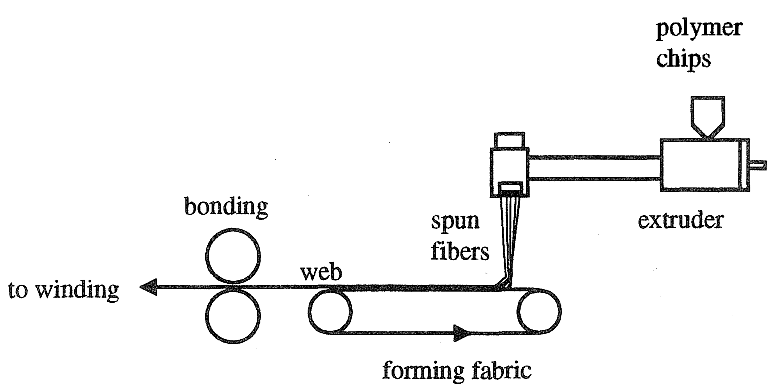


FIGURE 2.103. Schematic of spunbonding process.

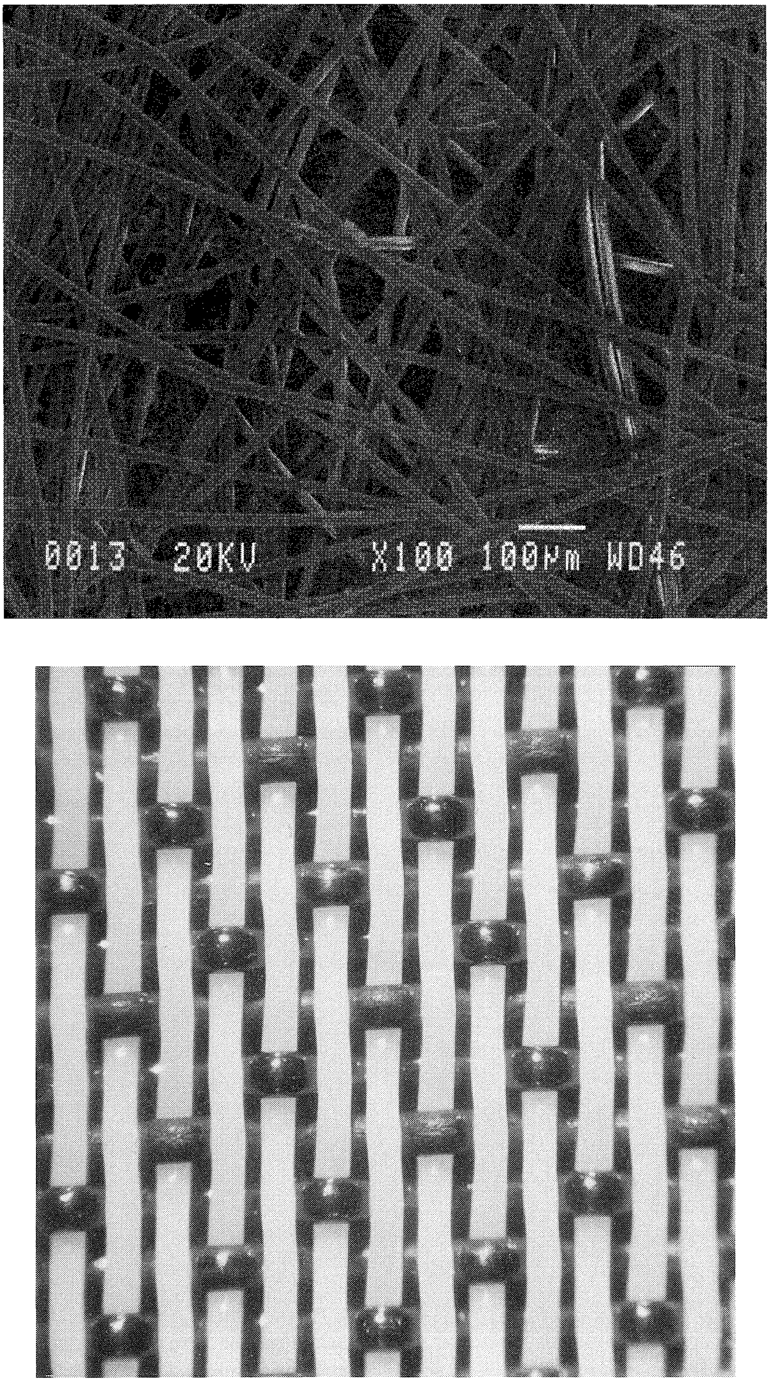


FIGURE 2.104. Polypropylene spunbonded medical product (top) and forming fabric used to make it.

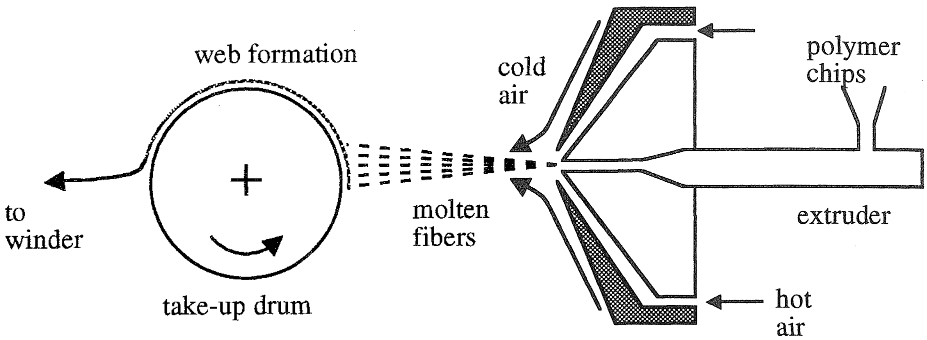


FIGURE 2.105. Schematic of meltblowing process.

- good fiber support
- high air permeability
- two layer designs
- medium-fine mesh range
- good seam joints
- anti-static properties on some applications

Spunbonded web characteristics are low weight, high stiffness and high tensile and tear strength.

Meltblowing

In meltblowing process, granules of polymer are melted down and blown with hot, high velocity air through extruder die tips. The filaments are then stretched until they break and are deposited onto the moving forming fabric (Figure 2.105). The fibers break into short lengths, as compared to the continuous fibers in the spunbonding process and are also much finer in diameter. Thus, the web produced is often used for wipes and filters.

The forming fabric for the meltblowing process must provide adequate fiber support, release and good stability. Figure 2.106 shows the fabric view from machine side with elastomeric polymer showing through. Figure 2.107 shows meltblown elastomeric polymer formed on a forming fabric.

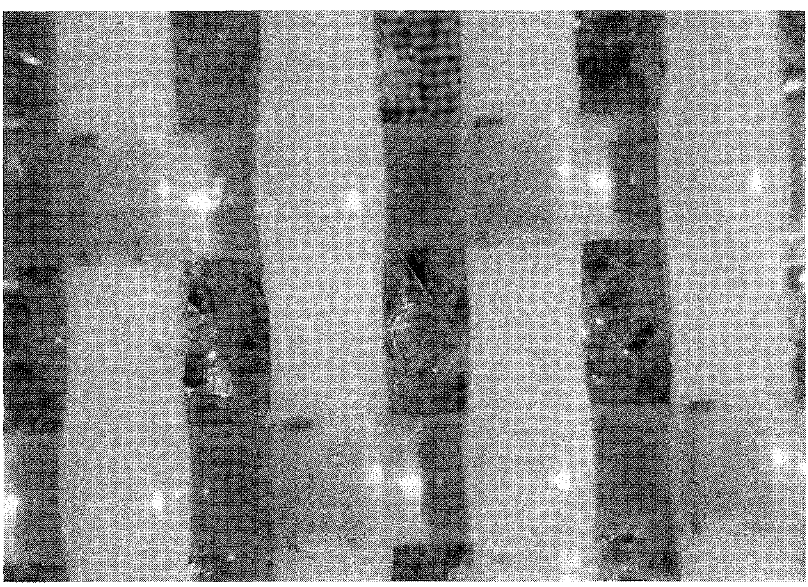


FIGURE 2.106. Machine side view of forming fabric and elastomeric meltblown polymer.

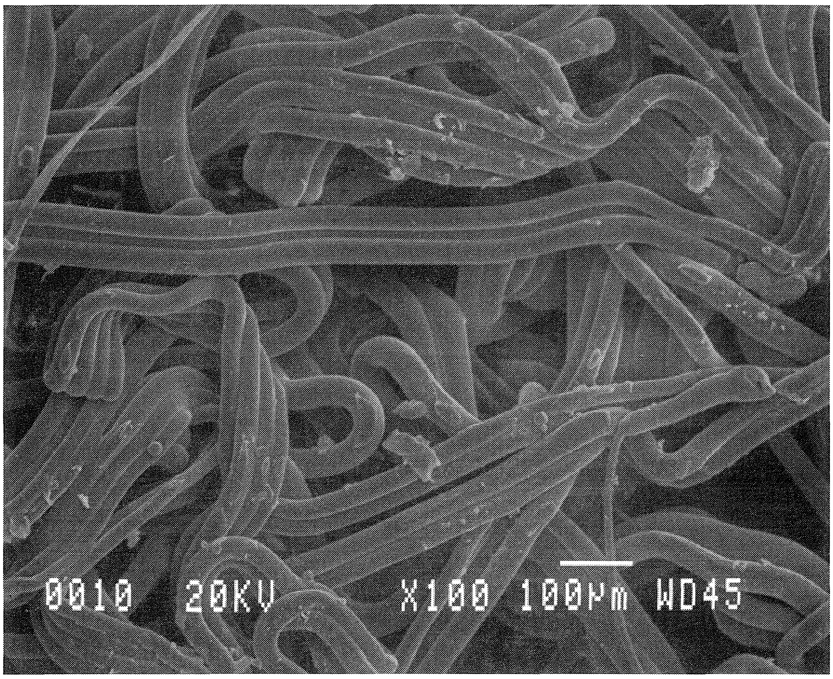


FIGURE 2.107. Photomicrograph of meltblown elastomeric polymer fabric formed on forming fabric.

Forming fabric characteristics for melt blowing process:

- large fabric design range
- single and two layer designs
- coarse-medium mesh range
- metal and synthetic fabrics
- good air permeability
- wire mark is less critical

Spunlacing (Hydroentanglement)

In this process, the finished product is produced by entangling fibers in a preformed web, generally carded, using high pressure, columnar water jets (Figure 2.108). As the jets penetrate the web and deflect from the forming fabric, some of the fluids splash back into the web with considerable force. Fiber segments are carried by the turbulent fluid and become entangled on a semimicron scale. In addition to bonding the web, spunlaced or hydraulic entanglement can also be used to impart a pattern to the web. Figure 2.109 shows a spunlaced rayon wipe and the fabric used to make it.

Forming fabric characteristics for spunlacing:

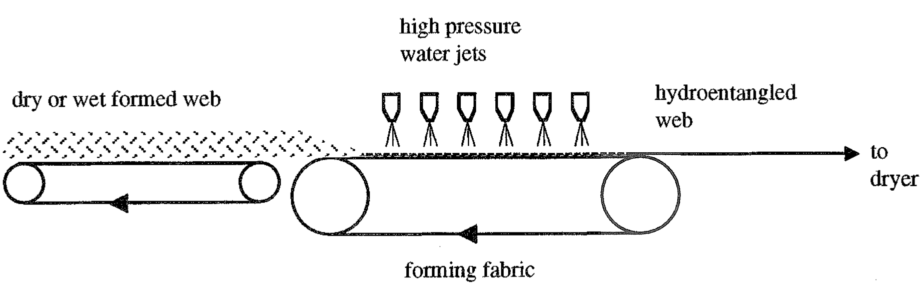


FIGURE 2.108. Schematic of spunlacing (hydroentanglement) process.

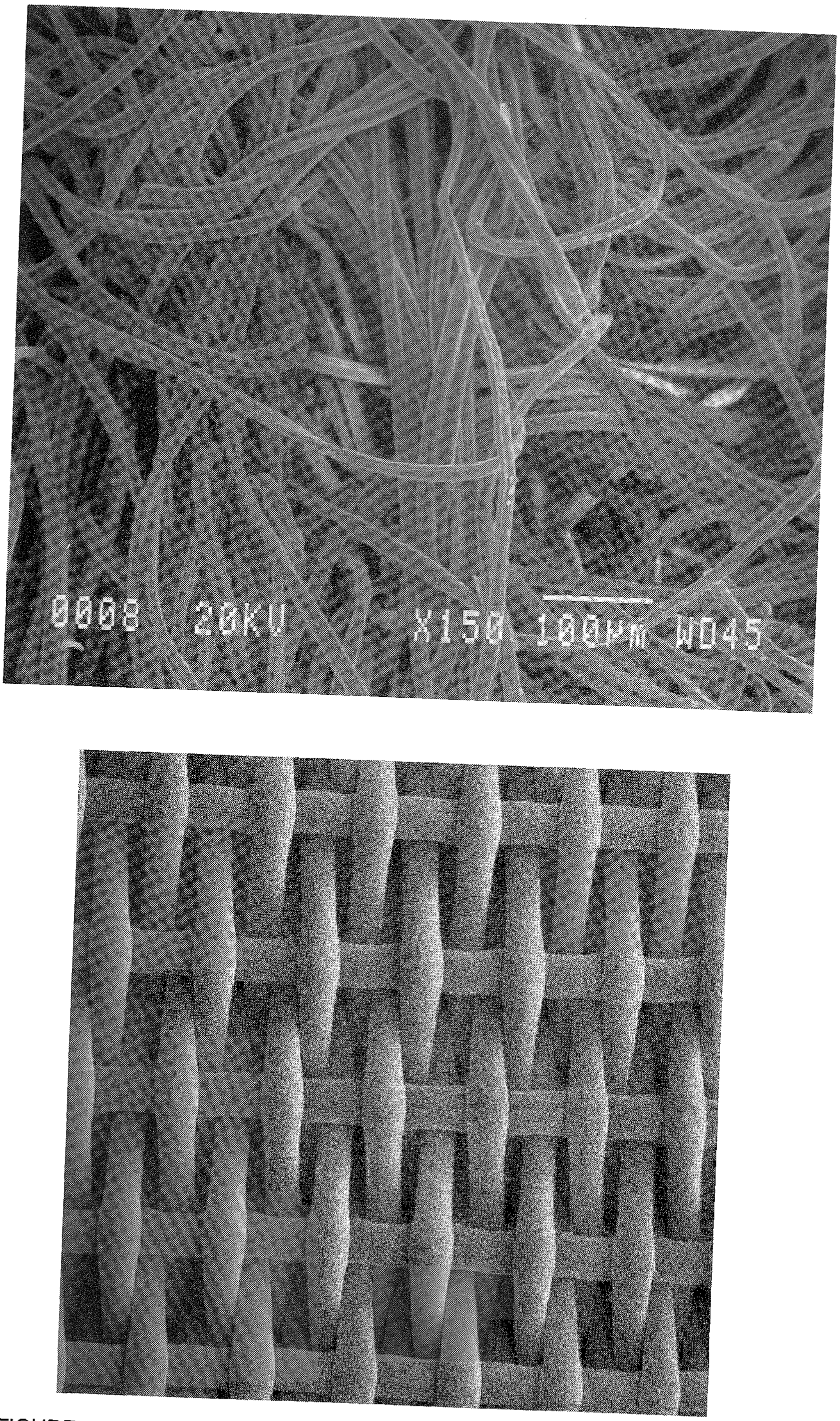


FIGURE 2.109. Spunlaced rayon wipe (top) and forming fabric used to make it.

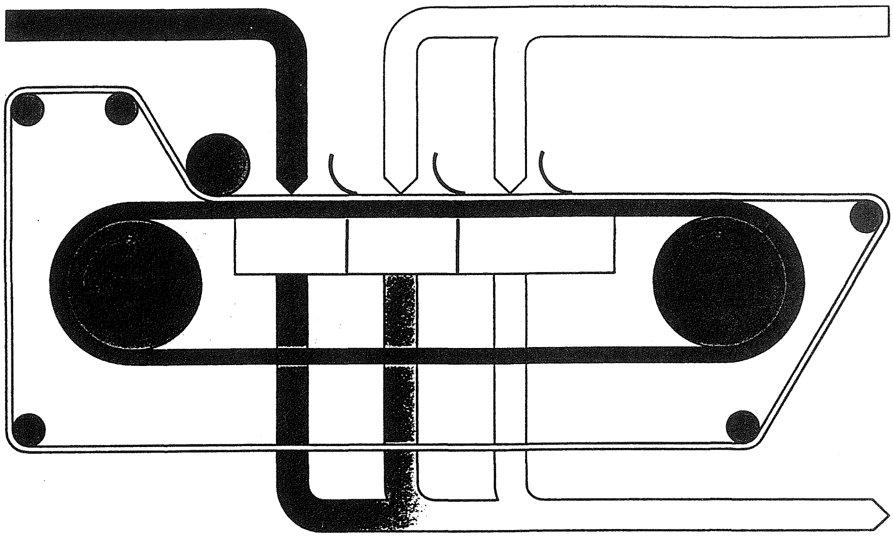


FIGURE 2.110. Schematic of filter belt application for cake washing (courtesy of Joy Manufacturing Company.)

- variety of designs
- stability
- good release
- fabric uniformity
- seam is critical

2.6.4 *Special Applications*

Fabrics with structures similar to paper machine clothing are also used in end-use applications other than papermaking. These fabrics are usually called "specialty fabrics." Examples of application areas for these fabrics are food processing, drying, dewatering and textile printing. Figure 2.110 shows the schematic application of a filter belt in cake washing.

Specialty fabrics are similar to forming fabrics or dryer fabrics (Chapter 4) for their structure and functions. The common characteristic of specialty fabrics is their coarse structure. They are woven under higher tensions than forming or dryer fabrics. Heatsetting is also done at higher tensions. The air permeabilities of common specialty fabrics range from 300 cfm to 1300 cfm.

Figure 2.111 shows the schematic application of a specialty fabric in textile drying

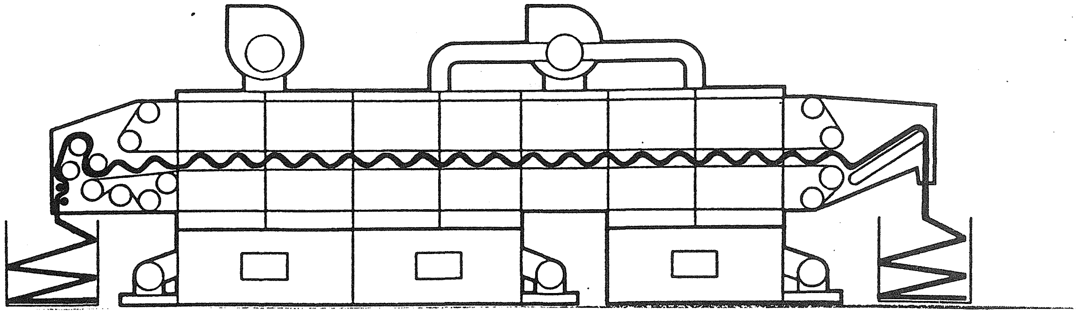


FIGURE 2.111. Schematic of textile drying application using a specialty fabric (courtesy of Santex AG Maschinenbau).

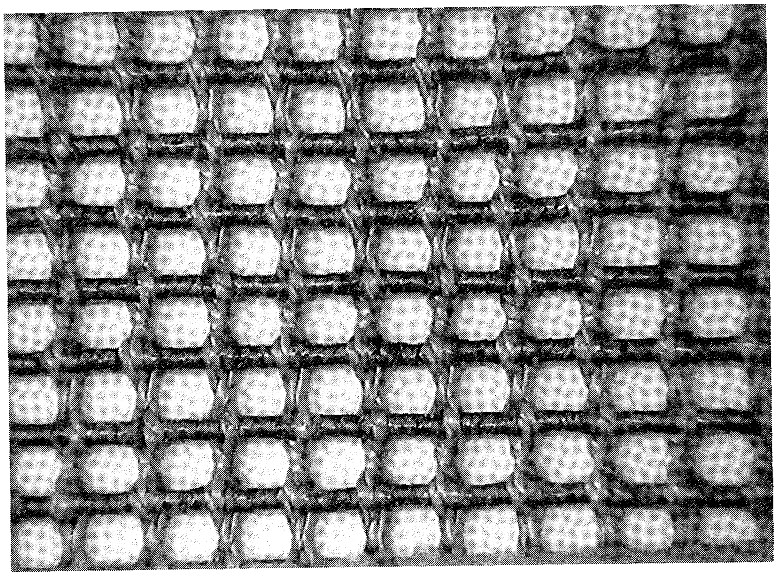


FIGURE 2.112. Leno design dryer fabric.

application. Figures 2.112 and 2.113 show two fabric designs that may be used for this type of application. Figure 2.113 is a leno design made of 100% Nomex® warp and Nomex® wrapped fiberglass filling. In the fabric shown in Figure 2.113, the warp yarns are grouped together to balance the weave and keep the fabric open. The fabric is made with 100% PPS monofilament.

Figure 2.114 shows a herringbone design that is used in recycled paper stock filtration. The design repeats on 12 warp yarns. Figure

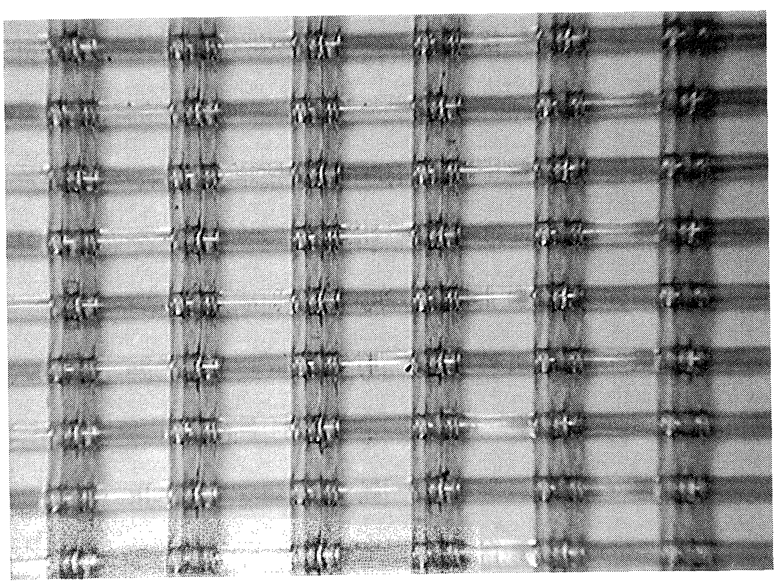


FIGURE 2.113. Fabric design with grouped warp yarns.

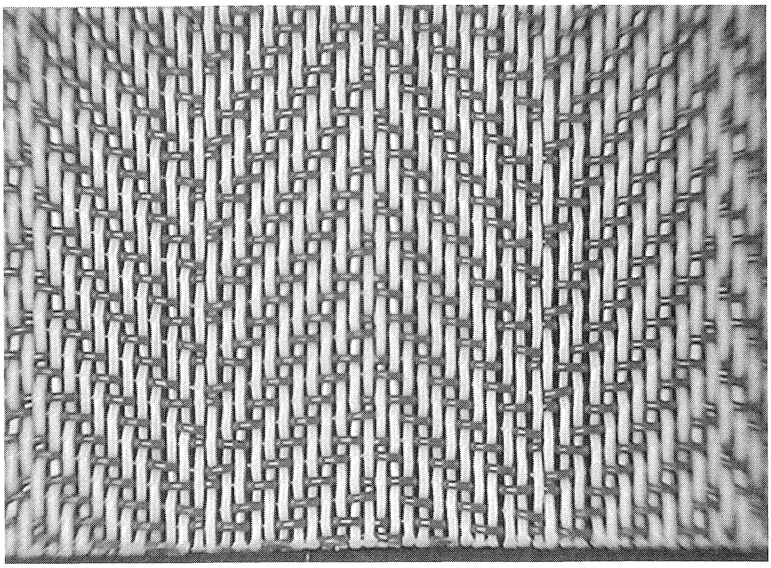


FIGURE 2.114. Herringbone point draw design.

2.115 shows a specialty fabric that is used for drying in milk powder production.

Some of these fabrics are woven endless while others are woven flat and seamed later. Seam styles include single wire clipper seam, coil seam, etc.

Coil Fabric

Figure 2.116 shows a unique polyester monofilament fabric designed for wet and dry formed construction board products. The design allows maximum water removal. The

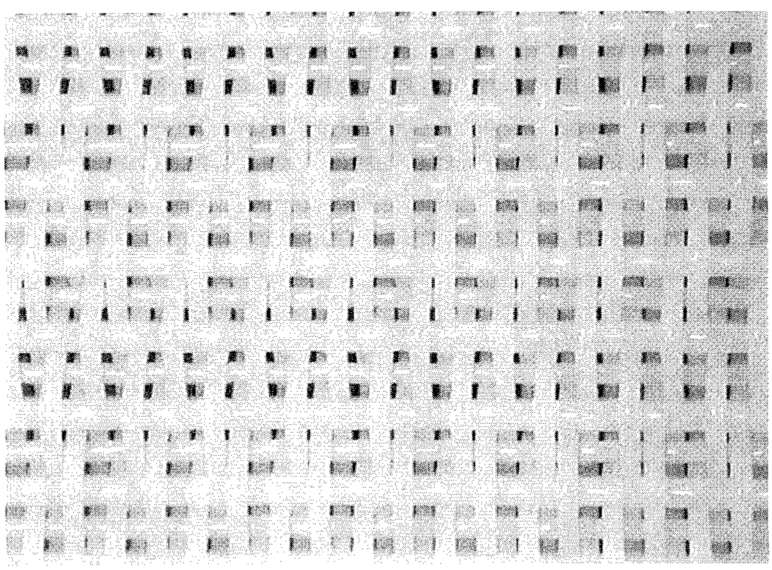


FIGURE 2.115. Fabric used in milk powder production.

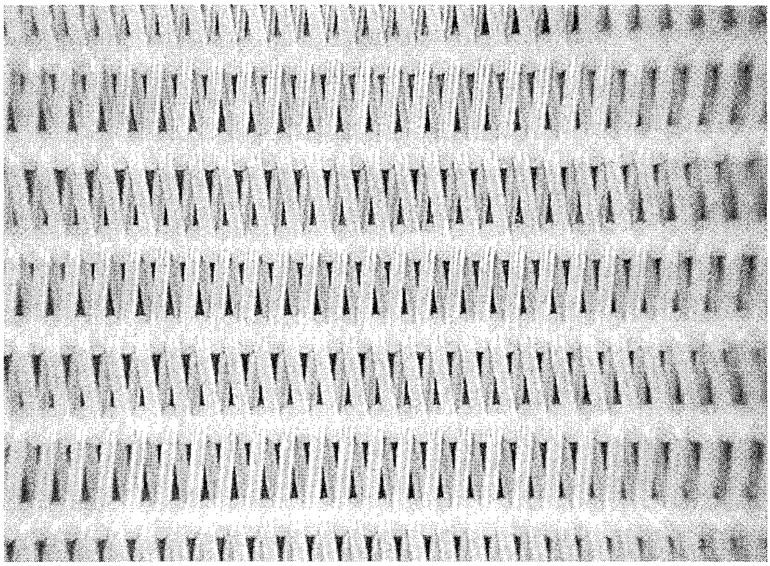


FIGURE 2.116. Coil polyester fabric.

durable coil construction offers good abrasion resistance and a smooth surface enhances doctoring properties for easy cake removal. The fabric can be supplied open ended without any type of conventional seam and is converted endless by merely inserting a synthetic filling yarn after meshing the ends.

Watermarks

Watermark is a pattern in paper made with a raised or indented design that comes in contact with the paper when it is approximately 90% water and 10% fibers. Watermarking has been known for almost 700 years. Watermarks first appeared in Italy around A.D. 1280 which depicted animals, fruits and flowers. Later two-color and light-and-shade watermarking were developed. One of the first commercial watermarks used in the United States was produced for Pennsylvania in 1777. Although it is not exactly known as to why watermarks were first used, they were probably used as a symbol of mysticism and religion. Today they are mostly used for aesthetic adornment and identification. Banks and insurance companies use them to prevent forgery.

Dandy rolls are used to make watermarks. With a wire mark dandy roll, while paper is still being formed, the stock passes under the dandy roll, and the design displaces some of the fibers to form the watermark pattern. As a result, the paper is more translucent and the watermark appears lighter than its surrounding area. Shaded mark dandy rolls utilize the intaglio process in which the design is depressed to form the watermark. As the dandy roll presses on the paper being formed, compression occurs everywhere but in the cutout design. Fibers accumulate more heavily in this depressed area, which results in a watermark slightly darker than its surrounding area.

2.7 Transfer of Sheet from Forming to Press Section

After the sheet is formed on the forming fabric, it is transferred to the press section for water removal and sheet consolidation. Figure 2.117 shows various configurations for sheet transfer from forming to press section.

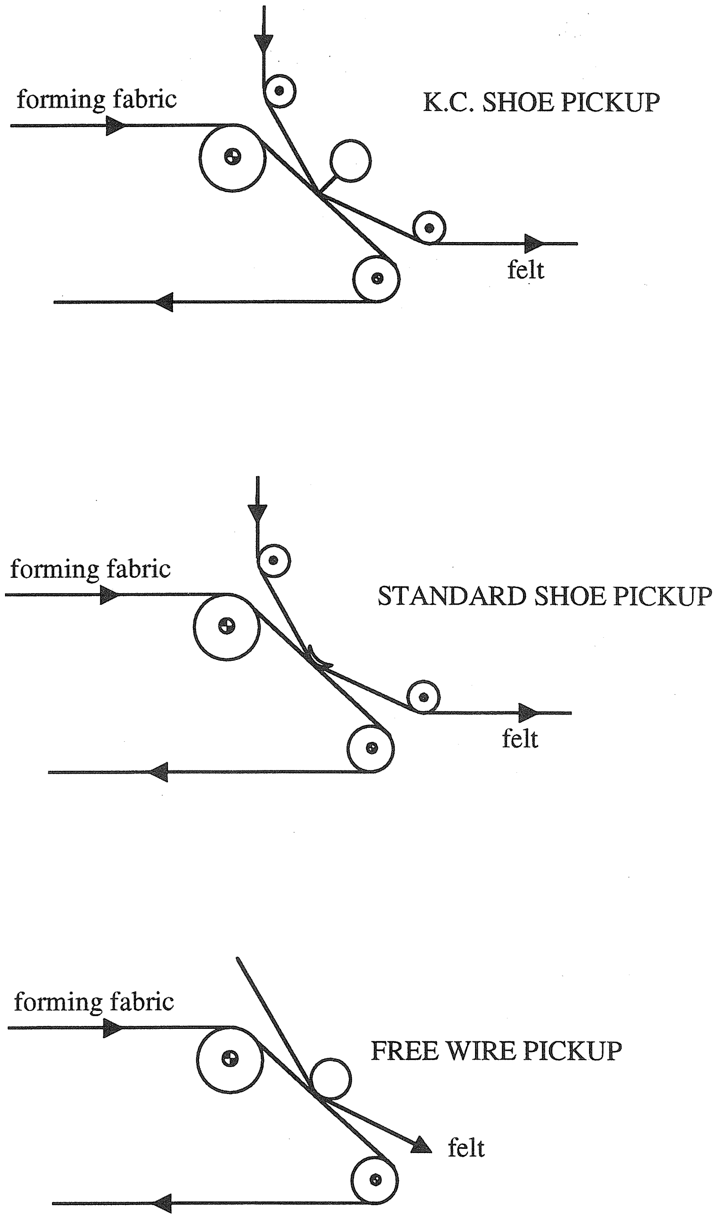


FIGURE 2.117. Sheet transfer from forming to press section.

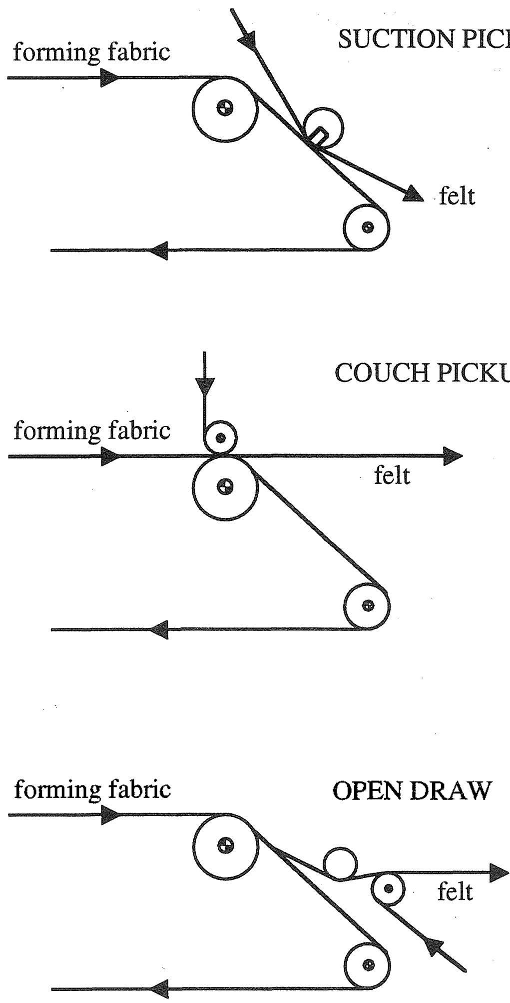


FIGURE 2.117 (continued). Sheet transfer from forming to press section.

2.8 References

1 Beach, E., *Fourdrinier Wet End,* Asten, Inc., 1995.

2 Graham, L. L., and Ring, G. J. F., Introduction to Pulp and Paper Properties and Technology, Course Notes, The Paper Science Department, University of Wisconsin/Stevens Point, 1991.

3 Parker, J. D., *The Sheet-Forming Process,* TAPPI, 1992.

4 *Forming Fabrics: Design and Applications,* Asten, Inc., 1995.

5 Grundstrom, K. J. et al., "High-Consistency Forming of Paper," *TAPPI,* 56(7), 1973.

6 Fellers, C., "The Significance of Structure for the Compression Behavior of Paperboard" in *Paper Structure and Properties,* Ed. J. A. Bristow and P. Kolseth, Marcel Dekker, Inc., New York, 1986.

7 Beach, E. M., "Forming Fabric Design, Effect on Coated Sheet Properties," *TAPPI,* May 1979, Vol. 62, No. 5.

8 Adanur, S., *Wellington Sears Handbook of Industrial Textiles,* Technomic Publishing Co., 1995.

9 American Society for Testing and Materials, *Annual Book of ASTM Standards* (Volumes 7.01 and 7.02), published yearly by the ASTM, Philadelphia, PA.

10 Resistat® Conductive Fiber, BASF Corporation, Williamsburg, Virginia, 1995.

11 Certificate of Quality System Assessment, Shakespeare Monofilament Division, 1994.

12 American Association of Textile Chemists and Colorists, *Technical Manual of the American Association of Textile Chemists and Colorists,* published yearly by AATCC, Research Triangle Park, NC.

13 Beran, R. L., "The Evaluation and Selection of Forming Fabrics," *TAPPI Journal,* Vol. 62, No. 4, 1979.

14 Johnson, D. B., "Retention and Drainage of Multi-Layer Fabrics," *Pulp & Paper Canada,* Vol. 87, 1986.

15 McCumsey, K., *Showering Recommendations,* Asten, Inc., 1994.

16 Thorp, B. A., "Principles of Twin-Wire and Multiple-Wire Formation and Drainage" in TAPPI 1991 Wet End Operations Short Course, Orlando, FL.

17 Smook, G. A., *Handbook for Pulp & Paper Technologists,* Joint Textbook Committee of the Paper Industry, TAPPI and Canadian Pulp and Paper Association, 1989.

18 B. A. Thorp, and M. J. Kocurek, Ed., *Pulp and Paper Manufacture, Vol. 7,* Paper Machine Operations, Joint Textbook Committee of the Paper Industry, TAPPI and Canadian Pulp and Paper Association, 1991.

19 TAPPI Wet End Operations Short Course, Orlando, FL, TAPPI Press, 1991.

20 Hollmark, H., "Mechanical Properties of Tissue" in *Handbook of Physical and Mechanical Testing of Paper and Paperboard,* Ed. R. E. Mark, Vol. 1, Marcel Dekker, Inc., 1983.

21 Broughton, R. M., and Brady, P. H., "Nonwoven Fabrics" in *Wellington Sears Handbook of Industrial Textiles,* Technomic Publishing Co., 1995.

2.9 Review Questions

1 Why is the formation of sheet so critical for paper properties? Explain.

2 Explain how sheet is formed. What are the major forces acting on fibers during formation?

3 Explain the similarities and dissimilarities between textile structures and paper structures.

4 How can you control the orientation of fibers on the forming section of the paper machine? Can you orient the fibers in any direction in three dimensions (XYZ)? Why?

5 What kind of fillers are used in papermaking? What is the purpose of using a filler?

6 Papermaking is an inefficient process in the sense that approximately for every 1 kg of paper, 99 kg of water is required. Could you think of any other more efficient way to form the sheet with or without water?

7 Explain the functions of forming fabrics in sheet manufacturing.

8 What are the advantages and disadvantages of flat versus endless weaving of forming fabrics? Explain.

9 What are the major forming fabric designs? Explain the major differences among them for the following properties:
- drainage capacity
- fiber support
- fabric life
- strength

10 How can you change the percent open area of the forming fabric without changing the hole size?

11 Explain the major steps in forming fabric manufacturing. Why are very heavy

weaving machines required to make forming fabrics? (hint: think of forces involved during weaving of forming fabric).

12 What are the advantages and disadvantages of polyester and nylon yarns in forming fabric structures? Would a 100% nylon forming fabric work? Why?

13 Explain the extrusion process.

14 Calculate the denier and the weight per kg of the following yarns:
- polyester diameter: 0.0055 inch
- nylon 6,6 diameter: 0.030 inch

15 What is the relation between fabric tensile strength and yarn tensile strength? What other factors affect the fabric tensile strength?

16 Can air-jet weaving be used to make forming fabrics? Why?

17 How is pick count changed in weaving of forming fabrics?

18 What are the reasons for preliminary heatsetting of forming fabrics? Explain the effects of heatsetting on fabric modulus with reasons.

19 How can you improve the seam strength of forming fabrics?

20 Can monoplane surface be achieved with proper weaving conditions? How?

21 Calculate the holes/cm^2, hole dimensions and % projected open area of the following single layer forming fabrics:

Mesh	Warp Diameter	Filling Diameter
12 × 14	0.03″	0.03″
24 × 19	0.024″	0.035″
63 × 49	0.0095″	0.0125″
86 × 55	0.0067″	0.009″
95 × 96	0.0055″	0.006″
103 × 107	0.004″	0.005″

22 How is fabric permeability measured? How effective is the fabric permeability in predicting the drainage of forming fabrics?

23 When selecting a forming fabric for a special application, what kind of paper and fabric properties are need to be considered?

24 Define contaminant resistance? Explain the ways to make forming fabrics more contaminant resistant?

25 What causes fabrics to bow and skew? What can be done to correct the problem?

26 Explain the following terms:
- pressure forming
- velocity forming
- rush/drag ratio

27 Why is turbulence is necessary in sheet formation?

28 In a gap roll former, if the tension on the outer fabric is 6 kg/cm, the forming roll radius is 75 cm, calculate the dewatering pressure (refer to Figure 2.87).

29 Explain the spunbonding process.